LeanForControl

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Chapter 1

Analysis toolbox

Facts about real functions, limits and integrals with no differential-equation content of their own. Each section says which later results consume it.

1.1 Continuity toolbox

Continuity gives the last root before a sign change and the first exit of a curve from an open region. First sphere crossing is the open-ball specialization of the general exit lemma.

Lemma 1 (Greatest zero of a sign-changing continuous function)**.** *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \le 0 < g(t_1)$. Then g has a* last *root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof. □

Lemma 2 (Boundary crossing by a continuous curve)**.** *A continuous curve on $[t_0, t_1]$, with $t_0 \le t_1$, starting in an open set D and ending outside D meets ∂D on that interval.*

Proof. The image of the interval is connected. If it avoided the boundary, meeting D would force the whole image to lie in D. □

Lemma 3 (First exit from an open set)**.** *If a continuous curve on $[t_0, T]$ starts in an open set D and is outside D at $t_1 \in [t_0, T]$, some $\tau \in [t_0, t_1]$ satisfies $\varphi(\tau) \in \partial D$ and $\varphi(t) \in D$ for $t \in [t_0, \tau)$.*

Proof. The nonempty compact set of exit times has a minimum. A boundary crossing by that time must occur at the minimum itself. □

Theorem 4 (First crossing of a sphere)**.** *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \le \|\varphi(t_1) - x_{\mathrm{eq}}\|$. Then there is a* first *hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ:*

$$\|\varphi(\tau) - x_{\mathrm{eq}}\| = \rho, \qquad \|\varphi(s) - x_{\mathrm{eq}}\| \le \rho \quad \forall\, s \in [t_0, \tau].$$

Proof. Apply lemma 3 to the open ball. Its boundary lies in the sphere; before the first exit the curve remains in the ball. □

A field cut off outside a set stays Lipschitz.

1.2 Limits superior

Real-valued lim sup is an infimum of upper bounds, so it is only well-behaved where the function is bounded. The facts collected here are stated over an arbitrary filter and have no differential-equation content; the Dini derivative of the next section is the only current consumer.

Lemma 5 (Limsup is unchanged by adding a null term)**.** *Let l be a nontrivial filter, let f be bounded above and below along l, and let $g \to 0$ along l. Then $\limsup_l (f+g) = \limsup_l f$.*

Proof. One inequality is subadditivity of lim sup together with $\limsup_l g = 0$. For the other, write $f = (f+g) + (-g)$ and apply the same argument, $-g$ also tending to 0; the boundedness of $f+g$ needed for this follows from that of f and g separately. □

1.3 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation the comparison lemma is in. It is defined here as a lim sup valued in $\mathbb{R}$ rather than in $[-\infty, +\infty]$, so most results below carry a hypothesis that the forward difference quotients are bounded: without it an unbounded quotient makes the lim sup an infimum over the empty set, which $\mathbb{R}$ reports as 0.

Definition 6 (Upper right Dini derivative)**.** The *upper right Dini derivative* of $f : \mathbb{R} \to \mathbb{R}$ at t is

$$D^+ f(t) \;=\; \limsup_{h \to 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

1.3.1 Relationship to the classical derivative

Lemma 7 (Forward difference quotients of a differentiable function)**.** *If f has right derivative L at t, then $\frac{f(t+h)-f(t)}{h} \to L$ as $h \to 0^+$.*

Proof. □

Lemma 8 (Dini derivative of a differentiable function)**.** *If f has right derivative L at t, then $D^+ f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

1.3.2 Characterization by difference quotients

Lemma 9 (Upper bound on the Dini derivative)**.** *Let the forward difference quotients of f at t be bounded above and cobounded below along $h \to 0^+$. Then $D^+ f(t) \le L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \to 0^+$.*

Proof. □

Lemma 10 (Lower bound on the Dini derivative)**.** *Under the same boundedness conditions as lemma 9, $L \le D^+ f(t)$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ exceeds $L - \varepsilon$ frequently as $h \to 0^+$.*

Proof. □

1.3.3 Linearity properties

Lemma 11 (Subadditivity of the Dini derivative)**.** *The upper right Dini derivative is subadditive: $D^+(f+g)(t) \le D^+ f(t) + D^+ g(t)$, provided the forward difference quotients of f are bounded above and below and those of g are bounded above and cobounded below.*

Proof. □

Lemma 12 (Adding a differentiable function)**.** *If the forward difference quotients of f at t are bounded above and below, and g is differentiable at t, then $D^+(f+g)(t) = D^+ f(t) + g'(t)$. Subadditivity becomes equality once one summand has a genuine derivative.*

Proof. □

Lemma 13 (Adding a linear function)**.** *If the forward difference quotients of f at t are bounded above and below, then $D^+(f + c\,\mathrm{id})(t) = D^+ f(t) + c$ — the case $g(s) = cs$ of lemma 12, and the form the comparison arguments use.*

Proof. □

Lemma 14 (Positive scalar multiple)**.** *For $c > 0$, and forward difference quotients of f at t bounded above and cobounded below, $D^+(cf)(t) = c\,D^+ f(t)$. Positivity is essential: a negative constant reverses the order and turns the* lim sup *into a* lim inf*.*

Proof. Multiplication by $c > 0$ is an order isomorphism of $\mathbb{R}$, and a lim sup commutes with one. Going through the order isomorphism rather than through the defining infimum avoids the degenerate empty-inf case. □

Lemma 15 (Negation and the lower Dini derivative)**.** *If the forward difference quotients of f at t are bounded above and below, then $-D^+(-f)(t) \le D^+ f(t)$. The left-hand side is the* lower *right Dini derivative, so this is the expected inequality $D_+ f(t) \le D^+ f(t)$ written in terms of D^+ alone.*

Proof. □

1.3.4 The comparison step

Lemma 16 (Dini derivative from a one-sided bound)**.** *Let $a < b$, let $z(a) = v(a)$ and $z(t) < v(t)$ for all $t \in (a, b]$, let z have right derivative d_z at a, and let the forward difference quotients of v at a be bounded above. Then $d_z \le D^+ v(a)$.*

Proof. On $(0, b-a)$ the hypotheses give $z(a+h) - z(a) \le v(a+h) - v(a)$, so the forward quotient of z is eventually below that of v. The quotient of z converges to d_z, so d_z = lim sup of it, and lim sup is monotone. □

Lemma 17 (Nonincreasing above a level)**.** *Let v be continuous on $[a,b]$ with right derivative $v'(t)$ at every $t \in [a,b)$, and suppose $v'(t) \le 0$ whenever $v(t) > \eta$. Then $v(t) \le \max\{v(a), \eta\}$ for all $t \in [a,b]$.*

Proof. Let $m = \max\{v(a), \eta\}$ and $\varepsilon > 0$. The barrier $B(t) = m + \varepsilon\,(1 + t - a)$ starts above v, and wherever v touches it, $v > \eta$, so $v' \le 0 < \varepsilon = B'$. The barrier comparison theorem gives $v \le B$ on $[a,b]$; let $\varepsilon \to 0$. □

1.4 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction of the chapter on comparison functions.

Lemma 18 (Difference of integrals over adjacent intervals)**.** *For μ continuous on $[a,t]$ and $s \in [a,t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Lemma 19 (Integral bound from a pointwise multiplicative bound)**.** *Let u take values in a normed space and v be real, both integrable on $[a,b]$ with $a \le b$. If $\|u(s)\| \le L\, v(s)$ for all $s \in [a,b]$, then*

$$\Big\|\int_a^b u\Big\| \le L \int_a^b v.$$

Proof. □

Lemma 20 (Integral bound from a constant pointwise bound)**.** *If $\|u(s)\| \le C$ for all $s \in [a,b]$ with $a \le b$, then*

$$\Big\|\int_a^b u\Big\| \le C\,(b-a).$$

Proof. □

Lemma 21 (Integral of a constant)**.** $\int_a^b C\,\mathrm{d}s = (b-a)\,C$ *for a real constant C.*

Proof. □

Lemma 22 (Interval integrability of a continuous composition)**.** *If f is jointly continuous and z is continuous on the segment between t_0 and t_1, then $s \mapsto f(s, z(s))$ is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Lemma 23 (Differentiating an integral in its upper limit)**.** *For μ continuous on $[a,b]$ and $t \in (a,b)$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=t} \int_a^x \mu(\tau)\,\mathrm{d}\tau \;=\; \mu(t).$$

Proof. □

Lemma 24 (Continuity of an integral in its upper limit)**.** *If f is integrable on $[a,t]$ with $a \le t$, then $s \mapsto \int_a^s f(\tau)\,\mathrm{d}\tau$ is continuous on $[a,t]$.*

Proof. □

Lemma 25 (Continuity of an integral in both endpoints)**.** *Let f be locally integrable on $(0,\infty)$, and let a, b be continuous on a set S with $a(\eta), b(\eta) > 0$ in S. Then*

$$\eta \;\longmapsto\; \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is continuous on S.

Proof. □

Lemma 26 (The sliding average of an antitone function is antitone)**.** *Let f be antitone and locally integrable on $(0,\infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \;\longmapsto\; \frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is antitone on S.

Proof. □

Lemma 27 (Sliding average of an eventually vanishing function)**.** *Let f be nonneg, antitone and locally integrable on $(0,\infty)$ with $f(s) \to 0$ as $s \to +\infty$, and let $a(\eta) < b(\eta)$ with $a(\eta) \to +\infty$ as $\eta \to +\infty$. Then*

$$\frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x \;\longrightarrow\; 0 \qquad (\eta \to +\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

1.5 Norm inequalities

1.6 One-sided Lipschitz maps

1.7 Positive definite functions

Positive definiteness in the sense of functions, not matrices: a function that vanishes at a point and is positive elsewhere. It is the shape every Lyapunov function and every comparison bound takes.

Definition 28 (Positive definite function)**.** A function W is *positive definite* on S about x_0 when it is continuous on S, $W(x_0) = 0$, and $W(x) > 0$ for every $x \in S \setminus \{x_0\}$.

Chapter 2

Ordinary differential equations

2.1 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 29 (Gronwall–Bellman inequality)**.** *Khalil, Lemma A.1. Let $\Lambda, \mu : [a,b] \to \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a,b] \to \mathbb{R}$ be continuous satisfying*

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \Lambda(s)\, \mu(s)\, e^{\int_s^t \mu(\tau)\,\mathrm{d}\tau}\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \geq 0$ is exactly the hypothesis. The transform $w = e^{-M} z$ has, by the product rule and lemma 23,

$$\dot{w} = e^{-M}\mu\,(y - z) = e^{-M}\mu\,(\Lambda - v) \;\leq\; e^{-M}\mu\Lambda,$$

the inequality because e^{-M}, μ and v are all nonnegative — this is the only place $\mu \geq 0$ is used, and it is essential: a sign change would reverse it. Integrating from a, where $w(a) = 0$, gives $w(t) \leq \int_a^t e^{-M(s)}\mu(s)\Lambda(s)\, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)}e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \leq \Lambda + z$ finishes. □

Lemma 30 (Gronwall–Bellman with a constant rate)**.** *Khalil, Lemma A.1 (the case $\Lambda(t) \equiv \Lambda$ constant). Let μ be continuous and nonnegative on $[a,b]$, and let y be continuous with*

$$y(t) \;\leq\; C + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \leq C\, e^{\int_a^t \mu(\tau)\,\mathrm{d}\tau}$ on $[a,b]$.

Proof. theorem 29 at $\Lambda \equiv C$ gives $y(t) \le C + \int_a^t C\mu(s) e^{\int_s^t \mu}\, ds$, and the integral evaluates in closed form: $s \mapsto -C\, e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $Ce^{M(t)} - C$. The two C's cancel. □

Lemma 31 (Gronwall–Bellman with constant data)**.** *Khalil, Lemma A.1 (the case Λ and $\mu \ge 0$ both constant). Let $\mu \ge 0$ be constant and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Then $y(t) \le C\, e^{\mu(t-a)}$ on $[a, b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. lemma 30 with $\int_a^t \mu = \mu(t-a)$. □

2.2 Existence, uniqueness and continuous dependence

Definition 32 (Integral solution)**.** A function x, defined on the segment between t_0 and t_1, is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) \;=\; x_0 + \int_{t_0}^t F(s,\, x(s))\, \mathrm{d}s$$

for every t between t_0 and t_1.

Lemma 33 (Re-anchoring an integral solution)**.** *Let x be a continuous integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1, with F jointly continuous. Then for any s, r on that segment, x is an integral solution on the segment between s and r, with initial value $x(s)$:*

$$x(t) \;=\; x(s) + \int_s^t F(w, x(w))\, \mathrm{d}w \qquad \text{for } t \text{ between } s \text{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s, which is legitimate because $w \mapsto F(w, x(w))$ is interval-integrable on each piece. □

Lemma 34 (Time reflection of an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1. Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0, of the reflected field*

$$(r, y) \;\longmapsto\; -F(t_0 + t_1 - r,\; y)$$

on the segment between t_1 and t_0.

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

2.2.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Lemma 35 (Integral curves are integral solutions)**.** *Let x be an integral curve of F on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) \;=\; x(t_0) + \int_{t_0}^{t} F(s, x(s))\,\mathrm{d}s.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. □

Lemma 36 (Integral solutions are integral curves)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. □

Lemma 37 (Integral and differential forms agree)**.** *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1, the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are lemma 36 and lemma 35. □

Lemma 38 (Integral and differential forms agree, forward time)**.** *lemma 37 in forward time: for $t_0 \le t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. □

Lemma 39 (Integral form for an autonomous field)**.** *Let g be a continuous* autonomous *field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \le t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of lemma 38, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. □

2.2.2 Time invariance and local existence

A time-invariant field has no preferred time origin: a solution can be shifted to start at time 0.

Lemma 40 (Time invariance)**.** *If x is an integral curve of an autonomous field g on a set s, then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt.*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated *equation instead.*

Proof. □

Lemma 41 (Shifting a solution segment to the origin)**.** *lemma 40 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Lemma 42 (Local existence for a C^1 field)**.** *Let g be an autonomous field that is C^1 at x_0. Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1.*

The anchor 0 is a construction choice rather than a restriction — by lemma 40 a segment from any other anchor follows.

Reference: the Picard–Lindelöf local existence theorem.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 43 (Picard–Lindelöf on a compact interval)**.** *Let $f : \mathbb{R} \to E \to E$ be jointly continuous and K-Lipschitz in the state variable, uniformly in time, on a Banach space E. Then for every $t_0 \le t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1: it forces linear growth in the state, ruling out finite-time blowup.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2 (Global Existence and Uniqueness), proved in his Appendix C.1. Difference: Khalil assumes f piecewise continuous in t; here it is jointly continuous.*

Proof. Mathlib's Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \le L$ on a ball of radius a about x_0 with $L\,\Delta t \le a$. For a K-Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M\Delta t \le a(1 - K\Delta t)$ — solvable in a exactly when $K\Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \le 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \le M + Ka = M(1 + 2Kh) \le 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$ from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \ge t_1 - t_0$ covers $[t_0, t_1]$. □

Lemma 44 (Right derivatives of an integral curve)**.** *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s, i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s.*

Proof. □

Lemma 45 (Gronwall separation of two solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then*

$$d(\alpha(t), \beta(t)) \le d(\alpha(t_0), \beta(t_0))\, e^{K(t-t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Uniqueness (lemma 46) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, in the unperturbed case: two solutions of the* same *field, so the perturbation bound is zero.*

Proof. □

Lemma 46 (Uniqueness of solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), the uniqueness half of Theorem 3.2.*

Proof. The initial distance in lemma 45 is zero, so the bound forces the distance to vanish throughout. □

Theorem 47 (Picard–Lindelöf on the forward ray)**.** *Let f be jointly continuous and K-Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot\alpha = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.*

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, whose t_1 is arbitrary, extended to the forward ray.*

Proof. For each n take a solution α_n on $[t_0, t_0 + n]$ (theorem 43). By uniqueness (lemma 46) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{\lceil t - t_0 \rceil}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0 + n]$. Near any $t \ge t_0$ choose $n > t - t_0$; then $[t_0, t_0 + n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n. □

Theorem 48 (Picard–Lindelöf, scalar integral form)**.** *Let $g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ be jointly continuous and L-Lipschitz in its state argument, uniformly in time, and let $t_0 \le t_1$. Then for every x_0 there is a z, continuous on $[t_0, t_1]$, that is an integral solution of $\dot z = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1)$.*

This is the scalar specialization of theorem 43; it discharges the perturbed-solution hypothesis that theorem 56 takes for generality.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, scalar case.*

Proof. Apply theorem 43 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (lemma 38); the right derivatives are lemma 44. □

2.2.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 49 (Continuous dependence on initial states and parameters)**.** *Khalil, Theorem 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on the segment between t_0 and t_1, let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \le \mu$. Then for every t on that segment,*

$$\|y(t) - z(t)\| \;\le\; \|y_0 - z_0\|\, e^{L|t-t_0|} \;+\; \frac{\mu}{L}(e^{L|t-t_0|} - 1).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z, never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \;\le\; \|y_0 - z_0\| + \mu(\tau - t_0) + L\int_{t_0}^{\tau} \|y(s) - z(s)\|\,\mathrm{d}s,$$

the Lipschitz hypothesis bounding the f-difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape lemma 31 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: lemma 34 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. □

The same mechanism bounds a single solution from below: an equilibrium cannot be approached faster than exponentially.

Theorem 50 (Continuous dependence on parameters)**.** *Khalil, Theorem 3.5. If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \le \alpha$ and $\|z_0 - y_0\| \le \alpha$, and $\alpha(1 + 1/L)e^{L(t_1-t_0)} \le \varepsilon$, then $\|y(t) - z(t)\| \le \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Specialise theorem 49 with $\mu = \alpha$ and $\|y_0 - z_0\| \le \alpha$: for $t \in [t_0, t_1]$ it gives $\|y(t) - z(t)\| \le \alpha e^{L(t-t_0)} + \frac{\alpha}{L}(e^{L(t-t_0)} - 1) = \alpha(1 + 1/L)e^{L(t-t_0)} - \alpha/L$. Drop the $-\alpha/L$ and bound $e^{L(t-t_0)}$ by $e^{L(t_1-t_0)}$ to get at most ε. □

2.3 Differentiability in the initial state

For a C^1 vector field the solution at a fixed time depends differentiably on its initial state, with derivative the solution of the variational equation along it. Appending time to the state turns a time-varying system into an autonomous one, so this also gives differentiability in the initial time.

Lemma 51 (Local Lipschitz dependence on the initial state)**.** *Let $g : E \to E$ be C^1 on a finite-dimensional space and γ a solution of $\dot z = g(z)$ on $[0,T]$. There are $K \geq 0$ and $\rho > 0$ such that any two solutions β, β' on $[0,T]$ starting within ρ of $\gamma(0)$ satisfy $\|\beta(s) - \beta'(s)\| \leq \|\beta(0) - \beta'(0)\|\, e^{Ks}$ for $s \in [0,T]$.*

Proof. Let K bound $\|Dg\|$ on the compact unit tube around γ, so that g is K-Lipschitz on each unit ball $\overline{B}(\gamma(t), 1)$. With $\rho = e^{-KT}/2$, a solution starting within ρ of $\gamma(0)$ stays in the tube: at the first time it reached distance 1 from γ, Gronwall's inequality on the tube would put it within $\rho e^{KT} = 1/2$. Inside the tube, Gronwall's inequality compares any two such solutions. □

Lemma 52 (Existence of the variational solution)**.** *Let $g : E \to E$ be C^1 on a finite-dimensional space and γ a solution of $\dot z = g(z)$ on $[0,T]$, $T \geq 0$. The variational equation $\dot\Phi = Dg(\gamma(s))\,\Phi$, $\Phi(0) = I$, has a solution on $[0,T]$.*

Proof. Freeze the coefficient outside $[0,T]$: $A(s) = Dg(\gamma(\pi(s)))$ with π the projection onto $[0,T]$. It is continuous and bounded, so $\Phi \mapsto A(s)\Phi$ is globally Lipschitz, uniformly in s, and Picard–Lindelöf (theorem 43) gives a solution; on $[0,T]$ the frozen coefficient is the original one. □

Theorem 53 (First-order sensitivity estimate)**.** *Let $g : E \to E$ be C^1 on a finite-dimensional space, γ a solution of $\dot z = g(z)$ on $[0,T]$, and Φ a solution of the variational equation $\dot\Phi = Dg(\gamma(s))\Phi$, $\Phi(0) = I$. For every $\varepsilon > 0$ there is $\rho > 0$ such that every solution β on $[0,T]$ with $\|\beta(0) - \gamma(0)\| < \rho$ satisfies*

$$\|\beta(s) - \gamma(s) - \Phi(s)(\beta(0) - \gamma(0))\| \leq \varepsilon\, \|\beta(0) - \gamma(0)\| \qquad \forall\, s \in [0,T].$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 3.3.*

Proof. Let $h = \beta(0) - \gamma(0)$ and $e = \beta - \gamma - \Phi h$, so $e(0) = 0$ and

$$\dot e = Dg(\gamma)\, e + \big(g(\beta) - g(\gamma) - Dg(\gamma)(\beta - \gamma)\big).$$

By lemma 51, $\|\beta - \gamma\| \leq \|h\| e^{KT}$, which is below the radius of **??** for a small η; so the bracket is at most $\eta e^{KT}\|h\|$, and $\|Dg(\gamma)\| \leq K'$. Gronwall's inequality gives $\|e(s)\| \leq \eta e^{KT}\|h\|\, G$ with G depending only on K' and T; choose η with $\eta e^{KT} G \leq \varepsilon$. □

Theorem 54 (Differentiability of solutions in the initial state)**.** *Let $g : E \to E$ be C^1 on a finite-dimensional space, and for every $z \in E$ let $\Psi(\cdot, z)$ be a solution of $\dot z = g(z)$ on $[0,T]$ with $\Psi(0,z) = z$. Let Φ solve the variational equation along $\Psi(\cdot, z_0)$. Then for $s \in [0,T]$, $z \mapsto \Psi(s,z)$ is differentiable at z_0 with derivative $\Phi(s)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 3.3.*

Proof. theorem 53 is the little-o estimate defining the derivative. □

2.4 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Lemma 55 (Integral form from a pointwise derivative)**.** *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^{s} f(\tau, u(\tau))\, \mathrm{d}\tau.$$

.

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. □

Theorem 56 (Comparison lemma)**.** *Khalil, Lemma 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \le u_0$, satisfies the Dini subsolution inequality $D^+v(t) \le f(t, v(t))$, and has forward difference quotients bounded above at each t;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \le u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are taken as a hypothesis for generality, so that callers with solutions already in hand need not re-derive them. Under the hypotheses already listed they always exist: $f + \lambda$ is Lipschitz exactly when f is, so theorem 48 discharges (3).

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \le z_\lambda$ on $[t_0, t_1]$. Otherwise $v(t_{\mathrm{bad}}) > z_\lambda(t_{\mathrm{bad}})$ for some t_{bad}, and since $v(t_0) \le z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\mathrm{bad}}$, with $v > z_\lambda$ on $(a, t_{\mathrm{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \le D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \;\le\; D^+v(a) \;\le\; f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 50 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \le \varepsilon/2$ on $[t_0, t_1]$. Then $v \le z_\lambda \le u + \varepsilon$. As ε was arbitrary, $v \le u$. □

2.5 Lie derivatives

The rate of change of a function $V(t, x)$ along the integral curves of a time-dependent vector field, with no differential-equation content beyond the chain rule: what every Lyapunov, Chetaev and invariance argument differentiates.

Definition 57 (Lie derivative along a vector field)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and a time-dependent vector field $v : \mathbb{R} \times E \to E$, the *Lie derivative of V along v* is

$$L_v V(t, x) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, v(t, x),$$

the derivative of V at (t, x) in the direction $(1, v(t, x))$: the Lie derivative of V, as a function on $\mathbb{R} \times E$, along the autonomous field $(t, x) \mapsto (1, v(t, x))$.

Lemma 58 (Lie derivative of a time-invariant function)**.** *If V does not depend on t and is differentiable at x, then $L_v V(t,x) = DV(x)\, v(t,x)$.*

Proof. □

Lemma 59 (Lie derivative of $V(x) + ct$)**.** *If V does not depend on t and is differentiable at x, then the Lie derivative of $(t,x) \mapsto V(x) + ct$ along v is $DV(x)\, v(t,x) + c$.*

Proof. □

Lemma 60 (Chain rule along an integral curve)**.** *Let γ be an integral curve of v on I, and V differentiable at $(t, \gamma(t))$ for $t \in I$. Then for $t \in I$, within I,*

$$\frac{d}{dt} V(t,\gamma(t)) = L_v V(t,\gamma(t))$$

(definition 57).

Proof. □

Lemma 61 (The Lie derivative along a solution)**.** *Let γ be an integral curve of v on $[t,\infty)$, V differentiable at $(\tau,\gamma(\tau))$ for $\tau \geq t$, and let $\tau \mapsto V(\tau,\gamma(\tau))$ have right derivative w at t. Then $L_v V(t,\gamma(t)) = w$.*

Proof. By lemma 60 the right derivative is also $L_v V(t,\gamma(t))$, and right derivatives are unique. □

Lemma 62 (Nonpositive Lie derivative along an integral curve)**.** *Let γ be an integral curve of v on an interval s, V differentiable at $(t,\gamma(t))$ for $t \in s$, and $L_v V(t,\gamma(t)) \leq 0$ at every interior point t of s. Then $t \mapsto V(t,\gamma(t))$ is nonincreasing on s.*

Proof. The mean value theorem, with the derivative given by lemma 60. □

Lemma 63 (Exponential bound from a linear Lie-derivative bound)**.** *Let γ be an integral curve of v on $[t_0,t_1]$, V differentiable at $(t,\gamma(t))$ for $t \in [t_0,t_1]$, and $c \in \mathbb{R}$ with $L_v V(t,\gamma(t)) \leq -c\, V(t,\gamma(t))$ for $t \in (t_0,t_1)$. Then $V(t,\gamma(t)) \leq e^{-c(t-t_0)}\, V(t_0,\gamma(t_0))$ on $[t_0,t_1]$.*

Proof. The derivative of $e^{ct} V(t,\gamma(t))$ is $e^{ct}(cV + L_v V) \leq 0$ (lemma 60), so it does not increase. □

Lemma 64 (Continuity of the Lie derivative)**.** *If V is C^1 and v is continuous, then $(t,x) \mapsto L_v V(t,x)$ is continuous.*

Proof. □

2.6 ω-limit sets

The points a curve or sequence returns arbitrarily close to, arbitrarily late: its cluster points at infinity, which is the ω-limit of topological dynamics for a single orbit. These are the topological facts behind LaSalle's invariance principle, in continuous and in discrete time alike: a continuous function converging along the orbit is constant on its ω-limit set, an orbit trapped in a closed set has its ω-limit set there, and an orbit trapped in a compact set converges to its ω-limit set.

Definition 65 (ω-limit set)**.** The *ω-limit set* of $\varphi : \tau \to X$, for τ a preordered index set such as $\mathbb{R}$ or $\mathbb{N}$, is the set of cluster points of φ as $t \to \infty$:

$$\omega(\varphi) = \bigcap_T \overline{\{\varphi(t) : t \geq T\}},$$

the points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \to \infty$.

Lemma 66 (Membership in the ω-limit set)**.** *$y \in \omega(\varphi)$ (definition 65) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.*

Proof. □

Lemma 67 (The ω-limit set as Mathlib's ω-limit)**.** *definition 65 is the ω-limit, in the sense of topological dynamics, of the one-member family $\{\varphi\}$.*

Proof. □

Lemma 68 (A limit of $g \circ \varphi$ is the value of g on $\omega(\varphi)$)**.** *Let $g : X \to Y$ be continuous, Y Hausdorff, and $g(\varphi(t)) \to L$ as $t \to \infty$. Then $g(y) = L$ for every $y \in \omega(\varphi)$ (definition 65).*

Proof. $g(y)$ is a cluster point of $g \circ \varphi$, which converges to L; in a Hausdorff space the two coincide. □

Lemma 69 (The ω-limit set of a curve trapped in a closed set)**.** *If Ω is closed and $\varphi(t) \in \Omega$ for all large t, then $\omega(\varphi) \subseteq \Omega$.*

Proof. A point outside Ω has the open neighbourhood $X \setminus \Omega$, which φ eventually avoids. □

Lemma 70 (Convergence to a set containing the ω-limit set)**.** *If $\varphi(t)$ lies in a compact set for all large t and $\omega(\varphi) \subseteq M$, then $\varphi(t) \to M$: for every open $W \supseteq M$, $\varphi(t) \in W$ for all large t.*

Proof. Mathlib's `eventually_mapsTo_of_isCompact_absorbing_of_isOpen_of_omegaLimit_subset` (lemma 67): a family eventually absorbed by a compact set is eventually inside every open set containing its ω-limit set. □

Chapter 3

Systems

The most general systems the library considers are $\dot{x} = F(t, x, u)$ in continuous time and $x^+ = F(k, x, u)$ in discrete time, each with an output $y = h(t, x, u)$. As for the linear systems of the next chapter, there is one object per time domain: the data has the same shape, but a continuous trajectory is an integral curve and a discrete one is a recursion.

Special cases are constructors on the general object, each with a predicate recognising it: *time invariance* drops t, and *autonomy* drops u. Autonomy is used here in the control-theory sense, *without input*; much of the dynamical-systems literature uses the word for time invariance instead.

3.1 Continuous time

Definition 71 (Continuous-time system)**.** A *continuous-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{R} \times E \times U \to E, \qquad h : \mathbb{R} \times E \times U \to Y,$$

to be read as $\dot{x} = F(t, x, u)$, $y = h(t, x, u)$. The structure is only the data.

Definition 72 (Time-invariant continuous-time system)**.** The continuous-time system $\dot{x} = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 73 (Time invariance, continuous-time system)**.** A continuous-time system is *time-invariant* when $F(t, \cdot, \cdot)$ and $h(t, \cdot, \cdot)$ are the same for every t.

Lemma 74 (Time-invariant continuous-time systems are time-invariant)**.** *The system of definition 72 is time-invariant (definition 73).*

Proof. □

Lemma 75 (Time-invariant continuous-time systems come from time-free data)**.** *A time-invariant continuous-time system (definition 73) is* timeInvariant *f g for some f and g. The converse of lemma 74.*

Proof. Read f and g off at $t = 0$; time invariance says every other time agrees with it. □

Definition 76 (Autonomous continuous-time system)**.** The continuous-time system $\dot{x} = f(t, x)$, $y = g(t, x)$, which has no input: its input space is the one-point type.

Definition 77 (Autonomy, continuous-time system)**.** A continuous-time system is *autonomous* when $F(t, x, u)$ and $h(t, x, u)$ do not depend on u.

Lemma 78 (Autonomous continuous-time systems are autonomous)**.** *The system of definition 76 is autonomous (definition 77).*

Proof. □

Lemma 79 (Autonomous continuous-time systems come from input-free data)**.** *If a continuous-time system is autonomous (definition 77) and its input space is nonempty, there are f and g with $F(t, x, u) = f(t, x)$ and $h(t, x, u) = g(t, x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 80 (The unforced system)**.** The autonomous system (definition 76) $\dot{x} = F(t, x, 0)$, $y = h(t, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 81 (Trajectories of the unforced system)**.** *A map is a trajectory of the unforced system (definition 80) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Well-posedness is kept apart from stability. The stability notions quantify over every solution, so they need no uniqueness; forward completeness enters only where a limit as $t \to \infty$ is taken. Theorems that need either property assume it.

Definition 82 (Forward completeness from a set)**.** A continuous-time system is *forward complete* from a set S at time t_0 under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 89) with $x(t_0) \in S$ agrees on that segment with a trajectory on $[t_0, \infty)$.

Definition 83 (Forward completeness)**.** A continuous-time system is *forward complete* under the input u when it is forward complete (definition 82) from every state at every initial time.

Definition 84 (Uniqueness of solutions)**.** A continuous-time system has *unique solutions* under the input u when any two trajectories on a segment $[t_0, t_1]$ (definition 89) that agree at t_0 agree on all of $[t_0, t_1]$.

Theorem 85 (Lipschitz systems have unique solutions)**.** *If $x \mapsto F(t, x, u(t))$ is K-Lipschitz for every t, the system has unique solutions under u (definition 84).*

Proof. Grönwall's inequality, in the form of Mathlib's `ODE_solution_unique_of_mem_Icc_right`: a derivative within $[t_0, t_1]$ at a point of $[t_0, t_1)$ is a right derivative there. □

Theorem 86 (Globally Lipschitz systems are forward complete)**.** *If $(t, x) \mapsto F(t, x, u(t))$ is continuous and K-Lipschitz in x uniformly in t, the system is forward complete under u (definition 83).*

Proof. From the endpoint $x(t_0)$ of a segment, Picard–Lindelöf gives a solution on $[t_0, \infty)$ (theorem 47), which agrees with the segment by uniqueness (theorem 85). □

3.1.1 The state transition map

The state transition map $\varphi(t; t_0, x_0)$ is the solution from x_0 at time t_0, evaluated at time t, on its maximal interval of existence. The state transition matrix of a linear system is its linear case.

Definition 87 (State equation of a continuous-time system)**.** Under an input signal $u : \mathbb{R} \to U$, the right-hand side of the state equation is the time-dependent vector field $(t, x) \mapsto F(t, x, u(t))$.

Definition 88 (Output equation of a continuous-time system)**.** Under an input signal u, the output equation is $(t, x) \mapsto h(t, x, u(t))$. An instantaneous relation: no derivative appears.

Definition 89 (Trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when

$$\dot{x}(t) = F(t, x(t), u(t)), \qquad \forall\, t \in I,$$

the derivative being taken within I. Stated on a set: a map can satisfy the equation on an interval without being defined for all time.

Definition 90 (Trajectory for all time of a continuous-time system)**.** A trajectory (definition 89) defined on all of $\mathbb{R}$.

Lemma 91 (Trajectories of a continuous-time system on the whole line)**.** *definition 89 at $I = \mathbb{R}$ is definition 90.*

Proof. □

Lemma 92 (Restriction of a trajectory of a continuous-time system)**.** *A trajectory for all time is a trajectory on every set.*

Proof. □

Inputs that switch — steps, sample-and-hold, bang-bang — produce solutions with corners at the switching times. With the input taken right-continuous, such a solution still has a right derivative everywhere, given by the new value of the input; these right trajectories are the solution concept for input-to-state stability, and every trajectory is one.

A set is forward invariant when trajectories that start in it stay in it.

Definition 93 (Forward invariant set)**.** A set $S \subseteq E$ is *forward invariant* (or positively invariant) for s under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 89) that starts in S stays in S:

$$x(t_0) \in S \;\Rightarrow\; x(t) \in S \quad \forall\, t \in [t_0, t_1].$$

The derivative of a function along the system is the rate at which it changes along trajectories. It takes the input value as an argument: stability uses the input-free system, input-to-state notions a general input.

Definition 94 (Derivative along the system)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and an input value $u \in U$, the *derivative of V along the system $\dot{x} = F(t, x, u)$* is

$$\dot{V}(t, x, u) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, F(t, x, u),$$

the Lie derivative of V along the field $F(\cdot, \cdot, u)$ (definition 57).

The derivative form above excludes the solutions produced by discontinuous inputs, which have corners where the input jumps. The integral form admits them, and is the one input-driven notions such as input-to-state stability are stated with; for continuous right-hand sides the two agree.

Definition 95 (Carathéodory trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$, with E a Banach space, is a *Carathéodory trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when, for all $a \leq b$ in I, the map $\tau \mapsto F(\tau, x(\tau), u(\tau))$ is integrable on $[a, b]$ and

$$x(b) - x(a) = \int_a^b F(\tau, x(\tau), u(\tau))\, \mathrm{d}\tau.$$

Unlike definition 89 it does not require x to be differentiable, so it admits the solutions produced by discontinuous or merely measurable inputs, which input-to-state stability quantifies over.

Definition 96 (Carathéodory trajectory for all time)**.** A Carathéodory trajectory (definition 95) on all of $\mathbb{R}$.

Lemma 97 (Restriction of a Carathéodory trajectory)**.** *A Carathéodory trajectory on I is one on every subset of I; in particular one for all time is one on every set.*

Proof. □

Theorem 98 (Carathéodory and classical trajectories agree on a segment)**.** *Let $t_0 \leq t_1$ and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on $[t_0, t_1]$. Then x is a Carathéodory trajectory on $[t_0, t_1]$ (definition 95) if and only if it is a trajectory there (definition 89).*

Proof. Both are equivalent to the integral equation anchored at t_0 (definition 32), by lemma 37. The anchored equation gives the equation between any $a \leq b$ by subtracting its values at b and a. □

Theorem 99 (Carathéodory and classical trajectories agree on an interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval (order-connected) and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on I. Then x is a Carathéodory trajectory on I (definition 95) if and only if it is a trajectory there (definition 89).*

Order-connectedness cannot be dropped: on $\{0\} \cup [1, 2]$ the integral form relates $x(1)$ to $x(0)$, the derivative form does not.

Proof. Both notions reduce to closed segments inside I, where they agree by theorem 98. For the integral form, the equation between $a \leq b$ only involves $[a, b] \subseteq I$. For the derivative form, at $t \in I$ choose $a \leq t \leq b$ in I, taking $a < t$ if I has a point left of t and $a = t$ otherwise, and likewise for b. Then $[a, b] \subseteq I$ agrees with I near t, so a derivative within $[a, b]$ at t is one within I. □

3.2 Discrete time

Definition 100 (Discrete-time system)**.** A *discrete-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{N} \times E \times U \to E, \qquad h : \mathbb{N} \times E \times U \to Y,$$

to be read as $x^+ = F(k, x, u)$, $y = h(k, x, u)$. The same data as definition 71 with time in $\mathbb{N}$; what differs is the reading, a shift rather than a derivative.

Definition 101 (Time-invariant discrete-time system)**.** The discrete-time system $x^+ = f(x,u)$, $y = g(x,u)$, whose equations do not depend on time.

Definition 102 (Time invariance, discrete-time system)**.** A discrete-time system is *time-invariant* when $F(k,\cdot,\cdot)$ and $h(k,\cdot,\cdot)$ are the same for every k.

Lemma 103 (Time-invariant discrete-time systems are time-invariant)**.** *The system of definition 101 is time-invariant (definition 102).*

Proof. □

Lemma 104 (Time-invariant discrete-time systems come from time-free data)**.** *A time-invariant discrete-time system (definition 102) is* timeInvariant *f g for some f and g. The converse of lemma 103.*

Proof. Read f and g off at $k = 0$; time invariance says every other time agrees with it. □

Definition 105 (Autonomous discrete-time system)**.** The discrete-time system $x^+ = f(k,x)$, $y = g(k,x)$, which has no input: its input space is the one-point type.

Definition 106 (Autonomy, discrete-time system)**.** A discrete-time system is *autonomous* when $F(k,x,u)$ and $h(k,x,u)$ do not depend on u.

Lemma 107 (Autonomous discrete-time systems are autonomous)**.** *The system of definition 105 is autonomous (definition 106).*

Proof. □

Lemma 108 (Autonomous discrete-time systems come from input-free data)**.** *If a discrete-time system is autonomous (definition 106) and its input space is nonempty, there are f and g with $F(k,x,u) = f(k,x)$ and $h(k,x,u) = g(k,x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 109 (State equation of a discrete-time system)**.** Under an input sequence $u : \mathbb{N} \to U$, the right-hand side of the state equation is the time-dependent next-state map $(k,x) \mapsto F(k,x,u(k))$. The same expression as definition 87; what differs is what a trajectory does with it, a shift rather than a derivative.

Definition 110 (Output equation of a discrete-time system)**.** Under an input sequence u, the output equation is $(k,x) \mapsto h(k,x,u(k))$.

Definition 111 (Trajectory of a discrete-time system)**.** A sequence $x : \mathbb{N} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when

$$x(k+1) = F(k,x(k),u(k)), \qquad \forall\, k \in I.$$

Membership constrains the step *out of* k, so $I = [k_0,\infty)$ is the forward recursion from k_0, and the horizon $[k_0,k_1]$ corresponds to $I = [k_0,k_1)$.

Definition 112 (Trajectory for all time of a discrete-time system)**.** A trajectory (definition 111) satisfying the recursion at every $k \in \mathbb{N}$.

Lemma 113 (Trajectories of a discrete-time system on the whole domain)**.** *definition 111 at $I = \mathbb{N}$ is definition 112.*

Proof. □

Lemma 114 (Restriction of a trajectory of a discrete-time system)**.** *A trajectory at every time is a trajectory on every set.*

Proof. □

Unlike continuous time, a trajectory needs no hypothesis to exist or to be unique: the recursion defines it from any initial time and state for all later times. So no discrete-time solution escapes in finite time, and nothing like forward completeness is ever needed.

Lemma 115 (Trajectories of a discrete-time system exist)**.** *For every input u, initial time k_0 and state x_0 there is a trajectory (definition 111) on $[k_0, \infty)$ with $x(k_0) = x_0$.*

Proof. Define it by the recursion itself. □

Lemma 116 (Trajectories of a discrete-time system are unique)**.** *Two trajectories (definition 111) on $[k_0, \infty)$ under the same input with $x(k_0) = y(k_0)$ satisfy $x(k) = y(k)$ for every $k \geq k_0$.*

Proof. Induction on k. □

Lemma 117 (Trajectories of a time-invariant discrete-time system are iterates)**.** *If $F(k, x, u(k)) = f(x)$ for all k and x, every trajectory (definition 111) on $[k_0, \infty)$ satisfies $x(k_0 + n) = f^n(x(k_0))$ for every $n \in \mathbb{N}$.*

Proof. Induction on n. □

Definition 118 (The unforced discrete-time system)**.** The autonomous system (definition 105) $x^+ = F(k, x, 0)$, $y = h(k, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 119 (Trajectories of the unforced discrete-time system)**.** *A sequence is a trajectory of the unforced system (definition 118) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Chapter 4

Linear systems

4.1 The linear system objects

Linear systems are a special case of the general systems of the Systems chapter: the continuous-time system with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$, and its discrete-time counterpart. The objects below carry only the coefficient data. Each is the input of a constructor into the general system, and the semantics — state equation, output equation, trajectories — are defined through that constructor, so every result about general systems applies to linear ones unchanged.

There are two objects per time domain: a *linear time-varying* (LTV) system with coefficient maps $A(t), B(t), C(t), D(t)$, and a *linear time-invariant* (LTI) system with constant matrices. An LTI system is not a different kind of system: it builds the LTV system with constant coefficients, and the general system it describes is the time-invariant system of the Systems chapter, by definition.

4.1.1 Continuous-time LTV systems

Definition 120 (Continuous-time linear time-varying system)**.** A *continuous-time linear time-varying (LTV) system* with states indexed by X, inputs by U, outputs by Y and coefficients in $\mathbb{K}$ is the data of four matrix-valued maps on $\mathbb{R}$,

$$A : \mathbb{R} \to \mathbb{K}^{X\times X}, \quad B : \mathbb{R} \to \mathbb{K}^{X\times U}, \quad C : \mathbb{R} \to \mathbb{K}^{Y\times X}, \quad D : \mathbb{R} \to \mathbb{K}^{Y\times U},$$

describing the continuous-time system (definition 71) $\dot{x} = A(t)x + B(t)u$, $y = C(t)x + D(t)u$ (definition 121).

Definition 121 (The system of an LTV system)**.** The continuous-time system (definition 71) with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$.

Definition 122 (State equation of an LTV system)**.** The right-hand side of the state equation, $(t,z) \mapsto A(t)\,z + B(t)\,u(t)$: the vector field (definition 87) of definition 121.

Lemma 123 (The unforced state equation of an LTV system)**.** *At zero input the state equation (definition 122) reduces to* $(t,z) \mapsto A(t)\,z$.

Proof. □

Lemma 124 (The state equation of an LTV system at the origin)**.** *At $z = 0$ the state equation (definition 122) contributes only the input term, $A(t)\,0 + B(t)u(t) = B(t)u(t)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 125 (Output equation of an LTV system)**.** The output equation, $(t, z) \mapsto C(t)\,z + D(t)\,u(t)$: the output map (definition 88) of definition 121.

Definition 126 (Trajectory of an LTV system)**.** A map x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when it is a trajectory (definition 89) of definition 121:

$$\dot{x}(t) = A(t)\,x(t) + B(t)\,u(t), \qquad \forall\, t \in I.$$

Definition 127 (Trajectory for all time of an LTV system)**.** A trajectory (definition 90) for all time of definition 121.

Definition 128 (Carathéodory trajectory of an LTV system)**.** A Carathéodory trajectory (definition 95) of definition 121.

Definition 129 (Time invariance of an LTV system)**.** A continuous-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 130 (Time-invariant LTV systems are time-invariant systems)**.** *If an LTV system is time-invariant (definition 129), so is the system it describes (definition 73).*

Proof. □

4.1.2 Continuous-time LTI systems

Definition 131 (Continuous-time linear time-invariant system)**.** A *continuous-time linear time-invariant (LTI) system* is the data of four matrices $A \in \mathbb{K}^{X\times X}$, $B \in \mathbb{K}^{X\times U}$, $C \in \mathbb{K}^{Y\times X}$, $D \in \mathbb{K}^{Y\times U}$, describing the LTV system (definition 120) with these constant coefficients, $\dot{x} = Ax + Bu$, $y = Cx + Du$.

Definition 132 (The LTV system of an LTI system)**.** The LTV system (definition 120) whose coefficients are the constant maps $t \mapsto A$, $t \mapsto B$, $t \mapsto C$, $t \mapsto D$.

Lemma 133 (LTI systems are time-invariant LTV systems)**.** *The LTV system of an LTI system (definition 132) is time-invariant (definition 129).*

Proof. □

Lemma 134 (Time-invariant LTV systems come from LTI systems)**.** *A time-invariant LTV system (definition 129) is the LTV system of some LTI system. The converse of lemma 133.*

Proof. Read the coefficients off at $t = 0$; time invariance says every other time agrees with it. □

Definition 135 (The system of an LTI system)**.** The continuous-time system (definition 71) of the LTV system of s (definition 121, definition 132).

Lemma 136 (An LTI system is a linear time-invariant system)**.** *The system of an LTI system (definition 135) is the time-invariant system (definition 72) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

4.1.3 Discrete-time LTV systems

Definition 137 (Discrete-time linear time-varying system)**.** A *discrete-time linear time-varying (LTV) system* is the same data as definition 120 with the coefficients indexed by $\mathbb{N}$ instead of $\mathbb{R}$, describing the discrete-time system (definition 100) $x(k+1) = A(k)x(k) + B(k)u(k)$, $y(k) = C(k)x(k) + D(k)u(k)$ (definition 138).

The forward time domain $\mathbb{N}$ rather than $\mathbb{Z}$ matches the discrete solution theory, whose state transition matrix is an ordered product over $[t_0, t)$.

Definition 138 (The system of a discrete-time LTV system)**.** The discrete-time system (definition 100) with $F(k, x, u) = A(k)x + B(k)u$ and $h(k, x, u) = C(k)x + D(k)u$.

Definition 139 (State equation of a discrete-time LTV system)**.** The right-hand side of the state equation, $(k, z) \mapsto A(k)\, z + B(k)\, u(k)$: the next-state map (definition 109) of definition 138.

Lemma 140 (The unforced state equation of a discrete-time LTV system)**.** *At zero input the state equation (definition 139) reduces to $(k, z) \mapsto A(k)\, z$.*

Proof. □

Lemma 141 (The state equation of a discrete-time LTV system at the origin)**.** *At $z = 0$ the state equation (definition 139) contributes only the input term, $A(k)\, 0 + B(k)u(k) = B(k)u(k)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 142 (Output equation of a discrete-time LTV system)**.** The output equation, $(k, z) \mapsto C(k)\, z + D(k)\, u(k)$: the output map (definition 110) of definition 138.

Definition 143 (Trajectory of a discrete-time LTV system)**.** A sequence x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when it is a trajectory (definition 111) of definition 138:

$$x(k+1) = A(k)\, x(k) + B(k)\, u(k), \qquad \forall\, k \in I.$$

Definition 144 (Trajectory for all time of a discrete-time LTV system)**.** A trajectory (definition 112) for all time of definition 138.

Definition 145 (Time invariance of a discrete-time LTV system)**.** A discrete-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 146 (Time-invariant discrete-time LTV systems are time-invariant systems)**.** *If a discrete-time LTV system is time-invariant (definition 145), so is the system it describes (definition 102).*

Proof. □

4.1.4 Discrete-time LTI systems

Definition 147 (Discrete-time linear time-invariant system)**.** A *discrete-time linear time-invariant (LTI) system* is the data of four matrices A, B, C, D, describing the discrete-time LTV system (definition 137) with these constant coefficients, $x(k+1) = Ax(k) + Bu(k)$, $y(k) = Cx(k) + Du(k)$.

Definition 148 (The LTV system of a discrete-time LTI system)**.** The discrete-time LTV system (definition 137) whose coefficients are the constant maps $k \mapsto A$, $k \mapsto B$, $k \mapsto C$, $k \mapsto D$.

Lemma 149 (Discrete-time LTI systems are time-invariant LTV systems)**.** *The LTV system of a discrete-time LTI system (definition 148) is time-invariant (definition 145).*

Proof. □

Lemma 150 (Time-invariant discrete-time LTV systems come from LTI systems)**.** *A time-invariant discrete-time LTV system (definition 145) is the LTV system of some LTI system. The converse of lemma 149.*

Proof. Read the coefficients off at $k = 0$. □

Definition 151 (The system of a discrete-time LTI system)**.** The discrete-time system (definition 100) of the LTV system of s (definition 138, definition 148).

Lemma 152 (A discrete-time LTI system is a linear time-invariant system)**.** *The system of a discrete-time LTI system (definition 151) is the time-invariant system (definition 101) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

4.2 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypotheses. Throughout this section A is continuous and satisfies $\|A(s)\| \le M$ for all s on the segment between the two times named in each statement. Both are needed: continuity to differentiate the series term by term, the bound to make it converge at all. Statements below do not repeat them.

Definition 153 (Peano–Baker term)**.** The k-th term of the Peano–Baker series for a time-varying state matrix $A(t)$:

$$P_0(t, t_0) := I, \qquad P_{k+1}(t, t_0) := \int_{t_0}^{t} A(s)\, P_k(s, t_0)\, \mathrm{d}s.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5 (Peano-Baker series).

Definition 154 (State transition matrix)**.** The *state transition matrix* $\Phi(t, t_0)$, given by the Peano–Baker series

$$\Phi(t, t_0) := \sum_{k=0}^{\infty} P_k(t, t_0).$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.1 (Peano-Baker series).

Lemma 155 (Continuity of the Peano–Baker terms)**.** *Each term $P_k(t, t_0)$ of the Peano–Baker series is continuous in t, for A continuous.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: P_0 is constant, and P_{k+1} is the primitive of a continuous integrand (definition 153), hence continuous by the fundamental theorem of calculus. □

Lemma 156 (Factorial bound on the Peano–Baker terms)**.** *If $\|A(s)\| \le M$ for all s between t_0 and t_1, then*

$$\|P_k(t, t_0)\| \;\le\; \|I\| \cdot \frac{(M|t - t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: bound the integrand $A(s)P_k(s, t_0)$ pointwise on the segment between t_0 and t, using submultiplicativity of the matrix norm and the induction hypothesis. Split on whether $t_0 \le t$ or $t \le t_0$ and integrate the majorant via the power rule, flipping the integral's orientation (`intervalIntegral.integral_symm`) in the backward case. □

Lemma 157 (Factorial bound on a subinterval)**.** *The bound in lemma 156, uniformized to the worst case $t = t_1$:*

$$\|P_k(t, t_0)\| \;\le\; \|I\| \cdot \frac{(M|t_1 - t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Monotonicity of $t \mapsto (M|t - t_0|)^k$ on the segment between t_0 and t_1, combined with lemma 156. □

Lemma 158 (Summability of the Peano–Baker series)**.** *The Peano–Baker series $\sum_{k=0}^{\infty} P_k(t, t_0)$ converges absolutely, for A bounded between t_0 and t (in either order).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Compare against the exponential series $\sum_k (M|t - t_0|)^k / k!$ via lemma 156 and the comparison test. □

Lemma 159 (Continuity of the state transition matrix)**.** *$\Phi(\cdot, t_0)$ is continuous on the segment between t_0 and t_1 (including both endpoints), for A continuous and bounded by M there.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Each term $P_k(\cdot, t_0)$ is continuous (lemma 155) with a summable, point-independent bound (lemma 157); apply the Weierstrass M-test for continuity of a series. □

Theorem 160 (The state transition matrix solves the matrix ODE)**.** *The state transition matrix solves the matrix ODE*

$$\dot{\Phi}(t, t_0) = A(t)\, \Phi(t, t_0)$$

for t strictly between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Differentiate the Peano–Baker series term by term, using lemma 156 as the uniform bound needed for term-by-term differentiation of a series. The $k = 0$ term is constant, so peel it off and match the shifted series' derivative against $A(t)\Phi(t, t_0)$ using linearity of the sum under left multiplication by $A(t)$. □

Theorem 161 (The state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.
Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Every term of the Peano–Baker series beyond $k = 0$ integrates over the degenerate interval $[t_0, t_0]$ and vanishes. □

Theorem 162 (Peano–Baker series: existence)**.** *Hespanha, Theorem 5.1.* $x(t) := \Phi(t, t_0)\, x_0$ *solves*

$$\dot{x}(t) = A(t)\, x(t).$$

Proof. Push theorem 160 through the fixed linear map $M \mapsto Mx_0$ (continuous, since the domain is finite-dimensional). □

Theorem 163 (Matrix-valued solutions)**.** *For a constant matrix* C, $Y(t) := \Phi(t, t_0)\, C$ *solves*

$$\dot{Y}(t) = A(t)\, Y(t).$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Right-multiply theorem 160 by the constant C (`HasDerivAt.mul_const`), then reassociate. □

Lemma 164 (Integral form, vector-valued)**.** $x(t) := \Phi(t, t_0)\, x_0$ *satisfies the integral equation*

$$x(t) = x_0 + \int_{t_0}^{t} A(s)\, x(s)\, \mathrm{d}s$$

for every t *between* t_0 *and* t_1.
Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 162, using $\Phi(t_0, t_0) = I$ (theorem 161) to fix the initial value. □

Lemma 165 (Integral form, matrix-valued)**.** *For a constant matrix* C, $Y(t) := \Phi(t, t_0)\, C$ *satisfies the integral equation*

$$Y(t) = C + \int_{t_0}^{t} A(s)\, Y(s)\, \mathrm{d}s$$

for every t *between* t_0 *and* t_1.
Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 163, using $\Phi(t_0, t_0) = I$ (theorem 161) to fix the initial value. □

4.2.1 Uniqueness

Theorem 166 (Peano–Baker series: uniqueness)**.** *Hespanha, Theorem 5.1. Any solution z of $\dot{x} = A(t)\,x$, $x(t_0) = x_0$, for t between t_0 and t_1, coincides with $x(t) := \Phi(t, t_0)\,x_0$.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value x_0, the resulting bound on $\|x(t) - z(t)\|$ collapses to 0. □

Theorem 167 (Columns of the state transition matrix)**.** *Hespanha, P5.2. For every fixed i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $\dot{x} = A(t)\,x$, $x(t_0) = e_i$, where e_i is the i-th standard basis vector.*

Proof. Restatement of theorem 166 at $x_0 := e_i$, using $\Phi(t, t_0)\,e_i = (\Phi(t, t_0))_{\cdot,i}$ (`Matrix.mulVec_single_one`). □

Theorem 168 (Uniqueness of matrix-valued solutions)**.** *Any solution Y of $\dot{Y} = A(t)\,Y$, $Y(t_0) = C$, for t between t_0 and t_1, coincides with $\Phi(t, t_0)\,C$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value C, the resulting bound on $\|Y(t) - Z(t)\|$ collapses to 0. □

4.2.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

Theorem 169 (Semigroup property)**.** *Hespanha, P5.3. For A continuous and bounded on $[\tau, t]$, and any $\tau \le s < t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Proof. $\Phi(\cdot, \tau)$, re-anchored at s, is an integral solution of the matrix ODE on the segment between s and t with initial value $\Phi(s, \tau)$, hence coincides with $\Phi(\cdot, s)\,\Phi(s, \tau)$ there by theorem 168. □

Theorem 170 (Invertibility of the state transition matrix)**.** *Hespanha, P5.4. For A continuous and bounded on the segment between τ and t, $\Phi(t, \tau)$ is nonsingular and*

$$\Phi(t, \tau)^{-1} = \Phi(\tau, t).$$

Proof. From theorem 169, generalized to arbitrary order by re-anchoring $\Phi(\cdot, \tau)$ at t and invoking uniqueness there, $\Phi(\tau, t)\Phi(t, \tau) = \Phi(\tau, \tau) = I$, so $\Phi(t, \tau)$ and $\Phi(\tau, t)$ are mutual inverses. □

Lemma 171 (Composition on a subinterval)**.** *For every z on the segment between a and b, $\Phi(z, a) = \Phi(z, b)\,\Phi(b, a)$. Composition with the* first *argument free.*

Proof. $\Phi(\cdot, a)$, re-anchored at b, is an integral solution of $\dot{Y} = A(r)Y$ with value $\Phi(b, a)$ at b; so is $\Phi(\cdot, b)\Phi(b, a)$. They agree by theorem 168. □

Theorem 172 (Semigroup property, order-free)**.** *Hespanha, P5.3. For* any *s on the segment between τ and t,*

$$\Phi(t,s)\,\Phi(s,\tau)=\Phi(t,\tau).$$

The $\tau\le s<t$ ordering in theorem 169 matches the textbook's literal statement but is not a genuine requirement; the variation-of-constants proof needs the unordered form, where $s=t_0$ is fixed and τ ranges over the whole segment.

Proof. lemma 171 gives $\Phi(s,\tau)=\Phi(s,t)\Phi(t,\tau)$; left-multiply by $\Phi(t,s)$ and cancel using $\Phi(t,s)\Phi(s,t)=I$ (theorem 170). □

Lemma 173 (Change of base time)**.** *For every τ on the segment between t_0 and t, $\Phi(t,t_0)\,\Phi(t_0,\tau)=\Phi(t,\tau)$: composition through a fixed* base *point. This differs from theorem 172 in which of the three times is required to lie between the others — here the two fixed integration endpoints are the outer ones and τ is free, which is the shape variation of constants needs.*

Proof. theorem 172 with τ itself as the middle point gives $\Phi(t,\tau)\Phi(\tau,t_0)=\Phi(t,t_0)$; right-multiply by $\Phi(t_0,\tau)$ and cancel $\Phi(\tau,t_0)\Phi(t_0,\tau)=I$. □

Lemma 174 (Continuity in the initial time)**.** *With the first argument held fixed, $\tau\mapsto\Phi(t,\tau)$ is continuous on the segment between t and t_1 — the mirror of lemma 159, which varies the first argument. Needed wherever $\Phi(t,\cdot)$ is an integrand, as in variation of constants.*

Proof. By theorem 170, $\Phi(t,\tau)=\Phi(\tau,t)^{-1}$, so the map is $\tau\mapsto\Phi(\tau,t)$ — continuous in its first argument — followed by matrix inversion, which is continuous at every unit. □

Theorem 175 (Derivative in the initial time)**.** *With the first argument held fixed, the state transition matrix is differentiable in its second argument, with*

$$\frac{\partial}{\partial t}\Phi(\tau,t)=-\Phi(\tau,t)\,A(t)$$

for t strictly between τ and t_1 — the mirror of theorem 160, which differentiates in the first argument.

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.12.*

Proof. By theorem 170, $\Phi(\tau,t)=\Phi(t,\tau)^{-1}$. Differentiate $t\mapsto\Phi(t,\tau)$ by theorem 160 and the inverse by $d(M^{-1})=-M^{-1}\,dM\,M^{-1}$: $-\Phi(\tau,t)\,A(t)\,\Phi(t,\tau)\,\Phi(\tau,t)=-\Phi(\tau,t)\,A(t)$. □

Lemma 176 (Exponential growth bound)**.** *If $\|A(s)\|\le M$ on the segment between t_0 and t, then*

$$\|\Phi(t,t_0)\|\le\|I\|\,e^{M|t-t_0|}.$$

Proof. Sum the term bounds of lemma 156: they are $\|I\|$ times the terms of the exponential series. □

When A is bounded on all of $\mathbb{R}$, the transition matrix is defined and differentiable everywhere, and the composition law and both derivatives hold with no ordering of the times.

Theorem 177 (Semigroup property for a globally bounded system)**.** *If A is continuous and bounded on $\mathbb{R}$, then $\Phi(t,s)\,\Phi(s,\tau)=\Phi(t,\tau)$ for all t,s,τ.*

Proof. One of the three times lies between the other two. If it is s, this is theorem 172. Otherwise compose through the middle time and cancel $\Phi(\tau, s)\Phi(s, \tau) = I$ (theorem 170). □

Theorem 178 (Derivative in the current time, globally bounded system)**.** *If A is continuous and bounded on $\mathbb{R}$, then for all τ and t, $\frac{\partial}{\partial t}\Phi(t, \tau) = A(t)\,\Phi(t, \tau)$.*

Proof. Write $\Phi(\sigma, \tau) = \Phi(\sigma, t-1)\,\Phi(t-1, \tau)$ (theorem 177); t is strictly inside a segment from $t-1$, where theorem 160 applies. □

Theorem 179 (Derivative in the initial time, globally bounded system)**.** *If A is continuous and bounded on $\mathbb{R}$, then for all τ and t, $\frac{\partial}{\partial t}\Phi(\tau, t) = -\Phi(\tau, t)\,A(t)$.*

Proof. Write $\Phi(\tau, \sigma) = \Phi(\tau, t-1)\,\Phi(t-1, \sigma)$ (theorem 177) and apply theorem 175 to the second factor. □

4.2.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ. The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

Theorem 180 (Variation of constants: existence)**.** *Hespanha, Theorem 5.2. For continuous A, B and u, the function*

$$x(t) := \Phi(t, t_0)x_0 + \int_{t_0}^{t} \Phi(t, \tau)B(\tau)u(\tau)\,d\tau$$

satisfies $\dot{x}(t) = A(t)x(t) + B(t)u(t)$ at every interior t.

Proof. The obstacle is that t appears in both the upper limit and the integrand. Factor it out: with $w(t) := x_0 + \int_{t_0}^{t} \Phi(t_0, \tau)B(\tau)u(\tau)\,d\tau$, whose integrand does not depend on t, the fundamental theorem of calculus gives w directly, and lemma 173 shows $x(t) = \Phi(t, t_0)w(t)$. Differentiating that product and cancelling $\Phi(t, t_0)\Phi(t_0, t) = I$ leaves the claim. Φ is thus only ever integrated in its second argument, never differentiated there. □

Lemma 181 (The forced response at the initial time)**.** *The forced response (definition 183) takes the value x_0 at $t = t_0$: the forcing integral is over the degenerate interval and $\Phi(t_0, t_0) = I$. Together with theorem 180 this makes it a solution of the initial value problem, not merely of the differential equation.*

Proof. □

Theorem 182 (Variation of constants: uniqueness)**.** *Hespanha, Theorem 5.2. Any two continuous integral solutions of the forced system $\dot{x} = A(t)x + B(t)u(t)$ with the same initial value x_0 agree on the segment between t_0 and t_1.*

Proof. For fixed t the right-hand side $v \mapsto A(t)v + B(t)u(t)$ is affine in v, hence Lipschitz with constant $\max(M, 1)$ where $\|A\| \le M$. Continuous dependence on initial conditions with zero perturbation then bounds $\|z_1(t) - z_2(t)\|$ by 0. The forcing term $B(t)u(t)$ is common to both solutions and cancels in the difference, so it never enters the Lipschitz estimate. □

4.2.4 The forced response as a trajectory

The solution theory above is stated on the coefficient maps A and B, which is where the analysis lives. The results below restate it for the system object of definition 120: the variation-of-constants formula is a *trajectory* in the sense of definition 127, which the stability theory quantifies over.

Definition 183 (Forced response)**.** The *forced response* of $\dot{x} = A(t)x + B(t)u(t)$ from the state x_0 at time t_0, given by the variation-of-constants formula

$$x(t) = \Phi(t, t_0)\, x_0 + \int_{t_0}^{t} \Phi(t, \tau)\, B(\tau)\, u(\tau)\, \mathrm{d}\tau,$$

with Φ the state transition matrix (definition 154).

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.2.

Definition 184 (Homogeneous response)**.** The *homogeneous response* of $\dot{x} = A(t)x$ from the state x_0 at time t_0,

$$x(t) = \Phi(t, t_0)\, x_0,$$

with Φ the state transition matrix (definition 154). This is the forced response (definition 183) with the input switched off, and the object Lyapunov stability is stated about.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Definition 8.1.

Lemma 185 (The homogeneous response at the initial time)**.** *The homogeneous response (definition 184) takes the value x_0 at $t = t_0$, since $\Phi(t_0, t_0) = I$.*

Proof. □

Lemma 186 (Continuity of the forced response)**.** *For continuous A, B and u, the forced response (definition 183) is continuous on the closed segment between t_0 and t_1, endpoints included.*

Proof. The integrating-factor identity rewrites the response as $\Phi(z, t_0)\, w(z)$ with $w(z) = x_0 + \int_{t_0}^{z} \Phi(t_0, \tau) B(\tau) u(\tau)\, \mathrm{d}\tau$, whose integrand does not depend on z — only the limit does. Continuity of w is then the ordinary continuity of a primitive, and $\Phi(\cdot, t_0)$ is continuous by lemma 159. □

Theorem 187 (Variation of constants, integral form)**.** *The forced response (definition 183) satisfies*

$$x(t) = x_0 + \int_{t_0}^{t} \left(A(s)x(s) + B(s)u(s)\right) \mathrm{d}s$$

for every t on the closed segment between t_0 and t_1.

This is theorem 180 upgraded to include the endpoints, which the differential form cannot reach: its window has t_0 as an endpoint for every choice of t_1, so t_0 is never interior to it.

Proof. The fundamental theorem of calculus, applied to the derivative on the open interval (theorem 180) together with continuity on the closed one (lemma 186), with lemma 181 fixing the initial value. □

Theorem 188 (The forced response is a trajectory)**.** *Let s be a continuous-time linear system with continuous coefficients A and B, and let u be a continuous input. Then the forced response (definition 183) from any state x_0 at any time t_0 is a trajectory of s (definition 127) — it satisfies the state equation at every $t \in \mathbb{R}$, with a two-sided derivative.*

Restriction to an interval, in particular to the ray $[t_0, \infty)$ that the stability predicates quantify over, is lemma 92.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.2.*

Proof. For $t \neq t_0$, apply theorem 180 on a window having t in its interior; since the window is unordered this covers $t < t_0$ as well as $t > t_0$, and continuity of A supplies the bound on the compact segment.

At $t = t_0$ no such window exists, because the formula is anchored there and t_0 is an endpoint of every window. Instead take the integral form (theorem 187) on $[t_0, t_0 + 1]$ and on $[t_0 - 1, t_0]$; each yields a one-sided derivative at t_0 (lemma 36), and the two glue to a two-sided one. □

Lemma 189 (Zero input gives the homogeneous response)**.** *With $u = 0$ the forced response (definition 183) is the homogeneous response (definition 184): $\Phi(t, t_0)x_0 + \int_{t_0}^{t} \Phi(t, \tau)B(\tau) \cdot 0 \, \mathrm{d}\tau = \Phi(t, t_0)x_0$.*

Proof. □

Theorem 190 (The homogeneous response is a trajectory)**.** *For a continuous-time linear system with continuous coefficients, the homogeneous response (definition 184) from any state x_0 at any time t_0 is a trajectory (definition 127) under zero input.*

Definition 8.1 is stated about this map, so this is the form in which the solution theory reaches the stability theory.

Proof. theorem 188 at a system with no inputs, where the forcing integral vanishes (lemma 189). The dummy system is an artefact of the proof: only A appears in the statement, and no hypothesis on B is needed. □

Theorem 191 (Solutions are homogeneous responses)**.** *Let A be continuous and let x solve $\dot{x} = A(t)x$ on $[t_0, t_1]$. Then*

$$x(t) = \Phi(t, t_0)\, x(t_0), \qquad \forall\, t \in [t_0, t_1].$$

Proof. On $[t_0, t]$ the solution satisfies the integral equation (lemma 35), so uniqueness for the Peano–Baker solution (theorem 166) identifies it with $\Phi(\cdot, t_0)x(t_0)$ there. □

Theorem 192 (The homogeneous response is a trajectory on the forward ray)**.** *The homogeneous response (definition 184) is a trajectory of the unforced system (definition 126) on $[t_0, \infty)$.*

Proof. theorem 190, restricted to the ray by lemma 92. □

Theorem 193 (Trajectories of the unforced system are homogeneous responses)**.** *Let s have continuous A and let φ be a trajectory of the unforced system on $[t_0, t_1]$ (definition 126). Then $\varphi(t) = \Phi(t, t_0)\, \varphi(t_0)$ for every $t \in [t_0, t_1]$.*

Proof. theorem 191, after reducing the unforced state equation by lemma 123. □

Theorem 194 (Linear systems are forward complete)**.** *If A is continuous, the unforced system (definition 80) of definition 121 is forward complete (definition 83).*

Proof. A trajectory on $[t_0, t_1]$ agrees there with the homogeneous response from its initial state (theorem 193), which is a trajectory on $[t_0, \infty)$ (theorem 192). □

4.2.5 The time-invariant case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha's (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about exp and A alone.

Lemma 195 (Peano–Baker terms for a constant matrix)**.** *For a constant state matrix A, the k-th Peano–Baker term collapses to*

$$\Phi_k(t, t_0) = \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Induction on k. The $k = 0$ term is the identity. For the step, $\Phi_{k+1}(t, t_0) = \int_{t_0}^{t} A\,\Phi_k(s, t_0)\,ds$; substituting the inductive hypothesis pulls A^{k+1} out of the integral and leaves $\int_{t_0}^{t} (s - t_0)^k ds = (t - t_0)^{k+1}/(k + 1)$, which combines with $1/k!$ to give $1/(k + 1)!$. □

Lemma 196 (The LTI state transition matrix as a series)**.** *For a constant state matrix A, the state transition matrix is the power series*

$$\Phi(t, t_0) = \sum_{k=0}^{\infty} \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Sum lemma 195 termwise; the Peano–Baker series converges for any continuous state matrix, a constant one included. □

Theorem 197 (The LTI state transition matrix is $e^{A(t-t_0)}$)**.** *For a constant state matrix A, the state transition matrix is the matrix exponential:*

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

This is the bridge that lets every property already proved for Φ in the time-varying setting — existence, uniqueness, the semigroup law, invertibility, variation of constants — be reused for LTI systems by specialization rather than reproved.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.2).*

Proof. Both sides are power series in A: lemma 196 for the left, the defining series of exp for the right. Match them termwise. □

Theorem 198 (Uniqueness of the LTI state response)**.** *Let A be constant and let z be a continuous integral solution of $\dot{x} = Ax$, $x(t_0) = x_0$, on $[t_0, t_1]$. Then $z(t) = e^{A(t-t_0)}x_0$ for every t in that interval.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.1.*

Proof. The time-varying uniqueness theorem applied at the constant state matrix $A(\cdot) \equiv A$, whose bound $\|A(t)\| \leq \|A\|$ is immediate, then rewritten through theorem 197. There is no new ODE content: uniqueness for LTI is uniqueness for LTV at a constant matrix. □

Theorem 199 (Columns of the LTI state transition matrix)**.** *Let A be constant. For each i, the i-th column of $e^{A(t-t_0)}$ is the unique continuous integral solution of $\dot{x} = Ax$ with $x(t_0) = e_i$, the i-th standard basis vector. Equivalently, the columns of the matrix exponential are the solutions from the standard basis.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.2.*

Proof. theorem 198 at $x_0 = e_i$, in the column form supplied by the time-varying column-uniqueness theorem. □

4.2.6 The forced response of a time-invariant system

For a time-invariant system the state transition matrix is the matrix exponential, so the variation-of-constants formula collapses to the closed form every textbook writes.

Lemma 200 (Forced response of a time-invariant system)**.** *For constant A and B the forced response (definition 183) is*

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6.*

Proof. theorem 197 identifies Φ with the matrix exponential, applied once to the homogeneous term and once under the integral sign. □

Theorem 201 (The LTI response is a trajectory)**.** *For an LTI system (definition 131) with continuous input u, the closed form*

$$t \mapsto e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau$$

is a trajectory (definition 127) of that system.

Proof. theorem 188 at the LTV system of the LTI system (definition 132), whose coefficient maps are constant and so continuous, rewritten by lemma 200. □

Lemma 202 (Variation of constants: output equation)**.** *(Hespanha, equation (5.8)). With x as in theorem 180, the output $y(t) = C(t)x(t) + D(t)u(t)$ splits into the homogeneous response and the forced response:*

$$y(t) = C(t)\Phi(t, t_0)x_0 + \int_{t_0}^{t} C(t)\Phi(t, \tau)B(\tau)u(\tau)\, d\tau + D(t)u(t).$$

Proof. $v \mapsto C(t)v$ is a continuous linear map and so commutes with the interval integral; the integrand is continuous by lemma 174. □

4.3 The matrix exponential and complexification

4.3.1 Complexification

Spectral arguments must happen over $\mathbb{C}$, because $\mathbb{R}$ is not algebraically closed, while the systems of interest have real coefficients. Entrywise complexification is the bridge, and these record what it commutes with. The one with content is the transfer of positive definiteness: a real positive-definite matrix remains positive definite against *complex* vectors, which is what lets a Lyapunov certificate be evaluated against an eigenvector.

Definition 203 (Complexification of a real matrix)**.** For a real matrix A, its *complexification* $A_{\mathbb{C}}$ is obtained by applying $\mathbb{R} \hookrightarrow \mathbb{C}$ to each entry.

Lemma 204 (Positive definiteness survives complexification)**.** *If a real matrix P is positive definite, so is its complexification $P_{\mathbb{C}}$, as a Hermitian matrix over $\mathbb{C}$.*

Proof. Write a complex vector as $v = a + ib$ with a, b real. Then $v^* P_{\mathbb{C}} v = a^\mathsf{T} P a + b^\mathsf{T} P b$: the imaginary part cancels because P is symmetric, and the real part is positive because $v \neq 0$ forces a or b to be nonzero. □

Over a field the spectrum of a matrix is its set of eigenvalues, which is how a condition on eigenpairs meets the spectral radius.

Lemma 205 (The spectrum of a matrix is its set of eigenvalues)**.** *For a square matrix M over a field K, $\mu \in \sigma(M)$ if and only if $Mv = \mu v$ for some $v \neq 0$.*

Proof. □

4.3.2 Spectral facts about the exponential

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A. The eigenpair lemma is the half of spectral mapping that is pure power-series algebra, and is what turns an eigenvalue condition on A into a statement about the growth of e^{At}.

Lemma 206 (Complexification commutes with the matrix exponential)**.** *Entrywise complexification commutes with the matrix exponential: for a real matrix A,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{\ }$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and exp commutes with any such. □

Lemma 207 (The exponential acts on an eigenvector by exponentiating its eigenvalue)**.** *Let M be a square matrix and let (μ, v) be an eigenpair of M, so that $Mv = \mu v$. Then*

$$e^M v = e^\mu \, v.$$

Reference: standard power-series functional calculus for the exponential.

Proof. Induction gives $M^k v = \mu^k v$ for every k. The map $N \mapsto Nv$ is linear and continuous, so it commutes with the sum defining $e^M = \sum_k (k!)^{-1} M^k$, giving $e^M v = \sum_k (k!)^{-1} \mu^k v = e^\mu v$. □

Lemma 208 (A scalar multiple scales the eigenvalues)**.** *If $Av = \mu v$ then $(cA)v = (c\mu)v$.*

Proof. □

Lemma 209 (Powers power the eigenvalues)**.** *If $Av = \mu v$ then $A^k v = \mu^k v$ for every k.*

Proof. □

Lemma 210 (A scalar shift shifts the eigenvalues)**.** *For a square matrix A over a commutative ring and scalars α, μ, a vector v satisfies $(A + \alpha I)v = (\mu + \alpha)v$ if and only if $Av = \mu v$.*

Proof. □

Lemma 211 (The kernel form of an eigenpair)**.** *$(\mu I - A)v = 0$ if and only if $Av = \mu v$.*

Proof. □

Lemma 212 (Negative spectrum gives a contractive exponential)**.** *If every eigenpair (μ, v) of the complexification of a real matrix A, with $v \neq 0$, has $\mathrm{Re}(\mu) < 0$, then $r(e^{A_{\mathbb{C}}}) < 1$.*

Reference: standard spectral mapping for the matrix exponential.

Proof. Reverse spectral mapping: every spectral value of $e^{A_{\mathbb{C}}}$ is e^μ for an eigenvalue μ of $A_{\mathbb{C}}$, and $|e^\mu| = e^{\mathrm{Re}\,\mu} < 1$. □

Theorem 213 (Semigroup property of the matrix exponential)**.** *Hespanha, P6.3. For every $t, \tau \in \mathbb{R}$,*

$$e^{At} e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^X e^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. □

Theorem 214 (The matrix exponential as a finite polynomial)**.** *Hespanha, P6.5. There are scalar functions $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t)\, A^i \qquad \text{for every } t.$$

The matrix exponential — an infinite series in A — is a finite *polynomial in A of degree less than n, at the cost of coefficients that are transcendental in t.*

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \leq CM^k$, dominating each α_i by a scalar exponential series. That growth bound is the crux of the proof and is established by iteration, not by spectral theory: "multiply by X, reduce modulo the characteristic polynomial" is a fixed linear operation on the n-dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. □

4.4 The discrete-time state transition matrix

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \le s \le t$ instead of holding in any order.

Definition 215 (Discrete state transition matrix)**.** The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2)\cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \le t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \le t_0$ is the empty product I. The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, equation (5.11).

Lemma 216 (The discrete state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Property P5.5.*

Proof. The defining product is over the empty range $[t_0, t_0)$. □

Lemma 217 (One-step recursion)**.** *For $t \ge t_0$, $\Phi(t+1, t_0) = A(t)\,\Phi(t, t_0)$. With lemma 216 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.*

Proof. Peel the leading factor $A(t)$ off the defining product. □

Theorem 218 (Discrete state transition matrix: uniqueness)**.** *Hespanha, P5.5. If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \ge t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \ge t_0$.*

Proof. Induction on t from t_0: the base case is lemma 216 and the step chains the recursion against lemma 217. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. □

Theorem 219 (Uniqueness of the discrete state response)**.** *If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \ge t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \ge t_0$: the vector form of theorem 218.*

Proof. □

Theorem 220 (Columns of the discrete state transition matrix)**.** *Hespanha, P5.6. For each i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \ge t_0$.*

Proof. theorem 219 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot,i}$. □

Theorem 221 (Semigroup property, discrete)**.** *Hespanha, P5.7. For $\tau \le s \le t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \le s \le t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s, chaining lemma 217 at both s and τ. □

4.4.1 Discrete variation of constants

Lemma 222 (Discrete variation of constants at the initial time)**.** *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3 (discrete variation of constants).*

Proof. □

Theorem 223 (Discrete variation of constants: existence)**.** *For $t \geq t_0$, the formula of lemma 222 satisfies the forced recursion*

$$x(t+1) = A(t)\,x(t) + B(t)\,u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t, \tau+1)$ into $\Phi(t+1, \tau+1)$ termwise by lemma 217, applied once more to the leading $\Phi(t, t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1, t+1) = I$. □

Theorem 224 (Discrete variation of constants: uniqueness)**.** *Any two solutions of $z(t+1) = A(t)z(t) + B(t)u(t)$ with the same value at t_0 agree for all $t \geq t_0$.*

Proof. Induction on t from t_0. Unlike theorem 182, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

4.4.2 The discrete responses as trajectories

The variation-of-constants formula and its unforced part, named, and read on the system object (definition 137): they are exactly the trajectories of the system from a given initial state, which is the form in which the solution theory reaches the stability theory.

Definition 225 (Forced response, discrete time)**.** The *forced response* of $x(t+1) = A(t)x(t) + B(t)u(t)$ from the state x_0 at time t_0,

$$x(t) = \Phi(t, t_0)\,x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)\,B(\tau)\,u(\tau),$$

with Φ the discrete state transition matrix (definition 215). It solves the recursion forward from t_0 only.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Definition 226 (Homogeneous response, discrete time)**.** The *homogeneous response* of $x(t+1) = A(t)x(t)$ from the state x_0 at time t_0, $x(t) = \Phi(t, t_0)\,x_0$: the forced response (definition 225) with the input switched off.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Lemma 227 (The discrete forced response at the initial time)**.** *The forced response (definition 225) takes the value x_0 at $t = t_0$.*

Proof. lemma 222. □

Lemma 228 (The discrete forced response solves the recursion). *For $t \geq t_0$ the forced response (definition 225) satisfies $x(t+1) = A(t)x(t) + B(t)u(t)$.*

Proof. theorem 223. □

Lemma 229 (The discrete homogeneous response at the initial time). *The homogeneous response (definition 226) takes the value x_0 at $t = t_0$.*

Proof. $\Phi(t_0, t_0) = I$ (lemma 216). □

Lemma 230 (The discrete homogeneous response solves the unforced recursion). *For $t \geq t_0$ the homogeneous response (definition 226) satisfies $x(t+1) = A(t)x(t)$.*

Proof. $\Phi(t+1, t_0) = A(t)\Phi(t, t_0)$ (lemma 217). □

Theorem 231 (The discrete state transition matrix of an LTI system). *For a constant state matrix A, the discrete state transition matrix (definition 215) is $\Phi(t, t_0) = A^{t-t_0}$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3.*

Proof. $\Phi(t, t_0)$ is the ordered product of $t - t_0$ factors, all equal to A. □

Theorem 232 (The discrete forced response is a trajectory). *For a discrete-time LTV system s and input u, the forced response (definition 225) from x_0 at t_0 is a trajectory of s under u on $[t_0, \infty)$ (definition 143).*

Proof. lemma 228, which is the trajectory condition on the nose. □

Theorem 233 (Discrete trajectories are forced responses). *If x is a trajectory of s under u on $[t_0, \infty)$ (definition 143), then $x(t)$ is the forced response (definition 225) from $x(t_0)$ at t_0, for every $t \geq t_0$.*

Proof. Both satisfy the forced recursion from the same initial value, so they agree by theorem 224. □

Theorem 234 (The discrete homogeneous response is a trajectory). *The homogeneous response (definition 226) from x_0 at t_0 is a trajectory of the unforced system on $[t_0, \infty)$ (definition 143).*

Proof. lemma 230, after reducing the unforced state equation by lemma 140. □

Theorem 235 (Discrete unforced trajectories are homogeneous responses). *If x is a trajectory of the unforced system on $[t_0, \infty)$ (definition 143), then $x(t) = \Phi(t, t_0)\, x(t_0)$ for every $t \geq t_0$.*

Proof. theorem 219, after reducing the unforced state equation by lemma 140. □

Chapter 5

Comparison functions

Comparison functions are the vocabulary of stability estimates: a class $\mathcal{K}$ function bounds how far a trajectory can go, a class $\mathcal{L}$ function how fast it decays, and a class $\mathcal{KL}$ function both at once. This chapter builds them, and the tools that turn a cruder bound into one of them.

5.1 Class $\mathcal{K}$ and $\mathcal{K}_\infty$ functions

Definition 236 (Class $\mathcal{K}$ function)**.** A *class* $\mathcal{K}$ function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \to [0, b)$ with $\alpha(0) = 0$. All class K functions have an inverse $\alpha^{-1} : [0, b) \to [0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}$). Differences:

- Khalil writes the codomain as $[0, \infty)$; here the bound b is tracked in the type.
- The inverse is bundled as data rather than derived.

Definition 237 (Class $\mathcal{K}_\infty$ function)**.** A *class* $\mathcal{K}_\infty$ function is a continuous strictly increasing map $\alpha : [0, \infty) \to [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \to \infty$ as $r \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}_\infty$). The inverse is bundled as data rather than derived.

5.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 238 (Class $\mathcal{K}$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 239 (Restriction of a class $\mathcal{K}$ function)**.** If α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, \alpha(c))$.

Definition 240 (Restriction to a prescribed codomain)**.** definition 239 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, e)$.

Definition 241 (Inverse of a class $\mathcal{K}$ function)**.** Class $\mathcal{K}$ is closed under inversion. The inverse of a class K function is also class K.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, first bullet. Khalil's domain $[0, \alpha(a))$ is the codomain bound b carried by the type.

Definition 242 (Composition of class $\mathcal{K}$ functions)**.** Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and β is class $\mathcal{K}$ on $[0, b) \to [0, c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0, a) \to [0, c)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, third bullet.

5.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 243 (Class $\mathcal{K}_\infty$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \to \infty$ as $r \to \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 244 (Inverse of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1}.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, second bullet.

Definition 245 (Composition of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fourth bullet.

Definition 246 (Sum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 247 (Positive multiple of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 248 (Minimum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 249 (Powers of a class $\mathcal{K}_\infty$ function)**.** For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 250 (A class $\mathcal{K}_\infty$ function restricts to class $\mathcal{K}$)**.** Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \to [0, \alpha(a))$.

Lemma 251 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\omega : \mathbb{R} \to \mathbb{R}$ satisfy $\omega(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \le \alpha(r)$ for all $r \ge 0$.*

Proof. lemma 279 supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \to \infty$ and $\omega \le f$ on $[0, \infty)$; by definition 243 that f is class $\mathcal{K}_\infty$. □

5.2 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0,\infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0,\infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Lemma 252 (Surjectivity onto $(0,\infty)$ from the two boundary limits)**.** *Let $W : \mathbb{R} \to \mathbb{R}$ be continuous on $(0,\infty)$ with $W(\eta) \to +\infty$ as $\eta \to 0^+$ and $W(\eta) \to 0$ as $\eta \to +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Lemma 253 (The right inverse inverts)**.** *Under the surjectivity conditions of lemma 252, $W\,(\mathrm{invFunOn}\,W\,(0,\infty)\,s) = s$ for all $s > 0$.*

Proof. □

Lemma 254 (Positivity of the right inverse)**.** *Under the surjectivity conditions of lemma 252,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s > 0$ *for all $s > 0$.*

Proof. □

Lemma 255 (The inverse of a strictly antitone function is strictly antitone)**.** *If W is strictly antitone on $(0,\infty)$, then* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is strictly antitone on $(0,\infty)$.*

Proof. □

Lemma 256 (The inverse vanishes at $+\infty$)**.** *Under the conditions of lemma 252 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to 0$ *as $s \to +\infty$.*

Proof. □

Lemma 257 (Continuity of the inverse)**.** *Under the conditions of lemma 252 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is continuous on $(0,\infty)$.*

Proof. □

Lemma 258 (The inverse blows up at 0^+)**.** *Under the conditions of lemma 252 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to +\infty$ *as $s \to 0^+$.*

Proof. □

5.3 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 259 (Class $\mathcal{L}$ function)**.** A *class $\mathcal{L}$* function on $[0,\infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \to 0$ as $s \to \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 260 (Singular class $\mathcal{L}$ function)**.** A *singular class* $\mathcal{L}$ function on $(0,\infty)$ is continuous, strictly positive and antitone, with $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 261 (Class $\mathcal{KL}$ function)**.** A *class* $\mathcal{KL}$ function on $[0,a)\times[0,\infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r>0$ is strictly decreasing and tends to 0 as $s\to\infty$. It arises as the bound $\|x(t)\|\le\beta(\|x_0\|,t)$ in asymptotic stability estimates.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3. Difference: Khalil says $\beta(\cdot,s)$ is class $\mathcal{K}$; here that is spelled out as separate fields, so no class $\mathcal{K}$ instance is required for each s.

Definition 262 (Global class $\mathcal{KL}$ function)**.** A *global class* $\mathcal{KL}$ function is a map $\beta:[0,\infty)\times[0,\infty)\to\mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r\ge 0$, is antitone and tends to 0 as $s\to\infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3 with $a=\infty$.

5.3.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 263 (Class $\mathcal{KL}$ from a class $\mathcal{K}$ times a class $\mathcal{L}$)**.** If α is class $\mathcal{K}$ on $[0,a)$ and γ is class $\mathcal{L}$, then $\beta(r,s):=\alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0,a)$.

The cap is built in the form $\min(\alpha(r),\sqrt{\alpha(r)\,U(s)})$, so the decay of that particular expression is recorded first.

Lemma 264 (Decay of a geometric-mean cap)**.** *Let $c\ge 0$ and let $U:\mathbb{R}\to\mathbb{R}$ satisfy $U(s)\to 0$ as $s\to+\infty$. Then $\min(c,\sqrt{c\,U(s)})\to 0$ as $s\to+\infty$.*

Proof. □

Theorem 265 (Class $\mathcal{KL}$ from a singular class $\mathcal{L}$ cap)**.** *Let α be class $\mathcal{K}$ on $[0,a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$). Then*

$$\beta(r,s):=\begin{cases}\alpha(r) & s=0,\\ \min(\alpha(r),\sqrt{\alpha(r)U(s)}) & s>0,\end{cases}$$

is class $\mathcal{KL}$ on $[0,a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U, the decay in s being lemma 264. Continuity is checked in two regimes. For $s>0$ the min branch is a composition of continuous maps. At $(r,0)$, pick a buffer radius $r_1\in(r,a)$; since $U(s)\to\infty$ as $s\to 0^+$, on a neighbourhood we have $\alpha(r_1)<U(s)$, hence $\alpha(r')\le U(s)$ and so $\alpha(r')\le\sqrt{\alpha(r')U(s)}$ — the minimum is locally just α, which is continuous. □

Definition 266 (Post-composition with a class $\mathcal{K}$ function)**.** Let β be class $\mathcal{KL}$ on $[0,a)$ and α be class $\mathcal{K}$ on $[0,b)$, and suppose $\beta(r,s) < b$ for every $r \in [0,a)$ and $s \geq 0$. Then $(r,s) \mapsto \alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1). The range hypothesis is additional: α_1 is defined only on $[0,b)$, so β's values must land there, which Khalil leaves implicit.

Definition 267 (Post-composition with a class $\mathcal{K}_\infty$ function)**.** If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$. This is definition 266 with the range hypothesis discharged by α being defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet, with α_1 of class $\mathcal{K}_\infty$; no range hypothesis is then needed.

Definition 268 (Pre-composition in the radius argument)**.** If β is class $\mathcal{KL}$ on $[0,b)$ and α is class $\mathcal{K}$ on $[0,a) \to [0,b)$, then $(r,s) \mapsto \beta(\alpha(r),s)$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2).

Definition 269 (Restriction of a class $\mathcal{KL}$ function)**.** If β is class $\mathcal{KL}$ on $[0,a) \times [0,\infty)$ and $0 < a' \leq a$, then β is class $\mathcal{KL}$ on $[0,a') \times [0,\infty)$.

Definition 270 (Global class $\mathcal{KL}$ from a product)**.** If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r,s) := \alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 271 (Post-composition, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \alpha(\beta(r,s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1), in the global case $a = \infty$.

Definition 272 (Pre-composition in the radius argument, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \beta(\alpha(r),s)$ is global class $\mathcal{KL}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2), in the global case $a = \infty$.

Theorem 273 (Capping a class $\mathcal{KL}$ function by a class $\mathcal{K}$ function)**.** *If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}$ on $[0,a)$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is class $\mathcal{KL}$ on $[0,a)$.*

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s, so antitonicity in s survives the minimum; and $\min(\beta(r,s),\alpha(r)) \to \min(0,\alpha(r)) = 0$ as $s \to \infty$ because $\alpha \geq 0$. □

Definition 274 (Capping a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

5.3.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds. The first lemma below is the continuity criterion the construction rests on.

Lemma 275 (Joint continuity from separate continuity and monotonicity)**.** *Let $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be nondecreasing in its first argument, nonincreasing in its second, and continuous in each argument separately. Then f is jointly continuous.*

Proof. Near (r_0, s_0), monotonicity gives $f(r_-, s_+) \le f(r, s) \le f(r_+, s_-)$ whenever $r_- < r < r_+$ and $s_- < s < s_+$. Given $u > f(r_0, s_0)$, continuity of $f(r_0, \cdot)$ gives $s_- < s_0$ with $f(r_0, s_-) < u$, and then continuity of $f(\cdot, s_-)$ gives $r_+ > r_0$ with $f(r_+, s_-) < u$; so $f < u$ on the neighbourhood $(-\infty, r_+) \times (s_-, \infty)$. The lower bound is symmetric. □

Lemma 276 (Class $\mathcal{KL}$ majorant)**.** *Let $\psi : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ satisfy, for all $r, s \ge 0$:*

1. *$\psi(r, s) \ge 0$;*
2. *$\psi(0, s) = 0$;*
3. *$r \mapsto \psi(r, s)$ is monotone on $[0, \infty)$;*
4. *$s \mapsto \psi(r, s)$ is antitone on $[0, \infty)$;*
5. *$\psi(r, s) \to 0$ as $s \to \infty$;*
6. *$r \mapsto \psi(r, 0)$ is continuous at 0 within $[0, \infty)$.*

Then there is a global class $\mathcal{KL}$ function β with $\psi(r, s) \le \beta(r, s)$ for all $r, s \ge 0$.

Proof. Clamp both arguments of ψ at 0 and average it over a window in each variable: first $\psi_1(r, s) = \int_0^1 \psi(r, s - v)\, dv$, over $[s-1, s]$, then $\psi_2(r, s) = \int_1^2 \psi_1(ru, s)\, du$, over $[r, 2r]$. Each window sits on the side that makes the average dominate, so $\psi \le \psi_1 \le \psi_2$, and ψ_2 is still nondecreasing in r and nonincreasing in s.

- $\psi_1(r, \cdot)$ is a difference of primitives of $\psi(r, \cdot)$, so continuous; $\psi_2(r, \cdot)$ is then continuous by dominated convergence, the integrands being bounded by $\psi(2r, 0)$.
- For $r > 0$, $\psi_2(r, s) = r^{-1} \int_r^{2r} \psi_1(\cdot, s)$ is a quotient of primitives, so continuous; at $r \le 0$, $0 \le \psi_2(r, s) \le \psi(2 \max(r, 0), 0) \to \psi(0, 0) = 0$ by continuity of $\psi(\cdot, 0)$ at 0.

By lemma 275, ψ_2 is jointly continuous. Also $\psi_2(r, s) \le \psi(2r, s - 1) \to 0$. Finally $\beta(r, s) = \psi_2(r, s) + re^{-s}$ is class $\mathcal{KL}$, the second term making it strictly increasing in r. □

Lemma 277 (Global class $\mathcal{KL}$ from a decay family)**.** *Let α be class $\mathcal{K}_\infty$ and let U be a family of functions such that*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*

2. $U(r,\cdot)$ *is antitone on* $(0,\infty)$ *for each* $r>0$*;*

3. $U(r,s)\to 0$ *as* $s\to\infty$*, for each* $r>0$*;*

4. $r\mapsto U(r+1,s)$ *is monotone on* $[0,\infty)$ *for each* $s>0$*.*

Then there is a global class $\mathcal{KL}$ *function* β *such that*

1. $\beta(r,0)\to\infty$ *as* $r\to\infty$*;*

2. $\alpha(r)\le\beta(r,0)$ *for all* $r\ge 0$*;*

3. $\min(\alpha(r),\sqrt{\alpha(r)U(r+1,s)})\le\beta(r,s)$ *for all* $r\ge 0$ *and* $s>0$*.*

Proof. Take $\psi(r,s):=\alpha(r)$ for $s=0$ and $\min(\alpha(r),\sqrt{\alpha(r)U(r+1,s)})$ for $s>0$, the radius being shifted by 1 so that U is only ever evaluated at a strictly positive radius. Each hypothesis of lemma 276 holds for ψ: monotonicity in r from that of α and of $r\mapsto U(r+1,s)$, antitonicity in s from that of $U(r+1,\cdot)$, decay from lemma 264, and continuity at $r=0$ because $\psi(\cdot,0)=\alpha$. The resulting β dominates ψ, which gives the three conclusions; $\beta(r,0)\ge\alpha(r)\to\infty$ gives the first. No property of ψ beyond these bounds appears in the statement. □

5.4 Class $\mathcal{K}$ bounds for positive definite functions

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. It rests on two smoothing results, which replace a monotone bound by a continuous strictly increasing one.

Lemma 278 (Class $\mathcal{K}$ minorant of a positive monotone function)**.** *Let* $r>0$ *and let* $\psi:\mathbb{R}\to\mathbb{R}$ *satisfy* $\psi(0)=0$*,* $\psi(s)>0$ *for* $s\in(0,r]$*, and* ψ *nondecreasing on* $[0,r]$*. Then there are* $b>0$ *and a function* f *with* $f(0)=0$*,* $f(r)=b$*,* f *continuous and strictly increasing on* $[0,r]$*, and* $f(s)\le\psi(s)$ *for all* $s\in[0,r]$*.*

That is: a positive nondecreasing function admits a class $\mathcal{K}$ *minorant on any bounded interval. This is what turns a positive-definite* V *into the lower comparison bound of theorem 281.*

Proof. Take $f(s)=r^{-1}\int_0^s\psi$ (with the argument of ψ clamped to $[0,r]$, so that it is monotone, hence integrable, on all of $\mathbb{R}$). It is continuous as a primitive, and $f(0)=0$. For $s<s'$ in $[0,r]$, $f(s')-f(s)=r^{-1}\int_s^{s'}\psi>0$ since $\psi>0$ on (s,s'), so f is strictly increasing and $b=f(r)>0$. Finally, ψ is nondecreasing, so $\int_0^s\psi\le s\,\psi(s)\le r\,\psi(s)$, i.e. $f(s)\le\psi(s)$. □

Lemma 279 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let* $\varphi:\mathbb{R}\to\mathbb{R}$ *satisfy* $\varphi(0)=0$*, be monotone on* $[0,\infty)$*, and be continuous at* 0 *from the right. Then there is an* f *with* $f(0)=0$*, continuous and strictly increasing on* $[0,\infty)$*,* $f(s)\to\infty$ *as* $s\to\infty$*, and* $\varphi(s)\le f(s)$ *for all* $s\ge 0$*.*

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ *majorant. The unbounded domain is what distinguishes this from lemma 280, which it implies.*

Continuity at 0 *is necessary: the unit step has no continuous majorant vanishing at* 0*.*

Proof. Take $f(s) = s + g(s)$ with $g(s) = \int_1^2 \varphi(su)\,du$, the average of φ over $[s, 2s]$ (with φ clamped to $\varphi(\max(t, 0))$ so that it is monotone, hence integrable, on all of $\mathbb{R}$). Monotonicity of φ gives $\varphi(s) \le g(s) \le \varphi(2s)$ and that g is nondecreasing, so f is strictly increasing, $f \ge \varphi$, and $f(s) \ge s \to \infty$. For $s > 0$, $g(s) = s^{-1}(\Phi(2s) - \Phi(s))$ with Φ the (continuous) primitive of φ, so g is continuous there; at 0, $0 \le g(s) \le \varphi(2s) \to 0$ by continuity of φ at 0. □

Lemma 280 (Class $\mathcal{K}$ majorant on a bounded interval)**.** *Let $r > 0$ and let φ satisfy $\varphi(0) = 0$, be monotone on $[0, r]$, and be continuous at 0 from the right. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \le f(s)$ for all $s \in [0, r]$.*

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$, still vanishes at 0 and agrees with φ near 0 (so is still continuous there), and apply lemma 279. Restricting the resulting majorant to $[0, r]$ gives f; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. □

Theorem 281 (Class $\mathcal{K}$ sandwich bounds)**.** *Let $W : \mathbb{R}^n \to \mathbb{R}$ be positive definite (definition 28) about x_{eq} on a set containing $\overline{B}(x_{\text{eq}}, r)$. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with*

$$\alpha_1(\|x - x_{\text{eq}}\|) \;\le\; W(x) \;\le\; \alpha_2(\|x - x_{\text{eq}}\|) \qquad \forall\, \|x - x_{\text{eq}}\| < r.$$

The conclusion holds on the open *ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x - x_{\text{eq}}\| = r$.*

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x - x_{\text{eq}}\|$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.3 (proved in his Appendix C.4). Differences:*

- *The bound holds on the open ball, with α_1, α_2 class $\mathcal{K}$ on $[0, r)$; Khalil states it on $[0, r]$.*
- *The radially unbounded $\mathcal{K}_\infty$ case is not derived here.*

Proof. Translate to the origin, $V(y) = W(y + x_{\text{eq}})$. For the lower bound take $\psi(s) = \inf\{V(y) : s \le \|y\| \le r\}$, the infimum of V over the annulus of inner radius s: it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(y) : \|y\| \le s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 278 gives a class $\mathcal{K}$ minorant of ψ and lemma 280 a class $\mathcal{K}$ majorant of ϕ. Since $\psi(\|y\|) \le V(y) \le \phi(\|y\|)$ pointwise, these sandwich V. □

5.5 The decay system and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the comparison system $\dot{y} = -\alpha(y)$ enters: a Lyapunov function along a trajectory is a subsolution of it.

Definition 282 (The decay system of a class $\mathcal{K}$ function)**.** For α class $\mathcal{K}$ on $[0, a)$ (definition 236), the *decay system* is the scalar time-invariant system

$$\dot{y} = -\alpha(y),$$

as a continuous-time system (definition 71) with no input. Its meaningful states are $y \in [0, a)$.

Theorem 283 (The decay system has a class $\mathcal{KL}$ solution operator)**.** *Let α be class $\mathcal{K}$ on $[0, a)$ and at most linear near the origin, $\alpha(x) \le Lx$ on $(0, \text{base}]$. Then there is a class $\mathcal{KL}$ function σ (definition 261) with $\sigma(y_0, 0) = y_0$ such that for every $y_0 \in [0, a)$ and every t_0, $t \mapsto \sigma(y_0, t - t_0)$ is a trajectory on $[t_0, \infty)$ (definition 89) of the decay system $\dot{y} = -\alpha(y)$ (definition 282).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.4. Differences:*

- *Khalil assumes α locally Lipschitz; here the hypothesis is the linear bound $\alpha(x) \le Lx$ near the origin, which is what the construction uses.*
- *Uniqueness of the solution is not asserted.*

Proof. **Osgood's construction.** Let $\eta(y) = -\int_{\text{base}}^{y} dx/\alpha(x)$ be the time for the solution started at base to reach y. Since $\alpha > 0$ on $(0, a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \le Lx$ near 0, $\eta(y) \ge (\log \text{base} - \log y)/L \to +\infty$ as $y \to 0^+$, so the origin is not reached in finite time. Setting $\sigma(r, s) := \eta^{-1}(\eta(r) + s)$, extended by $\sigma(0, s) = 0$, the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1}, and differentiating $\eta(\sigma) = \eta(r) + s$ gives $\dot{\sigma} = -\alpha(\sigma)$ for $s > 0$. At $s = 0$ the one-sided derivative is the limit of these, $-\alpha(r)$, by continuity of σ and α. □

Lemma 284 (Lipschitz class $\mathcal{K}$ minorant)**.** *Let α be a class $\mathcal{K}$ function on $[0, a)$. Then there are $b' > 0$, $L > 0$ and a class $\mathcal{K}$ function $\beta : [0, a) \to [0, b')$ with $\beta(x) \le \alpha(x)$ for $x \in [0, a)$, $\beta(x) \le Lx$ for $x \ge 0$, and β L-Lipschitz on all of $\mathbb{R}$.*

The range of β cannot in general be the range $[0, b)$ of α: $\alpha(x) = 1 - \sqrt{1 - x}$ on $[0, 1)$ approaches 1 more slowly than any Lipschitz function can.

Proof. Extend α to a nondecreasing $g : \mathbb{R} \to [0, b]$ by 0 on the left and b from a on, and set $\beta(x) = a^{-1} \int_0^x g$, so $b' = \beta(a)$. Since $0 \le g \le b$, β is (b/a)-Lipschitz, and with $\beta(0) = 0$ this gives $\beta(x) \le (b/a)\, x$. Since $g > 0$ on $(0, \infty)$, β is strictly increasing, hence class $\mathcal{K}$ by definition 238. Finally g is nondecreasing, so for $x \in [0, a)$, $\int_0^x g \le x\, \alpha(x) \le a\, \alpha(x)$, i.e. $\beta(x) \le \alpha(x)$. □

Theorem 285 (Subsolutions of the decay system decay)**.** *Let α be class $\mathcal{K}$ on $[0, a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r, 0) \le r$ such that: whenever v is continuous on $[t_0, t]$, takes values in $[0, a)$ there, has bounded forward difference quotients, and is a Dini subsolution of the decay system (definition 282),*

$$D^+ v(s) \;\le\; -\alpha(v(s)),$$

it obeys $v(t) \le \sigma(v(t_0),\, t - t_0)$.

A differential inequality *with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.*

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta \le \alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x) \le Lx$ that theorem 283 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot{y} = -\beta(y)$ with $\sigma(r, 0) = r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0), 0)$, theorem 56 bounds v by it — the perturbed solutions that lemma requires coming from theorem 48, whose hypotheses hold because β is globally Lipschitz (lemma 284). □

Theorem 286 (Subsolutions above a level decay to it)**.** *Let α be class $\mathcal{K}$ on $[0,a)$, and σ the class $\mathcal{KL}$ function of theorem 285. Let v be continuous on $[t_0,t]$ with values in $[0,a)$, with right derivative v' on $[t_0,t)$ satisfying $v'(s)\le-\alpha(v(s))$ whenever $v(s)>\eta$. Then*

$$v(t)\le\max\{\sigma(v(t_0),t-t_0),\ \eta\}.$$

Proof. If $v>\eta$ on all of $[t_0,t)$, v is a subsolution of the decay system there and theorem 285 applies. Otherwise $v(\tau)\le\eta$ for some $\tau\in[t_0,t)$. Wherever $v>\eta$, $v'\le-\alpha(v)\le0$, so by lemma 17 on $[\tau,t]$, $v(t)\le\max\{v(\tau),\eta\}=\eta$. □

5.6 Massera's lemma

A class $\mathcal{K}$ function with class $\mathcal{K}$ derivative can flatten any decay rate enough to make a Lyapunov integral along the solutions converge, together with its derivative in the initial state.

Lemma 287 (Massera's lemma)**.** *Let $g:[0,\infty)\to\mathbb{R}$ be positive, continuous and strictly decreasing with $g(t)\to0$, and $h:[0,\infty)\to\mathbb{R}$ positive, continuous and nondecreasing. There is a class $\mathcal{K}_\infty$ function G (definition 237) whose derivative G' is class $\mathcal{K}_\infty$, and constants $k_1,k_2>0$, such that for every continuous u with $0\le u(t)\le g(t)$ the integrals below converge and*

$$\int_0^\infty G(u(t))\,dt\le k_1,\qquad\int_0^\infty G'(u(t))\,h(t)\,dt\le k_2.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma C.1. Differences: G and G' are class $\mathcal{K}_\infty$ rather than class $\mathcal{K}$ on $[0,\infty)$, since only their values on $[0,g(0)]$ matter; convergence of the integrals is stated.*

Proof. Let $\eta(t)=g(t)+1/t$: continuous and strictly decreasing on $(0,\infty)$, with $\eta\to\infty$ at 0^+, $\eta\to0$ at ∞, and $\eta>g$. Its inverse $\theta=\eta^{-1}$ (lemma 257) satisfies $\theta(s)\ge t$ whenever $0<s\le g(t)$. Let

$$G'(s)=\frac{e^{-\theta(s)}}{h(\theta(s))}+\max(s-g(0),0),\qquad G(r)=\int_0^r G'(s)\,ds,$$

so that $G'(s)\le e^{-t}/h(t)$ for $0\le s\le g(t)$. Then $G'(u(t))\,h(t)\le e^{-t}$ and $G(u(t))\le u(t)\,e^{-t}/h(t)\le(g(0)/h(0))\,e^{-t}$, so $k_1=g(0)/h(0)$ and $k_2=1$. Khalil's piecewise-linear η is replaced by $g(t)+1/t$, which lies above g for every $t>0$, so no splitting at t_1 is needed. □

Chapter 6

Stability of continuous-time systems

Stability notions for an equilibrium of an autonomous continuous-time system $\dot{x} = f(t, x)$ (definition 71); a system with input is first unforced (definition 80). Each notion is defined once, for time-varying systems, with the initial time $t_0 \geq 0$ explicit; a time-invariant system is a special case, and that its notions may be read with $t_0 = 0$ is a theorem rather than a second definition.

Bounds — stability, uniform convergence times, exponential estimates — quantify over trajectories on segments $[t_0, t_1]$, so that a solution escaping in finite time counts against stability instead of being silently excluded. Limits quantify over trajectories on rays $[t_0, \infty)$.

6.1 Definitions

Definition 288 (Stability)**.** The equilibrium x_{eq} of an autonomous continuous-time system $\dot{x} = f(t, x)$ is *stable* when for every $\varepsilon > 0$ and $t_0 \geq 0$ there is $\delta = \delta(\varepsilon, t_0) > 0$ such that every trajectory on a segment $[t_0, t_1]$ with $\|x(t_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(t) - x_{\text{eq}}\| < \varepsilon$ for all $t \in [t_0, t_1]$.

Quantifying over segments rather than over solutions on $[t_0, \infty)$ means a solution that escapes in finite time counts against stability instead of being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*stable*).

Definition 289 (Instability)**.** The equilibrium is *unstable* when it is not stable (definition 288).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 290 (Uniform stability)**.** The equilibrium is *uniformly stable* when δ in definition 288 can be chosen independently of t_0: $\delta = \delta(\varepsilon)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly stable*).

Definition 291 (Global uniform stability)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 290) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$. In particular every trajectory is bounded, so none escapes in finite time.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 292 (Local asymptotic stability)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 288) and for every $t_0 \geq 0$ there is $c = c(t_0) > 0$ such that the system is forward complete from the ball of radius c about x_{eq} at t_0 (definition 82) and every trajectory on $[t_0, \infty)$ starting in that ball satisfies $x(t) \to x_{\text{eq}}$. Forward completeness keeps solutions that escape in finite time from being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*asymptotically stable*).

Definition 293 (Global asymptotic stability)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 288), the system is forward complete from every state at every $t_0 \geq 0$ (definition 82), and every trajectory on $[t_0, \infty)$ satisfies $x(t) \to x_{\text{eq}}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Section 4.1.

Definition 294 (Uniform convergence time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies $\|x(t) - x_{\text{eq}}\| < \eta$ for $t \in [t_0 + T, t_1]$.

Definition 295 (Uniform asymptotic stability)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 290) and there is $c > 0$ from which convergence has a uniform convergence time (definition 294).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly asymptotically stable*).

Definition 296 (Global uniform asymptotic stability)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 291) and every $c > 0$ admits a uniform convergence time (definition 294).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 297 (Local exponential stability)**.** The equilibrium is *locally exponentially stable* when there are $c, k, \gamma > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(t) - x_{\text{eq}}\| \leq k \, \|x(t_0) - x_{\text{eq}}\| \, e^{-\gamma(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

The constants do not depend on t_0: exponential stability is uniform, as in the reference.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Definition 298 (Global exponential stability)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 297 holds for every initial state, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 299 (Uniform boundedness)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq \beta \qquad \forall\, t \in [t_0, t_1]$$

for every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$. The point x_c need not be an equilibrium. The bound applies to every segment a solution exists on, so finite escape violates it.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 300 (Global uniform boundedness)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 299 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 301 (Uniform ultimate boundedness)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $t_0 \geq 0$ the system is forward complete from the ball of radius c about x_c (definition 82), and for every $0 < a < c$ there is $T = T(a, b) \geq 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq b \qquad \forall\, t \geq t_0 + T$$

for every trajectory on $[t_0, \infty)$. The point x_c need not be an equilibrium. Khalil leaves forward completeness implicit; without it, a solution escaping before $t_0 + T$ would never be tested.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 302 (Global uniform ultimate boundedness)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when the system is forward complete from every state at every $t_0 \geq 0$ (definition 82) and the ultimate bound of definition 301 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

6.2 Implications

Lemma 303 (Instability from a fixed escape radius)**.** *Fix $\varepsilon > 0$. If for every $\delta > 0$ some trajectory on a segment $[0, T]$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 289).*

Proof. Stability at $t_0 = 0$ with tolerance ε gives a δ; the trajectory for that δ contradicts it. □

Lemma 304 (Uniform stability implies stability)**.** *A uniformly stable equilibrium (definition 290) is stable (definition 288).*

Proof. □

Lemma 305 (Global asymptotic stability implies local asymptotic stability)**.** *A globally asymptotically stable equilibrium (definition 293) is locally asymptotically stable (definition 292).*

Proof. □

Lemma 306 (Uniform asymptotic stability implies local asymptotic stability)**.** *A uniformly asymptotically stable equilibrium (definition 295) of a forward-complete system (definition 83) is locally asymptotically stable (definition 292). Forward completeness is needed: bounds on solution segments do not by themselves make the segments extend.*

Proof. Uniform stability gives stability; a uniform convergence time gives convergence of every trajectory on a ray, with the same radius c for every t_0. □

Lemma 307 (Global uniform asymptotic stability implies global asymptotic stability)**.** *A globally uniformly asymptotically stable equilibrium (definition 296) is globally asymptotically stable (definition 293), given forward completeness (definition 83).*

Proof. Every trajectory starts in some ball, and every ball has a uniform convergence time. □

Lemma 308 (Exponential stability implies uniform stability)**.** *A locally exponentially stable equilibrium (definition 297) is uniformly stable (definition 290).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta = \min(c, \varepsilon/k)$ works for every t_0. □

Lemma 309 (Exponential stability implies uniform asymptotic stability)**.** *A locally exponentially stable equilibrium (definition 297) is uniformly asymptotically stable (definition 295).*

Proof. Uniform stability is lemma 308. From the ball of radius c the distance after time T is below $kc\,e^{-\gamma T}$, which is below any $\eta > 0$ for T large, independently of t_0. □

Lemma 310 (Global exponential stability implies local exponential stability)**.** *A globally exponentially stable equilibrium (definition 298) is locally exponentially stable (definition 297).*

Proof. □

Lemma 311 (Global exponential stability implies global uniform asymptotic stability)**.** *A globally exponentially stable equilibrium (definition 298) is globally uniformly asymptotically stable (definition 296).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta(\varepsilon) = \varepsilon/k$ witnesses global uniform stability. For each radius c the estimate is locally exponential from that ball, which gives a uniform convergence time by lemma 309. □

Lemma 312 (Exponential stability under a change of coordinates)**.** *Let $e : E \to E'$ be continuous linear with $\|x\| \le a\|ex\|$ and $\|ex\| \le b\|x\|$, $a, b > 0$, and let $F'(t, x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})) = e(F(t, x))$. If x'_{eq} is locally exponentially stable for s' (definition 297), so is x_{eq} for s.*

Proof. If x is a trajectory of s, then $x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})$ is one of s'. Its estimate, with radius c/b and constant abk, is one for x. □

6.3 Comparison-function characterizations

The uniform notions are equivalent to comparison-function bounds on the trajectories: uniform stability to a class $\mathcal{K}$ bound, uniform asymptotic stability to a class $\mathcal{KL}$ bound. The hard direction is the decay in elapsed time. A uniform convergence time says that for each tolerance η there is a delay after which every trajectory from a ball is inside the η-ball. Read the other way round it bounds how far a trajectory can still be after a given elapsed time, but only through a function of η that may be discontinuous. The results below regularize it into a decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$: the time-decay half of a class $\mathcal{KL}$ bound, whose other half is a uniform class $\mathcal{K}$ bound.

Definition 313 (Uniform class $\mathcal{K}$ bound)**.** The trajectories of s *have the uniform class $\mathcal{K}$ bound* $\alpha : [0, a) \to [0, b)$ about x_{eq} when

$$\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in [t_0, t_1]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.19).

Definition 314 (Uniform class $\mathcal{K}_\infty$ bound)**.** The trajectories of s *have the uniform class* $\mathcal{K}_\infty$ *bound* α about x_{eq} when $\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.20).

Lemma 315 (Local uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from the ball of radius c (definition 294) and a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 313). Let $0 < a \le c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 260) such that*

$$\|x(t) - x_{\mathrm{eq}}\| \le U(t - t_0)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in (t_0, t_1]$.

The bound depends on the elapsed time alone: neither on t_0, nor on the trajectory, nor on where in the a-ball it started.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5.*

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η-ball; it is finite by hypothesis and antitone in η, but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r)\, ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η, hence tends to $+\infty$ as $\eta \to 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \to 0$ as $\eta \to \infty$. Inverting it (lemma 254, lemma 255, lemma 256, lemma 257, lemma 258) gives $U = \mathrm{invFunOn}\, W(r, \cdot)\, (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \ge \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$-ball, which is the claimed bound. □

Lemma 316 (Global uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from every ball (definition 294) and a uniform class $\mathcal{K}_\infty$ bound (definition 314). Then there is a family $U : (0, \infty) \times (0, \infty) \to \mathbb{R}$ with*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r, \cdot)$ antitone on $(0, \infty)$ for each $r > 0$;*
3. *$U(r, s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r + 1, s)$ monotone on $[0, \infty)$ for each $s > 0$;*
5. *$\|x(t) - x_{\mathrm{eq}}\| \le U(r, t - t_0)$ for every $r > 0$, every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < r$, and every $t \in (t_0, t_1]$.*

Conditions (1)–(4) are exactly the hypotheses of lemma 277; (5) is the decay bound. Unlike the local case (lemma 315) no single envelope can serve every initial state, so the radius remains a parameter.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5 (global case).*

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (a uniform convergence time from the r-ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r,\cdot)$ to be the envelope of lemma 315 at that radius, namely the inverse of $W(r,\cdot)$. Conditions (1)–(3) and (5) are then that lemma's conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. □

Lemma 317 (Uniform convergence from a class $\mathcal{KL}$ bound)**.** *Let β be class $\mathcal{KL}$ on $[0,a)\times[0,\infty)$ (definition 261), and suppose every trajectory on a segment $[t_0,t_1]$, $t_0 \geq 0$, with $\|x(t_0)-x_{\text{eq}}\| < a$ satisfies $\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$ on the segment. Then for every $c < a$, trajectories from the ball of radius c have a uniform convergence time (definition 294).*

Proof. For $c < 0$ no trajectory starts in the ball. Otherwise every deviation is at most $\beta(c, t-t_0)$, by monotonicity of β in its first argument, and $\beta(c,\cdot) \to 0$. □

Lemma 318 (Class $\mathcal{K}$ characterization of stability)**.** *The equilibrium x_{eq} is stable (definition 288) if and only if for each $t_0 \geq 0$ there is a class $\mathcal{K}$ function α_{t_0} on $[0, a_{t_0})$ with $\|x(t)-x_{\text{eq}}\| \leq \alpha_{t_0}(\|x(t_0)-x_{\text{eq}}\|)$ for every trajectory on a segment $[t_0,t_1]$ with $\|x(t_0)-x_{\text{eq}}\| < a_{t_0}$, and every $t \in [t_0,t_1]$. A single α for all t_0 is uniform stability (lemma 319).*

Proof. The argument of lemma 319 with the initial time fixed. □

Lemma 319 (Class $\mathcal{K}$ characterization of uniform stability)**.** *The equilibrium x_{eq} is uniformly stable (definition 290) if and only if the trajectories have a uniform class $\mathcal{K}$ bound (definition 313): there is a class $\mathcal{K}$ function α on $[0,a)$ with*

$$\|x(t)-x_{\text{eq}}\| \leq \alpha(\|x(t_0)-x_{\text{eq}}\|)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0 \geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t \in [t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, first half.*

Proof. Forward: the worst deviation $\omega(r)$ reachable from the r-ball is nondecreasing, vanishes at 0 and is continuous there by uniform stability; a continuous strictly increasing majorant (lemma 280) is the class $\mathcal{K}$ bound. Backward: α is continuous at 0 with $\alpha(0) = 0$, so a small radius δ has $\alpha(\delta) < \varepsilon$. □

Lemma 320 (Class $\mathcal{KL}$ characterization of uniform asymptotic stability)**.** *The equilibrium x_{eq} is uniformly asymptotically stable (definition 295) if and only if there are a class $\mathcal{KL}$ function β on $[0,a)\times[0,\infty)$ (definition 261) with*

$$\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0 \geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t \in [t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, second half.*

Proof. Forward: uniform stability gives a class $\mathcal{K}$ bound α (lemma 319) and, with the uniform convergence time, a decay envelope U (lemma 315); $\beta = \min(\alpha, \sqrt{\alpha\, U})$ is class $\mathcal{KL}$ (theorem 265) and bounds the trajectories. Backward: $\beta(\cdot,0)$ is continuous at 0, which gives uniform stability, and lemma 317 gives the convergence time. □

Lemma 321 (Class $\mathcal{KL}$ characterization, global case)**.** *The equilibrium x_{eq} is globally uniformly asymptotically stable (definition 296) if and only if there is a $\beta : [0,\infty) \times [0,\infty) \to \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0,s) = 0$ for $s \geq 0$;*
2. *jointly continuous on $[0,\infty) \times [0,\infty)$;*
3. *strictly increasing in r for each fixed $s \geq 0$;*
4. *radially unbounded: $\beta(r,0) \to \infty$ as $r \to \infty$;*
5. *antitone in s for each fixed $r \geq 0$, with $\beta(r,s) \to 0$ as $s \to \infty$;*

such that $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \geq 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Condition (4) is what makes the characterization global*: without it β is class $\mathcal{KL}$ on a bounded radius and the statement collapses to the local one (lemma 320).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, global case.*

Proof. Forward: global uniform stability gives a class $\mathcal{K}_\infty$ bound, the convergence times from every ball a family of decay envelopes (lemma 316), and lemma 277 combines them. Backward: $\alpha = \beta(\cdot, 0)$ is class $\mathcal{K}_\infty$ and $\delta = \alpha^{-1}$ gives global uniform stability; from the r-ball every deviation is at most $\beta(r, \cdot) \to 0$. □

The same characterizations hold with the bounds on solutions over $[t_0, \infty)$. A bound on every segment restricts to those solutions, so that direction needs nothing more. The converse needs every segment from the ball to extend to such a solution: forward completeness from some ball about x_{eq} locally, and from every state globally.

Lemma 322 (Class $\mathcal{K}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 82). Then x_{eq} is uniformly stable if and only if there is a class $\mathcal{K}$ function α on $[0, a)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 319, with the bound moved between segments and solutions on $[t_0, \infty)$. Forward: restrict a solution to $[t_0, t]$. Converse: shrink the radius to $a' = \min(c, a/2)$, restricting α (definition 239); a segment starting within a' extends by completeness to a solution on $[t_0, \infty)$, which the bound covers. □

Lemma 323 (Class $\mathcal{KL}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 82). Then x_{eq} is uniformly asymptotically stable if and only if there is a class $\mathcal{KL}$ function β on $[0, a) \times [0, \infty)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 320, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 322, shrinking the radius to $\min(c, a)$ (definition 269). □

Lemma 324 (Class $\mathcal{KL}$ characterization for forward-complete systems, global case)**.** *Let the system be forward complete (definition 83). Then x_{eq} is globally uniformly asymptotically stable if and only if there is a β with the properties of lemma 321 and $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 321, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 322. □

6.4 Lyapunov functions

A Lyapunov function is a certificate: its existence implies a stability notion. The certificates are time-varying, $V(t, x)$, and named by what they certify. What separates them is which bounds hold, not whether time appears: a positive definite (definition 28) lower bound gives stability, and a positive definite upper bound as well (decrescence) is what makes the conclusion uniform. A time-invariant $V(x)$ is its own lower and upper bound, so it is a uniform certificate as soon as it decreases along the system.

Definition 325 (Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is a *Lyapunov function* on a neighbourhood D of x_{eq} when, for all $t \geq 0$ and $x \in D$, V is differentiable at (t, x) and

$$V(t, x_{\mathrm{eq}}) = 0, \qquad W_1(x) \leq V(t, x), \qquad \dot{V}(t, x) \leq 0,$$

for some W_1 positive definite on D (definition 28), with $\dot{V}$ the derivative along the system (definition 94). It certifies stability (definition 288).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 326 (Uniform Lyapunov function)**.** A Lyapunov function (definition 325) that is also *decrescent*: $V(t, x) \leq W_2(x)$ for all $t \geq 0$, $x \in D$, for some W_2 positive definite on D. The upper bound is what makes the conclusion uniform: it certifies uniform stability (definition 290).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 327 (Strict uniform Lyapunov function)**.** A uniform Lyapunov function (definition 326) with $\dot{V}(t, x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. It certifies uniform asymptotic stability (definition 295).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.9.

Definition 328 (Exponential Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is an *exponential Lyapunov function* with exponent $a > 0$ on a neighbourhood D of x_{eq} when there are $k_1, k_2, k_3 > 0$ with, for all $t \geq 0$ and $x \in D$, V differentiable at (t, x) and

$$k_1 \|x - x_{\mathrm{eq}}\|^a \leq V(t, x) \leq k_2 \|x - x_{\mathrm{eq}}\|^a, \qquad \dot{V}(t, x) \leq -k_3 \|x - x_{\mathrm{eq}}\|^a.$$

It certifies local exponential stability (definition 297).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.10.

Lemma 329 (Time-invariant Lyapunov functions are uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} (definition 28) and differentiable at its points, with $\nabla V(x) \cdot f(t, x) \leq 0$ for all $t \geq 0$, $x \in D$. Then $(t, x) \mapsto V(x)$ is a uniform Lyapunov function (definition 326).*

Proof. Take $W_1 = W_2 = V$. The derivative along the system of a function of x alone is its directional derivative along the vector field. □

Lemma 330 (Strict time-invariant Lyapunov functions are strict uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} and differentiable at its points, with $\nabla V(x) \cdot f(t,x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 327).*

Proof. As for lemma 329; $-W_3 \leq 0$ gives the non-strict decrease. □

Lemma 331 (Strict decrease suffices for a time-invariant system)**.** *Let the state equation be $\dot{x} = f(x)$ with f continuous and $f(x_{\text{eq}}) = 0$, and let $V : E \to \mathbb{R}$ be C^1 and positive definite on a neighbourhood D of x_{eq} with $\nabla V(x) \cdot f(x) < 0$ for $x \in D \setminus \{x_{\text{eq}}\}$. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 327).*

Proof. Take $W_3 = -\nabla V \cdot f$: continuous since V is C^1 and f is continuous, zero at x_{eq} since $f(x_{\text{eq}}) = 0$, and positive elsewhere on D; apply lemma 330. □

6.5 Lyapunov's direct method

For a time-varying V the derivative along the system acquires an explicit time term, and positive definiteness of $V(t,\cdot)$ on its own buys nothing uniform in t_0: V could flatten as $t \to \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity. Only trajectories on segments $[t_0, t_1]$ are used, so no solution is assumed to exist for all time.

Theorem 332 (Lyapunov's uniform stability theorem)**.** *If V is a uniform Lyapunov function for x_{eq} (definition 326) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly stable (definition 290).*

The sandwich $W_1(x) \leq V(t,x) \leq W_2(x)$ is what makes the conclusion uniform in t_0: W_1 and W_2 do not depend on t, so neither does the $\delta(\varepsilon)$ extracted from them.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.8.*

Proof. Given ε, work on a radius $r \leq \varepsilon$ whose closed ball lies in the region of the certificate. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 on it (theorem 281), fix a level d below both ranges, and take $\delta = \alpha_2^{-1}(d)$: starting within δ of x_{eq} forces $V(t_0, x(t_0)) < d$. To leave the r-ball the trajectory would first cross a sphere on which $W_1 > d$; up to that time V has not increased (lemma 62), so it is still below d, a contradiction. □

Theorem 333 (Lyapunov's uniform asymptotic stability theorem)**.** *If V is a strict uniform Lyapunov function for x_{eq} (definition 327) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly asymptotically stable (definition 295).*

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.9. Khalil's two "moreover" clauses are not part of this conclusion: the explicit class $\mathcal{KL}$ bound, which the proof constructs, and the global case.*

Proof. Uniform stability is theorem 332. For convergence, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3 on a ball in the region of the certificate, and the trapping radius c of that proof. Along

a trajectory from the c-ball, $\dot V \le -W_3(x) \le -\alpha_3(\|x - x_{\text{eq}}\|) \le -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, x(t))$ satisfies $D^+v \le -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 285 gives a class $\mathcal{KL}$ σ with $v(t) \le \sigma(v(t_0), t - t_0)$; unwinding the sandwich,

$$\|x(t) - x_{\text{eq}}\| \le \alpha_1^{-1}(\sigma(\alpha_2(\|x(t_0) - x_{\text{eq}}\|), t - t_0)),$$

a class $\mathcal{KL}$ bound. Conclude by lemma 317. □

In the global theorem compact sublevel sets of V are the hypothesis; for a continuous V they follow from radial unboundedness.

Lemma 334 (Sublevel sets of a radially unbounded function are compact)**.** *Let E be a proper normed space. If $V : E \to \mathbb{R}$ is continuous and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$), then every sublevel set $\{x \mid V(x) \le c\}$ is compact.*

Proof. Closed: it is $V^{-1}((-\infty, c])$, with V continuous. Bounded: radial unboundedness gives a compact K off which $V > c$, so the sublevel set lies in K. Compact: closed and bounded in a proper space. □

Theorem 335 (Lyapunov's global asymptotic stability theorem)**.** *Let the state equation be $\dot x = f(x)$ with f continuous, on $\mathbb{R}^n$. Let $V : \mathbb{R}^n \to \mathbb{R}$ be C^1, positive definite about x_{eq} on all of $\mathbb{R}^n$ (definition 28), with compact sublevel sets $\{V \le c\}$ (equivalently, radially unbounded) and $\nabla V(x) \cdot f(x) < 0$ for $x \ne x_{\text{eq}}$. Then x_{eq} is globally uniformly asymptotically stable (definition 296).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2.*

Proof. V does not increase along trajectories: $\nabla V \cdot f \le 0$ everywhere, since $\nabla V(x_{\text{eq}}) = 0$ at the minimum. So a trajectory starting below every value V takes outside the ε-ball never leaves it. Take $\delta(\varepsilon)$ the supremum of the radii $\rho \le \varepsilon$ whose closed ball lies strictly below those values: it is positive, since V is bounded below by a positive constant outside the ε-ball and small near x_{eq}, and it tends to ∞, since V is bounded on each ball and large far out. This is global uniform stability.

For convergence from the c-ball, trajectories stay in the compact set $\{V \le M\}$, $M = \max_{\overline{B}(x_{\text{eq}}, c)} V$, on which, outside the $\delta(\eta)$-ball, $\nabla V \cdot f \le -\gamma < 0$. So V falls at rate γ while the trajectory is outside, and since $V \ge 0$ it enters the $\delta(\eta)$-ball by time $t_0 + M/\gamma + 1$; from then on it stays in the η-ball. □

Theorem 336 (Lyapunov's exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} (definition 328), then x_{eq} is locally exponentially stable (definition 297): near x_{eq},*

$$\|x(t) - x_{\text{eq}}\| \le \left(\frac{k_2}{k_1}\right)^{1/a} \|x(t_0) - x_{\text{eq}}\|\, e^{-(k_3/(ak_2))(t - t_0)}.$$

The state space is any normed space.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10.*

Proof. The bounds give $\dot V \le -(k_3/k_2)V$, so $e^{(k_3/k_2)t}V(t, x(t))$ does not increase and $V(t, x(t)) \le V(t_0, x(t_0))\, e^{-(k_3/k_2)(t - t_0)}$ while the trajectory stays in D; the sandwich and an a-th root give the estimate. To stay in D, start within $r/(K + 1)$ of x_{eq}, where $\overline{B}(x_{\text{eq}}, r) \subseteq D$ and $K = (k_2/k_1)^{1/a}$: at the first time the trajectory reached the sphere of radius r the estimate would put it strictly inside. □

Theorem 337 (Lyapunov's global exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} on the whole space (definition 328 with $D = E$), then x_{eq} is globally exponentially stable (definition 298).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10, global case.*

Proof. The estimate of theorem 336 holds on every trajectory segment, since every segment stays in $D = E$. □

6.6 Converse Lyapunov theorems

Exponential stability has a quadratic Lyapunov function, built by integrating the squared distance along solutions over a fixed horizon. Its gradient is of the order of the distance to the equilibrium, which is what lets it absorb perturbations of the vector field.

Theorem 338 (Converse Lyapunov theorem for global exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times \mathbb{R}^n$ and $f(t, \cdot)$ L-Lipschitz for every $t \in \mathbb{R}$ (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$). If x_{eq} is globally exponentially stable (definition 298), there is an exponential Lyapunov function V with exponent 2 on $\mathbb{R}^n$ (definition 328) and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ for all $t \ge 0$ and $x \in \mathbb{R}^n$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14, global case. Differences:*

- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound, and is asked for all $t \in \mathbb{R}$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The constants are $c_1 = \delta e^{-2L\delta}$, $c_2 = \delta k^2$ and $c_4 = 2k\delta e^{L\delta}$, cruder than Khalil's.*

Proof. Solutions $\phi(\tau; t, x)$ exist from every (t, x), since f is globally Lipschitz in x. Let $V(t, x) = \int_0^\delta \|\phi(t+\sigma; t, x) - x_{\text{eq}}\|^2 \, d\sigma$ with $\delta = \ln(2k^2+1)/(2\gamma)$. The exponential estimate and the lower bound $\|\phi(\tau) - x_{\text{eq}}\|^2 \ge e^{-2L(\tau-t)}\|x - x_{\text{eq}}\|^2$ sandwich V. V is differentiable: appending time to the state, solutions are differentiable in their initial data (theorem 54) and locally Lipschitz in it (lemma 51), so the integral can be differentiated under the integral sign. Along a solution, uniqueness gives $V = \int_h^{h+\delta} \|\phi(t+\sigma) - x_{\text{eq}}\|^2 d\sigma$, so $\dot{V} = \|\phi(t+\delta) - x_{\text{eq}}\|^2 - \|x - x_{\text{eq}}\|^2 \le (k^2 e^{-2\gamma\delta} - 1)\|x - x_{\text{eq}}\|^2 \le -\frac{1}{2}\|x - x_{\text{eq}}\|^2$. Finally, solutions from nearby states separate at most like $e^{L\sigma}$, which makes $V(t, \cdot)$ locally Lipschitz with constant $\approx 2k\delta e^{L\delta}\|x - x_{\text{eq}}\|$. □

Theorem 339 (Converse Lyapunov theorem for exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, $f(t, x_{\text{eq}}) = 0$ for all t, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\text{eq}}\| < r\}$, and $f(t, \cdot)$ L-Lipschitz on D for all t (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$ there). Let $k, \gamma, r_0 > 0$ with $r_0 < r/k$, $D_0 = \{x : \|x - x_{\text{eq}}\| < r_0\}$, and suppose every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, with $x(t_0) \in D_0$ satisfies*

$$\|x(t) - x_{\text{eq}}\| \le k\,\|x(t_0) - x_{\text{eq}}\|\, e^{-\gamma(t-t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Then there is an exponential Lyapunov function V with exponent 2 on D_0 (definition 328), that is, with

$$c_1\|x - x_{\text{eq}}\|^2 \le V(t, x) \le c_2\|x - x_{\text{eq}}\|^2, \qquad \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \le -c_3\|x - x_{\text{eq}}\|^2$$

for $t \ge 0$ and $x \in D_0$, and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14. Differences:*

- *f is C^1 and Lipschitz for all $t \in \mathbb{R}$, not only $t \geq 0$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound.*
- *The estimate is assumed on trajectory segments.*

Proof. Cut the field off outside the ball: with χ a smooth bump equal to 1 on the ball of radius ρ_1, $kr_0 < \rho_1 < r$, and vanishing near the sphere of radius r, $\tilde{f} = \chi f$ is C^1 and globally Lipschitz in x, uniformly in t, and agrees with f on the ρ_1-ball. A solution of $\tilde{f}$ from D_0 never reaches the ρ_1-sphere: up to the first time it would, it is a solution of f, so Khalil's estimate keeps it within $kr_0 < \rho_1$. It is therefore a solution of f, and obeys the estimate for all time. The construction of theorem 338, run for $\tilde{f}$ from D_0, gives V; on D_0, where $\tilde{f} = f$, its derivative along $\tilde{f}$ is its derivative along f. □

6.7 Ultimate boundedness

A Lyapunov-like function whose derivative is negative only outside a ball does not certify stability, but it does bound the trajectories: they decay like a class $\mathcal{KL}$ function until they reach a neighbourhood of the ball, and stay there.

Theorem 340 (Ultimate boundedness theorem)**.** *Let the state space be $\mathbb{R}^n$, $\alpha_1 : [0, r) \to [0, b_1)$ and $\alpha_2 : [0, r) \to [0, b_2)$ class $\mathcal{K}$, and $V : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ differentiable at (t, x) and*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x) \leq -W_3(x) \quad \text{for } \mu \leq \|x - x_c\|$$

whenever $t \geq 0$ and $\|x - x_c\| < r$, where W_3 is positive definite about x_c on the closed r-ball (definition 28) and $0 \leq \mu < c$, where $c < r$ and $\alpha_2(c) < b_1$. Then there is a class $\mathcal{KL}$ function β on $[0, c)$, depending only on α_1, α_2, W_3 and c — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_c\| < c$ satisfies

$$\|x(t) - x_c\| \leq \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Khalil's estimates (4.42) and (4.43) are the two branches of the maximum.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18. Differences:*

- *The sandwich holds on an open ball; Khalil's level $\alpha_1(r)$ becomes b_1, and the initial states form the open c-ball.*
- *$\mu = 0$ is allowed.*
- *$\beta(\rho, s) \leq \alpha_1^{-1}(\alpha_2(\rho))$ is part of the conclusion. The global case is theorem 343.*

Proof. Let $v(t) = V(t, x(t))$, $\eta = \alpha_2(\mu)$ and $d = \alpha_2(c)$. If $v > \eta$ then $\alpha_2(\|x - x_c\|) \geq v > \alpha_2(\mu)$, so $\|x - x_c\| > \mu$ and $\dot{v} \leq -W_3 \leq 0$. The trajectory therefore never reaches the sphere $\alpha_1(\|x - x_c\|) = d$: up to the first time it did, v could only fall while above η, so it would be at most $\max\{v(t_0), \eta\} < d$ (lemma 17), while on the sphere $v \geq d$.

Bound W_3 below by a class $\mathcal{K}$ α_3 (theorem 281). Above η, $\dot{v} \leq -\alpha_3(\|x - x_c\|) \leq -\alpha_3(\alpha_2^{-1}(v))$, so theorem 286 gives $v(t) \leq \max\{\sigma(\alpha_2(\|x(t_0) - x_c\|), t - t_0), \eta\}$. Apply the increasing α_1^{-1} and set $\beta(\rho, s) = \alpha_1^{-1}(\sigma(\alpha_2(\rho), s))$. □

Corollary 341 (Uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 340, the trajectories are uniformly bounded around x_c (definition 299).*

Proof. From the c-ball, start within $a < c$: the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$, as β is increasing in its first argument and decreasing in its second. □

Corollary 342 (Uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 340 with $\mu > 0$, if the system is forward complete from the c-ball about x_c at every $t_0 \ge 0$ (definition 82), its trajectories are uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 301).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, estimate (4.43).*

Proof. The ultimate bound $b = \alpha_1^{-1}(\alpha_2(\mu))$ is positive, and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$. Restrict a trajectory on $[t_0, \infty)$ to $[t_0, t]$ and apply the estimate. □

Theorem 343 (Ultimate boundedness theorem, global case)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 28) and $\mu \ge 0$, with*

$$\alpha_1(\|x - x_c\|) \le V(t,x) \le \alpha_2(\|x - x_c\|), \qquad \dot V(t,x) \le -W_3(x) \quad \text{for } \mu \le \|x - x_c\|$$

for all $t \ge 0$ and $x \in \mathbb{R}^n$. Then there is a global class $\mathcal{KL}$ function β (definition 262), depending only on α_1, α_2 and W_3 — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, satisfies

$$\|x(t) - x_c\| \le \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, global case. Khalil's proof chooses one class $\mathcal{K}$ lower bound for W_3 on all of $\mathbb{R}^n$, which need not exist: W_3 may tend to 0 at infinity.*

Proof. For each $k \in \mathbb{N}$, apply theorem 340 with initial radius $c_k = k + 1$, on a ball large enough that $\alpha_2(c_k) < \alpha_1(R_k)$. If $\mu \ge \|x(t_0) - x_c\|$, the bound of β_k at an index with $c_k > \mu$ is at most $\alpha_1^{-1}(\alpha_2(\mu))$, so the trajectory never leaves the ultimate bound. It gives a class $\mathcal{KL}$ β_k with $\beta_k(\rho, s) \le \alpha_1^{-1}(\alpha_2(\rho))$. The sums $U(r,s) = \sum_{k \le \lceil r \rceil} \beta_k(k,s)$ are positive, decreasing to 0 in s, and increasing in r, and a trajectory from the r-ball that is outside the ultimate bound at time t is within $\beta_{\lceil r \rceil}(r, t - t_0) \le U(r+1, t-t_0)$ and within $\alpha(r) = \alpha_1^{-1}(\alpha_2(r))$ of x_c. lemma 277 turns the class $\mathcal{K}_\infty$ bound α and the family U into a global class $\mathcal{KL}$ β above $\min\{\alpha(r), \sqrt{\alpha(r)\, U(r+1, s)}\}$. □

Corollary 344 (Global uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 343, the trajectories are globally uniformly bounded around x_c (definition 300).*

Proof. Starting within a, the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$. □

Corollary 345 (Global uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 343 with $\mu > 0$, if the system is forward complete from every state at every $t_0 \ge 0$ (definition 82), its trajectories are globally uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 302).*

Proof. The ultimate bound b is positive and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$; apply the estimate on $[t_0, t]$. □

6.8 Input-to-state stability

Input-to-state stability bounds the state by a decaying function of the initial condition plus a gain of the size of the input. An ISS-Lyapunov function certifies it through the global ultimate boundedness theorem, applied once per input: its class $\mathcal{KL}$ function does not depend on the input.

Definition 346 (Input-to-state stability)**.** A continuous-time system with input is *input-to-state stable* about x_{eq} when there are a global class $\mathcal{KL}$ function β (definition 262) and a class $\mathcal{K}_\infty$ function γ (definition 237) such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, under any input u satisfies

$$\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0) + \gamma(M) \qquad \forall\, t \in [t_0, t_1]$$

whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.7. Differences:

- Any bound M on the input replaces its supremum, which is equivalent.
- γ is class $\mathcal{K}_\infty$, as in Sontag, rather than class $\mathcal{K}$; a class $\mathcal{K}$ gain γ is dominated by $\gamma + \text{id}$.
- Solutions are not required to exist for all time.

Theorem 347 (Input-to-state stability implies global uniform asymptotic stability)**.** *If s is input-to-state stable about x_{eq} (definition 346), then x_{eq} is a globally uniformly asymptotically stable equilibrium of the unforced system $\dot{x} = F(t, x, 0)$ (definition 296).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 4.9.*

Proof. With $u \equiv 0$ and $M = 0$, $\gamma(M) = 0$ and the estimate is $\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$. The bound $\beta(r, s) + re^{-s}$ is again global class $\mathcal{KL}$, now also radially unbounded, and lemma 321 concludes. □

Theorem 348 (Input-to-state estimate from an ISS-Lyapunov function)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 28), and $\rho : [0, \infty) \to [0, \infty)$ nondecreasing, with*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x, u) \leq -W_3(x) \quad \text{for } \|x - x_c\| \geq \rho(\|u\|)$$

for all $t \geq 0$, x and u. Then there is a global class $\mathcal{KL}$ function β such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, satisfies

$$\|x(t) - x_c\| \leq \beta(\|x(t_0) - x_c\|, t - t_0) + \alpha_1^{-1}(\alpha_2(\rho(M)))$$

for $t \in [t_0, t_1]$, whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, whose ρ is class $\mathcal{K}$; only its monotonicity is used.*

Proof. Fix the input and t. Freeze the input, clamped to $[t_0, t]$, into the system: $\dot{x} = F(\tau, x, u(\tau))$ is an unforced system with the same trajectories on $[t_0, t]$, and since $\|u(\tau)\| \leq M$ and ρ is nondecreasing, $\dot{V} \leq -W_3$ wherever $\|x - x_c\| \geq \rho(M)$. Apply theorem 343 with $\mu = \rho(M)$; its β does not depend on the system or on μ, so it is the same for every input. Finally $\max\{a, b\} \leq a + b$ for $a, b \geq 0$. □

Theorem 349 (ISS-Lyapunov theorem)**.** *Under the hypotheses of theorem 348 with ρ class $\mathcal{K}_\infty$, the system is input-to-state stable about x_c (definition 346) with gain $\gamma = \alpha_1^{-1} \circ \alpha_2 \circ \rho$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, with ρ of class $\mathcal{K}$; a class $\mathcal{K}$ ρ can be replaced by $\rho + \mathrm{id}$, which only weakens the hypothesis.*

Proof. theorem 348, with $\alpha_1^{-1} \circ \alpha_2 \circ \rho$ class $\mathcal{K}_\infty$ (definition 244, definition 245). □

6.9 LaSalle's invariance principle

When $\nabla V \cdot f \leq 0$ only, V may stall, and Lyapunov's theorems give stability but not convergence. LaSalle's principle recovers convergence for a time-invariant system: a trajectory confined to a compact set converges to its ω-limit set, on which V is constant, and forward invariance of that set turns constancy of V into $\nabla V \cdot f = 0$.

Lemma 350 (The ω-limit set is forward invariant)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, and let φ be a trajectory on $[t_0, \infty)$. Then its ω-limit set $\omega(\varphi)$ (definition 65) is forward invariant (definition 93).*

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.1.*

Proof. Let ψ be a trajectory on $[a, b]$ with $p = \psi(a) \in \omega(\varphi)$, and fix $t \in [a, b]$. Given $\varepsilon > 0$, put $C = e^{K(b-a)}$ and pick, by the definition of the ω-limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon/C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - a)$ solves the same time-invariant equation on $[a, b]$ with $u(a) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r + t - a) - \psi(t)\| = \|u(t) - \psi(t)\| \leq \|\varphi(r) - p\|\, e^{K(t-a)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - a$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity, which presumes a globally defined flow; comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Lemma 351 ($\dot{V}$ vanishes on the ω-limit set)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant, $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω, and φ a trajectory on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$. Then $\nabla V(y) \cdot f(y) = 0$ for every $y \in \omega(\varphi)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.4.*

Proof. $V(\varphi(t))$ is nonincreasing and bounded below on the compact Ω, so it converges to some L, and $V \equiv L$ on $\omega(\varphi)$ (lemma 68). Through $y \in \omega(\varphi)$ take a solution ψ on $[0, 1]$ (Picard–Lindelöf, theorem 47); it stays in $\omega(\varphi)$ by lemma 350, so $V \circ \psi \equiv L$ and its derivative at 0, which is $\nabla V(y) \cdot f(y)$, vanishes. □

Theorem 352 (LaSalle's invariance principle)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant (definition 93), and $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω. Put*

$$E = \{x \in \Omega : \nabla V(x) \cdot f(x) = 0\},$$

and let M contain every forward invariant subset of E. Then every trajectory φ on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$ satisfies $\varphi(t) \to M$ as $t \to \infty$.

Khalil takes M to be the largest invariant *subset of E. Here M must absorb every* forward *invariant subset, a larger family, so the hypothesis on M is correspondingly stronger: backward invariance of $\omega(\varphi)$ would need solutions through its points to extend backwards in time.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.4.*

Proof. The ω-limit set lies in Ω, which is closed and contains the trajectory (lemma 69), and $\nabla V \cdot f$ vanishes on it (lemma 351), so $\omega(\varphi) \subseteq E$. It is forward invariant (lemma 350), so $\omega(\varphi) \subseteq M$. A trajectory confined to a compact set converges to every set containing its ω-limit set (lemma 70). □

Theorem 353 (Barbashin's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on a neighbourhood D of x_{eq} (definition 28) with $\nabla V \cdot f \leq 0$ on D. Suppose that for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D, compact and forward invariant, and that $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x \in \Omega_c : \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is locally asymptotically stable (definition 292).*

The hypothesis on E is the one a user of LaSalle discharges: solve $\nabla V \cdot f = 0$ and check that no solution can stay there but the equilibrium.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.1.*

Proof. Stability: V is a uniform Lyapunov function (lemma 329, theorem 332). Forward completeness: f is Lipschitz (theorem 86). Convergence: near x_{eq}, $V < c$, so a trajectory starting there starts in Ω_c, and theorem 352 with $M = \{x_{\text{eq}}\}$ applies. □

Theorem 354 (Krasovskii's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on $\mathbb{R}^n$, with compact sublevel sets (equivalently, radially unbounded) and $\nabla V \cdot f \leq 0$ everywhere. Suppose that for every c, $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x : V(x) \leq c,\ \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is globally asymptotically stable (definition 293).*

Unlike Barbashin's theorem (theorem 353) no forward invariant set is assumed: every sublevel set is compact, and $\nabla V \cdot f \leq 0$ keeps a trajectory in the one through its initial state.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.2.*

Proof. Stability and forward completeness as in Barbashin's theorem. The sublevel set $\{V \leq V(x(t_0))\}$ is compact and, since V does not increase along trajectories, forward invariant; apply theorem 352 with $M = \{x_{\text{eq}}\}$. □

6.10 Instability

Instability is the failure of stability, and the simplest witness is a trajectory that leaves a fixed neighbourhood from arbitrarily close initial states (lemma 303). Chetaev's theorem produces such trajectories from a function that grows along the system near the equilibrium.

The geometric criterion uses a positive certificate with positive derivative in an open region and vanishing on its boundary, with these conditions local to a closed ball.

Definition 355 (Chetaev function)**.** Let s be an input-free continuous-time system. A Chetaev certificate consists of an open region D, a point $x_{\text{eq}} \in \partial D$, a radius $\rho > 0$, and a globally C^1 function V. For $\|x - x_{\text{eq}}\| \leq \rho$, $V(x) > 0$ in D and $V(x) = 0$ on ∂D; in D the derivative $\nabla V(x) \cdot F(t, x, ())$ is strictly positive for every $t \geq 0$.

Reference: Khalil, *Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.

Theorem 356 (Geometric Chetaev instability criterion)**.** *For an input-free system $\dot{x} = f(x)$ with f globally C^1, a Chetaev certificate at x_{eq} (definition 355) implies instability (definition 289).*

Reference: Khalil, Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.*

Proof. Choose a positive seed near the boundary point and continue a cutoff curve. Boundary vanishing retains it in the region. The Lie derivative has a positive minimum on its compact superlevel, forcing growth beyond the ball's certificate ceiling. □

The exponential criterion is its corollary on the full positive set $D = \{V > 0\}$.

Theorem 357 (Exponential Chetaev instability criterion)**.** *Suppose the vector field f and certificate V are C^1, V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable (definition 289) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*, 3rd ed. (Prentice Hall, 2002), Theorem 4.3, p. 125, for the method; this quantitative variant is not its statement.*

Proof. Take $D = \{V > 0\}$. The quadratic bound gives $V(x_{\text{eq}}) = 0$; the nearby positive seeds put x_{eq} in its closure. Continuity makes $V = 0$ on ∂D, and $\dot{V} \geq 2\alpha V > 0$ in D within the ball. Apply theorem 356. □

Chapter 7

Stability of discrete-time systems

The notions of the previous chapter for an autonomous discrete-time system $x^+ = f(k, x)$ (definition 100), under the same names. Discrete time is simpler in one respect: trajectories always exist and are unique (lemma 115, lemma 116), so every notion quantifies over trajectories on $[k_0, \infty)$ and no forward completeness is needed anywhere. Exponential estimates are geometric, ρ^{k-k_0} with $0 \leq \rho < 1$, in place of $e^{-\gamma(t-t_0)}$.

7.1 Definitions

Definition 358 (Stability of a discrete-time system)**.** The equilibrium x_{eq} of an autonomous discrete-time system $x^+ = f(k, x)$ is *stable* when for every $\varepsilon > 0$ and $k_0 \in \mathbb{N}$ there is $\delta = \delta(\varepsilon, k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(k) - x_{\text{eq}}\| < \varepsilon$ for all $k \geq k_0$.

The counterpart of definition 288. Trajectories on rays suffice: a discrete-time trajectory exists from every state for all later times (lemma 115).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4; Jungers and van de Wouw (2026), Definition 2.

Definition 359 (Instability of a discrete-time system)**.** The equilibrium is *unstable* when it is not stable (definition 358).

Definition 360 (Uniform stability of a discrete-time system)**.** The equilibrium is *uniformly stable* when δ in definition 358 can be chosen independently of k_0.

Definition 361 (Global uniform stability of a discrete-time system)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 360) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$.

Definition 362 (Local asymptotic stability of a discrete-time system)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 358) and for every k_0 there is $c = c(k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $x(k) \to x_{\text{eq}}$. Unlike definition 292 no forward completeness is asked for: discrete-time trajectories always exist.

Definition 363 (Global asymptotic stability of a discrete-time system)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 358) and every trajectory on $[k_0, \infty)$ satisfies $x(k) \to x_{\text{eq}}$.

Definition 364 (Uniform convergence time, discrete time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) \in \mathbb{N}$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $\|x(k) - x_{\text{eq}}\| < \eta$ for $k \geq k_0 + T$.

Definition 365 (Uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 360) and there is $c > 0$ from which convergence has a uniform convergence time (definition 364).

Definition 366 (Global uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 361) and every $c > 0$ admits a uniform convergence time (definition 364).

Definition 367 (Local exponential stability of a discrete-time system)**.** The equilibrium is *locally exponentially stable*, or *geometrically stable*, when there are $c, C > 0$ and $0 \leq \rho < 1$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(k) - x_{\text{eq}}\| \leq C \, \|x(k_0) - x_{\text{eq}}\| \, \rho^{k-k_0} \qquad \forall \, k \geq k_0.$$

The counterpart of definition 297, with $\rho = e^{-\gamma}$.

Reference: Jungers and van de Wouw (2026), Definition 2, estimate (6).

Definition 368 (Global exponential stability of a discrete-time system)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 367 holds for every initial state, with no radius c.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 369 (Uniform boundedness of a discrete-time system)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq \beta \qquad \forall \, k \geq k_0$$

for every trajectory on $[k_0, \infty)$. The counterpart of definition 299.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 370 (Global uniform boundedness of a discrete-time system)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 369 holds for every $a > 0$, with no radius c.

Definition 371 (Uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $T = T(a, b) \in \mathbb{N}$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq b \qquad \forall \, k \geq k_0 + T$$

for every trajectory on $[k_0, \infty)$. The point x_c need not be an equilibrium.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6; Jungers and van de Wouw (2026), Definition 3.

Definition 372 (Global uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when definition 371 holds for every $a > 0$, with no radius c.

7.2 Implications

Lemma 373 (A stable equilibrium is a fixed point)**.** *If x_{eq} is stable (definition 358), then $f(k, x_{\text{eq}}) = x_{\text{eq}}$ for every k.*

Proof. The trajectory from x_{eq} at time k stays within every $\varepsilon > 0$ of x_{eq}, so its value at $k+1$, which is $f(k, x_{\text{eq}})$, is at distance 0. □

Lemma 374 (Instability from a fixed escape radius, discrete time)**.** *Fix $\varepsilon > 0$ and k_0. If for every $\delta > 0$ some trajectory on $[k_0, \infty)$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 359).*

Proof. □

Lemma 375 (Uniform stability implies stability, discrete time)**.** *A uniformly stable equilibrium (definition 360) is stable (definition 358).*

Proof. □

Lemma 376 (Global asymptotic stability implies local, discrete time)**.** *A globally asymptotically stable equilibrium (definition 363) is locally asymptotically stable (definition 362).*

Proof. □

Lemma 377 (Uniform asymptotic stability implies local, discrete time)**.** *A uniformly asymptotically stable equilibrium (definition 365) is locally asymptotically stable (definition 362). Unlike lemma 306, no forward completeness is needed.*

Proof. □

Lemma 378 (Global uniform asymptotic stability implies global, discrete time)**.** *A globally uniformly asymptotically stable equilibrium (definition 366) is globally asymptotically stable (definition 363).*

Proof. □

Lemma 379 (Exponential stability implies uniform stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 367) is uniformly stable (definition 360).*

Proof. The estimate bounds the distance by C times the initial distance, so $\delta = \min(c, \varepsilon/C)$ works for every k_0. □

Lemma 380 (Exponential stability implies uniform asymptotic stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 367) is uniformly asymptotically stable (definition 365).*

Proof. Uniform stability is lemma 379. From the ball of radius c the distance after T steps is below $Cc\rho^T$, which is below any $\eta > 0$ for T large, independently of k_0. □

Lemma 381 (Global exponential stability implies local, discrete time)**.** *A globally exponentially stable equilibrium (definition 368) is locally exponentially stable (definition 367).*

Proof. □

Lemma 382 (Global exponential stability implies global uniform, discrete time)**.** *A globally exponentially stable equilibrium (definition 368) is globally uniformly asymptotically stable (definition 366).*

Proof. $\delta(\varepsilon) = \varepsilon/C$ witnesses global uniform stability, and from every ball the estimate gives a uniform convergence time as in lemma 380. □

7.3 Lyapunov's direct method

For a time-invariant system $x^+ = f(x)$ the decrease of a Lyapunov function along the system is the difference $\Delta V(x) = V(f(x)) - V(x)$. Unlike continuous time, the theorems need continuity of f for a reason of their own: a trajectory can jump over a sphere in one step, so the first-exit argument of continuous time is replaced by keeping one step from a small ball inside the region where V is controlled.

Theorem 383 (Lyapunov's stability theorem, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$ and f continuous at x_{eq}. Let $\mathcal{D}$ be a neighbourhood of x_{eq} and V positive definite on $\mathcal{D}$ about x_{eq} (definition 28), with*

$$\Delta V(x) = V(f(x)) - V(x) \leq 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is uniformly stable (definition 360).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but it cannot be dropped: on $\mathbb{R}$, $V(x) = x^2/(1+x^4)$ and $f(x) = 1/x^2$ for $0 < |x| < 1$ ($f(x) = x$ otherwise) satisfy everything else, yet $x_0 = 10^{-3}$ jumps to $x_1 = 10^6$.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 1.

Proof. Given ε, shrink it so that the closed ball of radius ε' lies in $\mathcal{D}$, and choose $s \leq \varepsilon'$ so that f maps the ball of radius s into the ball of radius ε'. On the compact annulus $s \leq \|x - x_{\mathrm{eq}}\| \leq \varepsilon'$, V has a positive minimum m; choose $\delta < s$ with $V < m$ on the ball of radius δ. By induction, a trajectory with $\|x(k_0) - x_{\mathrm{eq}}\| < \delta$ keeps $\|x(k) - x_{\mathrm{eq}}\| < s$ and $V(x(k)) < m$: the next state is within ε' by the choice of s, has $V(x(k+1)) \leq V(x(k)) < m$, and so cannot lie in the annulus. Nothing depends on k_0, since a trajectory from k_0 is the iterate sequence of $x(k_0)$ (lemma 117). □

Theorem 384 (Lyapunov's theorem for local asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$. Let $\mathcal{D}$ be a neighbourhood of x_{eq} on which f is continuous, and V positive definite on $\mathcal{D}$ about x_{eq} (definition 28), with*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is locally asymptotically stable (definition 362).

Continuity of f on $\mathcal{D}$ is not among the hypotheses of the reference, but it cannot be dropped. With $V(x) = |x|$ on $\mathbb{R}$, let f be odd with $f(0) = 0$ and $f(x) = (2^{-n-1} + x)/2$ on each interval $(2^{-n-1}, 2^{-n}]$: then $\Delta V < 0$ away from 0, but a trajectory starting in $(2^{-n-1}, 2^{-n}]$ converges to 2^{-n-1}, at every scale.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 2.

Proof. Stability is theorem 383. For attractivity, choose a closed ball $K \subseteq \mathcal{D}$ around x_{eq} and, by stability, a δ whose trajectories stay in K. Along such a trajectory $V(x(k))$ is nonincreasing and bounded below, so it converges to some L. A cluster point y of the trajectory, which exists in the compact set K, has $V(y) = L$; by continuity of f at y, $x(k+1) = f(x(k))$ clusters at $f(y)$, so $V(f(y)) = L$ too. Then $\Delta V(y) = 0$, so $y = x_{\text{eq}}$. A sequence in a compact set whose only cluster point is x_{eq} converges to it. □

Theorem 385 (Lyapunov's theorem for global asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with f continuous and $f(x_{\text{eq}}) = x_{\text{eq}}$. Let V be positive definite on E about x_{eq} (definition 28), with compact sublevel sets $\{V \leq c\}$ (equivalently, for continuous V, radially unbounded: lemma 334), and*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \neq x_{\text{eq}}.$$

Then x_{eq} is globally asymptotically stable (definition 363).

Continuity of f is not among the hypotheses of the reference but cannot be dropped: with $V(x) = |x|$ on $\mathbb{R}$, the map $f(x) = x/2$ for $|x| \leq 1$, $f(x) = \operatorname{sign}(x)(1+|x|)/2$ otherwise, has $\Delta V < 0$ away from 0, yet the trajectory from 2 converges to 1.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 3.

Proof. Stability is theorem 383 with $\mathcal{D} = E$. From any initial state, $V(x(k)) \leq V(x(k_0))$, so the trajectory stays in the compact sublevel set $\{V \leq V(x(k_0))\}$, and the ω-limit argument of theorem 384 applies there. □

Theorem 386 (Lyapunov's theorem for local exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$ with $f(x_{\text{eq}}) = x_{\text{eq}}$ and f continuous at x_{eq}, and let $\mathcal{D}$ be a neighbourhood of x_{eq}. Let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every $x \in \mathcal{D}$,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) \leq -c_3 V(x).$$

Then x_{eq} is locally exponentially stable (definition 367).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but without it a trajectory can leave $\mathcal{D}$ in a single step.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4.

Proof. Choose $r_0 > 0$ so that the ball $B(x_{\text{eq}}, r_0)$ lies in $\mathcal{D}$ and f maps it into $\mathcal{D}$, and let $C = (c_2/c_1)^{1/p} \geq 1$ and $\delta = r_0/(2C)$. By induction, a trajectory starting within δ stays in $B(x_{\text{eq}}, r_0)$ and satisfies $V(x(k)) \leq (1-c_3)^{k-k_0} V(x(k_0))$: the decrease applies at $x(k) \in \mathcal{D}$, and $x(k+1) = f(x(k)) \in \mathcal{D}$, so the sandwich gives $\|x(k+1) - x_{\text{eq}}\| \leq C\delta = r_0/2$. The geometric bound with $\rho = (1-c_3)^{1/p}$ then follows as in theorem 387. □

Theorem 387 (Lyapunov's theorem for global exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$, let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every x,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) = V(f(x)) - V(x) \leq -c_3 V(x).$$

Then x_{eq} is globally exponentially stable (definition 368), with $C = (c_2/c_1)^{1/p}$ and $\rho = (1-c_3)^{1/p}$. No continuity of f is needed.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4, on the whole space. They ask for $p \geq 1$; $p > 0$ suffices.

Proof. The decrease condition is $V(f(x)) \leq (1-c_3)V(x)$, so by induction $V(x(k)) \leq (1-c_3)^{k-k_0}V(x(k_0))$. Sandwiching,

$$c_1\|x(k) - x_{\mathrm{eq}}\|^p \leq (1-c_3)^{k-k_0}c_2\|x(k_0) - x_{\mathrm{eq}}\|^p,$$

and taking p-th roots gives the geometric bound. □

Chapter 8

Stability of linear systems

The stability notions of a linear system are those of definition 288 and the definitions after it, applied to the unforced system (definition 80): there is no separate notion of stability for linear systems. What linearity adds is that each notion becomes a condition on the state transition matrix, and for constant A a condition on its eigenvalues.

8.1 Linear time-varying systems

Hespanha states Lyapunov stability for the continuous-time linear time-varying system $\dot{x} = A(t)x$ as four conditions on its homogeneous state response $x(t) = \Phi(t, t_0)x_0$. None of them is taken as a definition here: each is proved *equivalent* to the corresponding stability predicate of the unforced system, so the linear track and the nonlinear track share one notion of stability rather than two.

Two things stand between the textbook conditions and the predicates, and both are settled in this section. The predicates quantify over trajectories where the textbook quantifies over initial conditions, which the trajectory characterization below closes in both directions; and the predicates bound the response from a *ball* where the textbook bounds it from every initial state, which homogeneity closes. The second is why a linear system has no nontrivial region of attraction, recorded as lemma 394.

8.1.1 Hespanha's Definition 8.1

Lemma 388 (Stability as a bound on the state transition matrix)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is a stable equilibrium (definition 288) of this system if and only if for every $t_0 \geq 0$ there is C with

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. ($\Rightarrow$) Apply stability at $\varepsilon = 1$ to get $\delta(1, t_0)$. Every response from the ball of radius δ stays inside the unit ball, and scaling turns that into the bound with $C = 2/\delta$.

($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$. A trajectory is its own response (theorem 191), so $\|\varphi(t)\| \leq C\|\varphi(t_0)\| < \varepsilon$. □

Theorem 389 (Lyapunov stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is a stable equilibrium (definition 288) of this system if and only if, for every $t_0 \geq 0$ and every initial condition x_0, the homogeneous state response $x(t) = \Phi(t, t_0)\, x_0$ is bounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. The state-transition form is lemma 388. A bound on Φ gives the response bound $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 390 (Uniform stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is a uniformly stable equilibrium (definition 290) of this system if and only if there is a single C with

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1), with the margin independent of t_0.*

Proof. Identical to lemma 388, with the margin δ and the bound C both carried outside the quantifier on t_0. □

Theorem 391 (Asymptotic stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is an asymptotically stable equilibrium (definition 292) of this system if and only if, for every $t_0 \geq 0$ and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $[t_0, \infty)$ and tends to 0 as $t \to \infty$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(2).*

Proof. The stability conjunct is theorem 389. For attractivity: an arbitrary x_0 is rescaled into the attractivity ball, and the limit scales back, since the response is linear in the initial state; conversely every trajectory is a response (theorem 191), so convergence of the responses is convergence of the trajectories, from any radius. □

Theorem 392 (Global exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is a globally exponentially stable equilibrium (definition 298) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. Both directions are the trajectory characterization plus homogeneity: the exponential estimate along every trajectory is the estimate applied to the initial state, and conversely the estimate on any ball scales to the estimate everywhere. □

Theorem 393 (Exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is an exponentially stable equilibrium (definition 297) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

The condition is the one in theorem 392: the radius in the local definition carries no information here.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. As in the global case; the radius c of the local definition is absorbed by the scaling argument, and $c = 1$ serves in the converse. □

Lemma 394 (Local exponential stability is global, for a linear system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is an exponentially stable equilibrium (definition 297) of this system if and only if it is globally exponentially stable (definition 298).

Proof. Both are equivalent to the same estimate, theorem 393 and theorem 392. □

For a linear system, uniform asymptotic stability already forces exponential decay of the state transition matrix: splitting time into windows of a uniform convergence time, each window contracts by a fixed factor.

Theorem 395 (Uniform asymptotic stability of a linear time-varying system is exponential)**.** *Consider the linear time-varying system $\dot{x} = A(t)\, x$ with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154). The origin is uniformly asymptotically stable (definition 295) if and only if there are $k, \gamma > 0$ with*

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. ($\Leftarrow$) The estimate is exponential stability (theorem 393), which is uniform asymptotic stability. ($\Rightarrow$) Uniform stability bounds $\|\Phi(t,t_0)\| \le C$ (theorem 390), and a uniform convergence time T from the c-ball to the $c/(2e)$-ball gives $\|\Phi(t,t_0)\| \le 1/e$ for $t \ge t_0 + T$, by homogeneity. Splitting $[t_0, t]$ into $N = \lfloor (t-t_0)/T \rfloor$ windows of length T and using $\Phi(t,t_0) = \Phi(t,t_0+T)\,\Phi(t_0+T,t_0)$ gives $\|\Phi(t,t_0)\| \le \max(C,1)\,e^{-N} \le e\max(C,1)\,e^{-(t-t_0)/T}$. □

Corollary 396 (Global uniform asymptotic stability of a linear time-varying system)**.** *Under the hypotheses of theorem 395, the origin is globally uniformly asymptotically stable (definition 296) if and only if $\Phi(t,t_0)$ decays exponentially, uniformly in $t_0 \ge 0$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. Global uniform asymptotic stability is uniform asymptotic stability, hence exponential decay (theorem 395); the decay is global exponential stability (theorem 392), which is global uniform asymptotic stability. □

The time-varying form of the Lyapunov equation.

Definition 397 (Differential Lyapunov equation)**.** For matrix-valued functions A, P and Q of time, P satisfies the *differential Lyapunov equation* for A with forcing Q when, for every $t \ge 0$, P is differentiable at t and

$$-\dot{P}(t) = P(t)A(t) + A(t)^{\mathsf{T}}P(t) + Q(t).$$

For constant matrices it is the continuous Lyapunov equation (definition 420).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), equation (4.28).

Lemma 398 (Constant solutions of the differential Lyapunov equation)**.** *For constant A, P and Q, the differential Lyapunov equation (definition 397) holds if and only if the continuous Lyapunov equation (definition 420) does.*

Proof. The derivative of a constant is 0, and derivatives are unique, so the differential equation says $0 = -(PA + A^{\mathsf{T}}P + Q)$. □

Theorem 399 (Converse Lyapunov theorem for linear time-varying systems)**.** *Consider $\dot{x} = A(t)x$ with A continuous and $\|A(t)\| \le L$, and let the origin be exponentially stable (definition 297). Let $Q(t)$ be continuous, symmetric, with $\|Q(t)\| \le q$ and $x^{\mathsf{T}}Q(t)x \ge q_1\|x\|^2$, $q_1 > 0$. Then there is a bounded symmetric $P(t)$ satisfying the differential Lyapunov equation (definition 397), and $V(t,x) = x^{\mathsf{T}}P(t)x$ is an exponential Lyapunov function with exponent 2 (definition 328); in particular $c_1\|x\|^2 \le x^{\mathsf{T}}P(t)x \le c_2\|x\|^2$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.12. Differences: the state space carries the sup norm, as everywhere for linear time-varying systems here.*

Proof. Let $P(t) = \int_t^\infty \Phi(\tau,t)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)\,d\tau$, written as $\Phi(0,t)^{\mathsf{T}}R(t)\Phi(0,t)$ with $R(t) = \int_t^\infty \Phi(\tau,0)^{\mathsf{T}}Q(\tau)\Phi(\tau,0)\,d\tau$, which converges by exponential decay of Φ; the same decay bounds $P(t)$. Then $\dot{R}(t) = -\Phi(t,0)^{\mathsf{T}}Q(t)\Phi(t,0)$ and $\partial_t\Phi(0,t) = -\Phi(0,t)A(t)$ (theorem 179), and the product rule gives $-\dot{P} = PA + A^{\mathsf{T}}P + Q$. Along a solution, $x^{\mathsf{T}}P(t)x = \int_t^\infty (\Phi(\tau,t)x)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)x\,d\tau$: exponential decay bounds it above, the growth bound (lemma 176) bounds it below on $[t, t+1]$, and its derivative along the solution is $-x^{\mathsf{T}}Q(t)x$. □

Theorem 400 (Instability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 154).

The origin is unstable (definition 289) for this system if and only if there are $t_0 \geq 0$ and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(4).*

Proof. The negation of theorem 389. □

8.2 Linear time-invariant systems

For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 197), and the stability conditions of the previous section collapse to a condition on the eigenvalues of A alone. This is Hespanha's Theorem 8.1.

Clauses (1) and (4) of that theorem — marginal stability and instability — ask that the Jordan blocks at imaginary-axis eigenvalues be 1×1. They are not recorded here: they need a Jordan normal form, which is a project rather than a lemma.

8.2.1 The Hurwitz condition and its rate-indexed refinement

For a real square matrix A, Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification, over an abstract index type. The rate-indexed predicate records a strict spectral margin. Ordinary Hurwitz stability is the zero-rate case, but is stated separately rather than derived, so that it unfolds directly to $\mathrm{Re}(\mu) < 0$.

Definition 401 (Hurwitz matrix)**.** A real square matrix A is *Hurwitz* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < 0$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Definition 402 (Hurwitz with a prescribed decay rate)**.** A real square matrix A is *Hurwitz with decay rate α* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < -\alpha$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Theorem 403 (Rate-Hurwitz via a spectral shift)**.** *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Reference: João P. Hespanha, Linear Systems Theory *(2nd ed.), continuous-time stability criterion. This spectral-shift corollary is proved directly from eigenpairs.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\mathrm{Re}(\mu) < -\alpha$ is equivalent to $\mathrm{Re}(\mu + \alpha) < 0$. □

Theorem 404 (A Hurwitz matrix has a positive spectral margin)**.** *If A is Hurwitz then there is $c > 0$ such that every eigenvalue μ of A satisfies $\mathrm{Re}(\mu) < -c$.*

Proof. The spectrum is compact and nonempty, so Re attains a maximum on it at some μ_0, which is an eigenvalue and hence has $\mathrm{Re}(\mu_0) < 0$. Take $c = -\mathrm{Re}(\mu_0)/2$. □

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Lemma 405 (A contractive power from a spectral radius below one)**.** *Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.*

Reference: Rudin, Functional Analysis *(Gelfand's spectral-radius formula).*

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \to r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. □

The two predicates are related by a spectral shift (theorem 403), and they agree at rate zero.

Lemma 406 (Hurwitz is rate zero)**.** *A real square matrix A is Hurwitz (definition 401) if and only if it is Hurwitz with decay rate 0 (definition 402).*

Proof. Both unfold to a condition on $\operatorname{Re}\mu$, the first to $\operatorname{Re}\mu < 0$ and the second to $\operatorname{Re}\mu < -0$. □

Lemma 407 (A certified decay rate may be weakened)**.** *If A is Hurwitz with decay rate α (definition 402) and $\beta \leq \alpha$, then A is Hurwitz with decay rate β.*

Proof. Each eigenvalue satisfies $\operatorname{Re}\mu < -\alpha \leq -\beta$. □

Lemma 408 (Zero-dimensional matrices are vacuously Hurwitz)**.** *If the index type is empty then every A is Hurwitz with every decay rate α (definition 402).*

This records the convention chosen for the degenerate case: the predicate quantifies over nonzero eigenvectors, and in dimension zero there are none.

Proof. There is no nonzero vector, so the hypothesis of the predicate is never met. □

Lemma 409 (The scalar example)**.** *For $\alpha < \gamma$, the 1×1 matrix $(-\gamma)$ is Hurwitz with decay rate α (definition 402).*

Included as a concrete check on the sign convention of the definition.

Proof. The only eigenvalue is $-\gamma$, and $\operatorname{Re}(-\gamma) = -\gamma < -\alpha$. □

The Hurwitz condition bounds each eigenvalue away from the imaginary axis one at a time; because the spectrum is compact, that upgrades to a uniform margin (theorem 404). This is what turns a qualitative hypothesis into a quantitative rate, and it is used twice below.

8.2.2 Equivalent analytic conditions

Nothing in this subsection mentions a system: the Hurwitz condition, decay of e^{At}, and an explicit exponential envelope for $\|e^{At}\|$ are shown to be the same condition on the matrix. Two generic facts about exponential envelopes are recorded first.

Lemma 410 (The decay envelope vanishes)**.** *For $\gamma > 0$ and any constants K, t_0,*

$$\lim_{t \to \infty} K\, e^{-\gamma(t - t_0)} = 0.$$

Proof. The affine map $t \mapsto -\gamma(t - t_0)$ tends to $-\infty$ because $-\gamma < 0$; compose with exp and scale by K. □

Lemma 411 (Squeeze against an exponentially decaying envelope)**.** *Let E be a normed space, $f : \mathbb{R} \to E$, and $\gamma > 0$. If eventually*

$$\|f(t)\| \le k\, e^{-\gamma(t-t_0)}\, c,$$

then $f(t) \to 0$ as $t \to \infty$.

Stated over an arbitrary normed space because it is applied to both matrix- and vector-valued quantities.

Proof. The squeeze theorem: $\|f\|$ is bounded below by 0 and above by an envelope that tends to 0 (lemma 410, after reassociating $k \cdot E \cdot c = (kc) \cdot E$). □

Theorem 412 (A Hurwitz matrix has an exponentially decaying exponential)**.** *If A is Hurwitz (definition 401) then there are $k > 0$ and $\gamma > 0$ with*

$$\|e^{At}\| \le k\, e^{-\gamma t}, \qquad \forall\, t \ge 0.$$

Proof. A contracting block. The spectral radius of $e^{A_{\mathbb{C}}}$ is below 1 (lemma 212), so by Gelfand's formula (lemma 405) some integer time $p > 0$ has $\|e^{Ap}\| < 1$. Fix c with $\|e^{Ap}\| \le c < 1$, kept away from 0 so that $\log c$ exists.

One block. $\|e^{Ar}\|$ is continuous, hence bounded by some $M \ge 1$ on the compact interval $[0, p]$.

Spreading. Write $t = qp + r$ with $q \in \mathbb{N}$ and $r \in [0, p)$. The exponents commute, so $e^{At} = e^{Ar}(e^{Ap})^q$, and submultiplicativity gives $\|e^{At}\| \le Mc^q$ by induction on q. Setting $\gamma = -\log c/p$ turns c^q into $e^{-\gamma qp}$, and $qp \ge t - p$ costs one further factor c^{-1}. The bound is $\|e^{At}\| \le (M/c)\, e^{-\gamma t}$.

The contraction is taken in the L^∞ operator norm, which is what makes it usable against $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$ in theorem 413.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1. □

Theorem 413 (Exponential decay of the state under a Hurwitz matrix)**.** *If A is Hurwitz (definition 401) then there are $k > 0$ and $\gamma > 0$ with*

$$\|e^{At}x\| \le k\, e^{-\gamma t}\, \|x\|, \qquad \forall\, t \ge 0,\ \forall\, x.$$

This is the form the Lyapunov stability predicates consume.

Proof. theorem 412 bounds the matrix; one application of $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$ transfers it to the state. This is the step at which the choice of the L^∞ operator norm earns its keep. □

Theorem 414 (Decay of the matrix exponential forces the Hurwitz condition)**.** *If $e^{At} \to 0$ as $t \to \infty$ then A is Hurwitz (definition 401).*

Stated from mere convergence rather than from an exponential bound, so that it serves clause (2) and clause (3) of Theorem 8.1 alike.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1.*

Proof. Suppose $\operatorname{Re}\mu \ge 0$ for some eigenpair (μ, v) with $v \ne 0$. Complexification is continuous, so $e_{\mathbb{C}}^{At}v \to 0$ as well. But v is an eigenvector of $tA_{\mathbb{C}}$ with eigenvalue $t\mu$, hence of its exponential (lemma 207), so

$$\|e_{\mathbb{C}}^{At}v\| = |e^{t\mu}|\, \|v\| = e^{t \operatorname{Re}\mu}\|v\| \ge \|v\| > 0$$

for every $t \ge 0$, contradicting convergence to 0. □

Theorem 415 (Pointwise decay of the state forces the Hurwitz condition)**.** *If $e^{At}x \to 0$ as $t \to \infty$ for every real state x, then A is Hurwitz (definition 401).*

This is the composite that both clauses of Theorem 8.1 reach for: the stability predicates speak about states, not about the matrix.

Proof. Feeding the hypothesis the standard basis vectors returns the columns of e^{At}, and convergence in the matrix space is entrywise, so $e^{At} \to 0$; apply theorem 414. □

Theorem 416 (Hurwitz is decay of the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 401) if and only if $e^{At} \to 0$ as $t \to \infty$.*

Proof. Forwards, theorem 412 supplies an exponentially decaying envelope and lemma 411 squeezes; backwards is theorem 414. □

Theorem 417 (Hurwitz is an exponential envelope for the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 401) if and only if there are $k > 0$ and $\gamma > 0$ with $\|e^{At}\| \le k\, e^{-\gamma t}$ for all $t \ge 0$.*

Recorded separately from theorem 416 because the two are what clauses (2) and (3) of Theorem 8.1 respectively need: bare convergence for asymptotic stability, an explicit rate for exponential stability.

Proof. Forwards is theorem 412. Backwards, the envelope forces $e^{At} \to 0$ (lemma 411), so theorem 416 applies. □

8.2.3 Hespanha's Theorem 8.1

Theorem 418 (Asymptotic stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an asymptotically stable equilibrium (definition 292) if and only if A is Hurwitz (definition 401), that is, every eigenvalue of A has strictly negative real part.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(2).*

Proof. For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 197), so theorem 391 turns the statement into one about $e^{At}x_0$.

If the response converges then A is Hurwitz (theorem 415). Conversely a Hurwitz A gives an exponentially decaying envelope for $\|e^{At}x_0\|$ (theorem 413); the envelope is bounded by $k\|x_0\|$, which supplies the stability half, and tends to zero (lemma 410), which supplies the attractivity half. □

Theorem 419 (Exponential stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an exponentially stable equilibrium (definition 297) if and only if A is Hurwitz (definition 401).

Together with theorem 418 this says the two notions coincide for a time-invariant system — a genuine feature of constant A, not of linearity: for time-varying $A(t)$ they differ, as $\dot{x} = -x/(1+t)$ shows.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(3).*

Proof. As for theorem 418, but against theorem 393, which already asks for an exponential envelope. The forward direction squeezes $\|e^{At}x\|$ to zero under that envelope and applies theorem 415; the converse is theorem 413 re-anchored from 0 to t_0. □

8.2.4 The Lyapunov equation

Definition 420 (Continuous Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *continuous Lyapunov equation* for A with forcing Q when

$$PA + A^{\mathsf{T}}P = -Q.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.2.

Definition 421 (Matrix quadratic form)**.** A real matrix P represents the quadratic form $x \mapsto x^{\mathsf{T}}Px$ on Euclidean state space.

Reference: standard quadratic Lyapunov-function construction.

Hespanha's Theorem 8.2 ties the spectral condition to solvability of $PA + A^{\mathsf{T}}P = -Q$. The direction a Lyapunov certificate is *used* in comes first: exhibiting a single positive-definite P proves the spectrum lies in the open left half-plane.

Theorem 422 (A Lyapunov certificate proves the spectrum lies in the open left half-plane)**.** *If P and Q are positive definite and $PM + M^{\mathsf{T}}P = -Q$, then every eigenvalue of M has strictly negative real part.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Complexify the equation and pair it against an eigenvector v of M with eigenvalue μ. The two terms contribute $\mu\, v^*Pv$ and $\bar{\mu}\, v^*Pv$, so the left-hand side is $2\,\mathrm{Re}(\mu)\, v^*Pv$ while the right-hand side is $-v^*Qv$. Both v^*Pv and v^*Qv are strictly positive (lemma 204), so $\mathrm{Re}(\mu) < 0$. □

Theorem 423 (The Lyapunov inequality certifies the Hurwitz condition)**.** *If P is positive definite and $-(PM + M^{\mathsf{T}}P)$ is positive definite, then M is Hurwitz (definition 401).*

This is Hespanha's Theorem 8.2, (5) ⇒ (3). It is stated through positive definiteness of the negated residual rather than through a matrix order relation, so that it needs no Loewner order.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Take the forcing to be minus the residual and apply theorem 422. □

The converse is a construction. One finite block of the integral $\int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$ is summed into a convergent discrete Lyapunov series, with convergence and uniqueness both coming from the contraction $\|e^{Am}\| < 1$. A Hurwitz matrix supplies that contraction, which is theorem 426.

Lemma 424 (A convergent sum of positive semidefinite matrices is positive semidefinite)**.** *If (M_i) is a summable family of real positive semidefinite matrices, then $\sum_i M_i$ is positive semidefinite.*

Proof. Transposition and the quadratic form $P \mapsto x^{\mathsf{T}}Px$ are continuous and linear, so they commute with the sum: the sum is symmetric, and its quadratic form is a sum of nonnegative terms. □

Theorem 425 (The Lyapunov equation from a contracting exponential)**.** *Suppose $\|e^{Am}\| < 1$ for some integer $m > 0$. Then for every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 420), and P is the unique matrix solution.*

Reference: Khalil, Nonlinear Systems.

Proof. Write $L(P) = PA + A^{\mathsf{T}}P$ for the Lyapunov operator, $C = e^{Am}$, and $T(W) = C^{\mathsf{T}}WC$ for the congruence (Stein) operator.

One block. Let $R = \int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$. Differentiating the integrand two ways and equating the derivatives by uniqueness gives the key identity $L(R) = T(Q) - Q$. R is positive definite because the integrand is, pointwise, and positive definiteness survives the integral.

Summing the blocks. $\|C\| < 1$ makes $\|T\| < 1$, so $\sum_k T^k R$ converges; call it P. It is positive definite, being a convergent sum of positive definite terms. Since A and C commute, L commutes with T, so $L(T^k R) = T^{k+1}Q - T^k Q$ and the series for $L(P)$ telescopes to $-Q$.

Uniqueness. A contraction has only the trivial fixed point, so L is injective; two solutions with the same forcing therefore coincide.

The contraction is taken in the Frobenius norm, the only place in the construction where $\|C^{\mathsf{T}}\| = \|C\|$ is needed. □

Theorem 426 (Lyapunov equation for a Hurwitz matrix)**.** *If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^{\mathsf{T}}P = -Q$, and this P is the unique matrix solution.*

omit [DecidableEq X] in Reference: Khalil, Nonlinear Systems*.*

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P; contraction proves convergence and uniqueness. □

Lemma 427 (From every forcing to one certificate)**.** *If every positive definite Q admits a positive definite solution of $PM + M^{\mathsf{T}}P = -Q$ (definition 420), then some positive definite P makes $-(PM + M^{\mathsf{T}}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (4) ⇒ (5)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Instantiate the forcing at $Q = I$; the residual is then $-I$, whose negation is the identity. The edge carries no mathematics beyond the instantiation. □

8.2.5 The rate form

The Lyapunov inequality $PA + A^{\mathsf{T}}P \preceq -2cP$ carries an explicit decay rate: along a trajectory it gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$. It is not one of Hespanha's clauses, but it is the shape every contraction and guaranteed-decay-rate condition is written in, so it is recorded as a sixth equivalent.

Theorem 428 (The rate form certifies the Hurwitz condition)**.** *If P is positive definite, $c > 0$, and $PA + A^{\mathsf{T}}P \preceq -2cP$ — that is, $-(PA + A^{\mathsf{T}}P) - 2cP$ is positive semidefinite — then A is Hurwitz (definition 401).*

Along a trajectory this inequality gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$, so it certifies not merely stability but a decay rate.

Proof. Split $-(PA + A^{\mathsf{T}}P)$ as $(-(PA + A^{\mathsf{T}}P) - 2cP) + 2cP$: positive semidefinite plus positive definite is positive definite, so theorem 423 applies. □

Theorem 429 (A Hurwitz matrix admits a rate certificate)**.** *If A is Hurwitz (definition 401) then there are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Proof. Solve the Lyapunov equation for the *shifted* matrix. A Hurwitz A has a positive spectral margin c (theorem 404), so $A+cI$ is still Hurwitz (theorem 403). Solving $P(A+cI)+(A+cI)^{\mathsf{T}}P = -I$ (theorem 426) and expanding gives $PA+A^{\mathsf{T}}P = -I-2cP$, so the residual $-(PA+A^{\mathsf{T}}P)-2cP$ is *exactly* the identity.

No domination bound between two positive definite matrices is needed: the residual is not merely bounded below, it is I. □

8.2.6 Theorem 8.2

Theorem 430 (Hurwitz is solvability of the Lyapunov equation for every forcing)**.** *A real square matrix A is Hurwitz (definition 401) if and only if for every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 420), unique among all matrix solutions.*

This is Hespanha's Theorem 8.2, (3) ⇔ (4)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards is theorem 426. Backwards, instantiate the forcing at $Q = I$ and apply theorem 422. □

Theorem 431 (Hurwitz is the existence of one Lyapunov certificate)**.** *A real square matrix A is Hurwitz (definition 401) if and only if some positive definite P makes $-(PA + A^{\mathsf{T}}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (3) ⇔ (5)*, and the form in which a Lyapunov certificate is actually supplied: one matrix, not a family.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards, theorem 426 produces a solution for each forcing and lemma 427 extracts one certificate from them; backwards is theorem 423. □

Theorem 432 (Hurwitz is the existence of a rate certificate)**.** *A real square matrix A is Hurwitz (definition 401) if and only if there are a positive definite P and a rate $c > 0$ with $PA+A^{\mathsf{T}}P \preceq -2cP$.*

This is clause 6 of theorem 433, which is not in Hespanha's list. It is the rate-carrying refinement of theorem 431, and the shape every contraction and guaranteed-decay-rate condition is written in.

Proof. theorem 429 and theorem 428. □

Theorem 433 (Hespanha's Theorem 8.2)**.** *For the time-invariant system $\dot{x} = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 292).*
2. *The origin is exponentially stable (definition 297).*
3. *A is Hurwitz (definition 401).*
4. *For every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 420), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(PA + A^{\mathsf{T}}P)$ positive definite.*
6. *There are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Clause 6 is not in Hespanha's list. It is the rate-carrying form of clause 5: along a trajectory it gives $\dot{V} \leq -2cV$ for $V = x^\mathsf{T} P x$, hence $V(t) \leq V(0)e^{-2ct}$, and it is the shape every contraction and guaranteed-decay-rate condition is written in.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Every clause is tied to clause 3 and the cycle is closed by transitivity: $1 \Leftrightarrow 3$ is theorem 418, $2 \Leftrightarrow 3$ is theorem 419, $3 \Leftrightarrow 4$ is theorem 430, $3 \Leftrightarrow 5$ is theorem 431, and $3 \Leftrightarrow 6$ is theorem 432. □

8.2.7 Quadratic forms along a linear vector field

What makes a solution of the Lyapunov equation a Lyapunov *function*: the forcing matrix is exactly the dissipation rate of the associated quadratic form.

Theorem 434 (A Lyapunov solution makes its quadratic form decrease)**.** *Let P solve $PA + A^\mathsf{T} P = -Q$ (definition 420). Then the derivative of the centered quadratic form $x \mapsto (x - x_{\mathrm{eq}})^\mathsf{T} P (x - x_{\mathrm{eq}})$ (definition 421) along the linear vector field $y \mapsto Ay$ is*

$$-\,(x - x_{\mathrm{eq}})^\mathsf{T} Q\,(x - x_{\mathrm{eq}}).$$

This is what makes a Lyapunov solution a Lyapunov function*: the forcing Q is exactly the dissipation rate.*

Proof. The derivative of the quadratic form in the direction Ay is $\langle y, Py' \rangle + \langle y', Py \rangle$ with $y' = Ay$, which is the form represented by $PA + A^\mathsf{T} P$. Transporting the equation through `toEuclideanCLM` — under which transpose becomes adjoint — rewrites that as $-Q$. □

Lemma 435 (Identity forcing gives decrease at the squared norm)**.** *For P solving $PA + A^\mathsf{T} P = -I$ (definition 420), the derivative of the centered quadratic form (definition 421) along $y \mapsto Ay$ is $-\|x - x_{\mathrm{eq}}\|^2$.*

Proof. theorem 434 at $Q = I$, where the quadratic form is the squared Euclidean norm. □

8.2.8 A quadratic certificate for an unstable eigenmode

The instability counterpart (theorem 442): an eigenvalue with positive real part yields a real quadratic Chetaev certificate. The construction solves a shifted Lyapunov–Sylvester equation, whose injectivity rests on a nonresonance condition on the spectrum — which is what the lemmas below supply.

Lemma 436 (A nonresonant bilinear form vanishes on a pair of generalized eigenspaces)**.** *Let T be an endomorphism of a complex vector space and B a bilinear form satisfying the Sylvester-type identity*

$$B(x, Ty) + B(Tx, y) = c\, B(x, y) \qquad \forall\, x, y.$$

If $\xi + \nu \neq c$, then B vanishes on $\ker(T - \xi)^k \times \ker(T - \nu)^l$ for all k, l.

Proof. Strong induction on $k + l$. The base cases are the zero generalized eigenspaces, where the vector is 0. Otherwise write $Tx = N_x + \xi x$ and $Ty = N_y + \nu y$ with N_x, N_y one order lower; the induction hypothesis kills $B(N_x, y)$ and $B(x, N_y)$, leaving $(\xi + \nu - c)B(x, y) = 0$, and the coefficient is nonzero by nonresonance. □

Lemma 437 (A nonzero generalized eigenvector witnesses its eigenvalue)**.** *If $x \neq 0$ lies in the maximal generalized eigenspace of T at ξ, then ξ is an eigenvalue of T.*

Proof. Membership gives $x \in \ker(T - \xi)^k$ for some k, necessarily $k \neq 0$ since $x \neq 0$; so that generalized eigenspace is nontrivial, and a nontrivial generalized eigenspace forces an eigenvalue. □

Lemma 438 (A nonresonant bilinear form vanishes identically)**.** *Let V be a finite-dimensional complex vector space, T an endomorphism, and B a bilinear form with $B(x, Ty) + B(Tx, y) = c\,B(x, y)$. If no pair of eigenvalues of T sums to c, then $B = 0$.*

This is the uniqueness half of solvability for the Sylvester-type equation: the associated operator has trivial kernel exactly when the spectrum is nonresonant.

Proof. Over $\mathbb{C}$ the generalized eigenspaces of T span V, so it suffices to check $B(x, y) = 0$ for x, y in generalized eigenspaces. A nonzero such vector witnesses its eigenvalue (lemma 437), so nonresonance applies and lemma 436 gives $B(x, y) = 0$. □

Lemma 439 (Right multiplication acts on the right argument)**.** *For square matrices H, A over a commutative semiring, $(HA)(x, y) = H(x, Ay)$ as bilinear forms.*

Proof. Both sides unfold to $x \cdot (HAy)$ by associativity of matrix-vector multiplication. □

Lemma 440 (Left multiplication by the transpose acts on the left argument)**.** *For square matrices H, A over a commutative semiring, $(A^{\mathsf{T}}H)(x, y) = H(Ax, y)$ as bilinear forms.*

Together with lemma 439 this is what turns the matrix equation $HA + A^{\mathsf{T}}H = cH$ into the Sylvester-type identity on bilinear forms that lemma 438 consumes.

Proof. Move the transpose across the dot product: $x \cdot (A^{\mathsf{T}}Hy) = (Ax) \cdot (Hy)$. □

Lemma 441 (A symmetric matrix represents a symmetric form)**.** *If H is a symmetric real matrix then $H(x, y) = H(y, x)$ as a bilinear form.*

Proof. Move the transpose across the dot product and apply $H^{\mathsf{T}} = H$. □

Theorem 442 (Quadratic instability certificate from an unstable eigenpair)**.** *A complex eigenpair of a real matrix A with $\operatorname{Re}\mu > 0$ yields a Hermitian real matrix H, a shift $\alpha > 0$, and a direction w such that $w^{\mathsf{T}}Hw > 0$ and*

$$HA + A^{\mathsf{T}}H - 2\alpha H$$

is positive definite.

Reference: Hahn, Stability of Motion*; Khalil,* Nonlinear Systems*.*

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. □

8.3 Discrete-time linear systems

For $x(t+1) = A(t)x(t)$ Hespanha's conditions are again theorems about the stability predicates of the unforced system, now those of definition 358 and the definitions after it. The state transition matrix is a finite product, so no regularity of A is needed, and trajectories always exist, so no forward completeness enters. For constant A the spectral condition is the *Schur* condition: every eigenvalue in the open unit disc.

8.3.1 Hespanha's Definition 8.2

Lemma 443 (Discrete-time stability as a bound on the state transition matrix)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$ with state transition matrix Φ (definition 215), the origin is stable (definition 358) if and only if for every t_0 there is $C \geq 0$ with*

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. ($\Rightarrow$) Stability at $\varepsilon = 1$ gives a margin δ; every response from the δ-ball stays in the unit ball, and homogeneity scales that to the bound with $C = 2/\delta$. ($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$: a trajectory is its own homogeneous response (theorem 235). □

Theorem 444 (Lyapunov stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is stable (definition 358) if and only if, for every t_0 and every initial condition x_0, the homogeneous response $x(t) = \Phi(t, t_0)\, x_0$ (definition 226) is bounded on $t \geq t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. The state-transition form is lemma 443. A bound on Φ bounds each response by $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 445 (Uniform stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is uniformly stable (definition 360) if and only if there is a single C with $\|\Phi(t, t_0)\, x\| \leq C\, \|x\|$ for all $t \geq t_0$ and all x.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1), with the margin independent of t_0.*

Proof. As lemma 443, with the margin and the bound both carried outside the quantifier on t_0. □

Theorem 446 (Asymptotic stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally asymptotically stable (definition 362) if and only if, for every t_0 and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $t \geq t_0$ and tends to 0.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(2).*

Proof. The stability conjunct is theorem 444. For attractivity an arbitrary x_0 is rescaled into the attractivity ball and the limit scales back; conversely every trajectory is a response (theorem 235). □

Theorem 447 (Global exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is globally exponentially stable (definition 368) if and only if there are $c > 0$ and $0 \leq \rho < 1$ with*

$$\|\Phi(t, t_0)\, x\| \leq c\, \rho^{t-t_0}\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. The estimate along every trajectory is the estimate applied to its initial state, since a trajectory is its own homogeneous response; conversely the response from x is a trajectory. □

Theorem 448 (Exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 367) if and only if there are $c > 0$ and $0 \le \rho < 1$ with $\|\Phi(t, t_0)\, x\| \le c\, \rho^{t-t_0}\, \|x\|$ for all $t \ge t_0$ and all x: the condition of theorem 447.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. As in the global case; the radius of the local definition is absorbed by the scaling argument, and radius 1 serves in the converse. □

Lemma 449 (Local exponential stability is global, for a discrete-time linear system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 367) if and only if it is globally exponentially stable (definition 368).*

Proof. Both are equivalent to the same estimate: theorem 448 and theorem 447. □

Theorem 450 (Instability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is unstable (definition 359) if and only if there are t_0 and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $t \ge t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(4).*

Proof. The negation of theorem 444. □

8.3.2 The Schur condition

Nothing in this subsection mentions a system: the Schur condition, decay of A^k, and a geometric envelope for $\|A^k\|$ are the same condition on the matrix. The proof runs through Gelfand's formula, as for the Hurwitz condition, with powers in place of the exponential.

Definition 451 (Schur matrix)**.** A real square matrix A is *Schur* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $|\mu| < 1$. The discrete-time counterpart of definition 401.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.3.

Theorem 452 (A Schur matrix has geometrically decaying powers)**.** *If A is Schur (definition 451) then there are $C > 0$ and $0 \le \rho < 1$ with*

$$\|A^k\| \le C\, \rho^k, \qquad \forall\, k \in \mathbb{N}.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. The spectral radius of $A_{\mathbb{C}}$ is below 1, so by Gelfand's formula (lemma 405) some power has $\|A^m\| < 1$; fix $\|A^m\| \le c < 1$ with $c > 0$, and let M bound $\|A^r\|$ for $r < m$. Writing $k = r + mq$ with $r < m$, submultiplicativity gives $\|A^k\| \le Mc^q$, and with $\rho = c^{1/m}$, $c^q = \rho^{mq} \le \rho^k/c$. □

Theorem 453 (Geometric decay of the state under a Schur matrix)**.** *If A is Schur (definition 451) then there are $C > 0$ and $0 \le \rho < 1$ with $\|A^k x\| \le C\, \rho^k\, \|x\|$ for every k and x.*

Proof. theorem 452 and $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$. □

Theorem 454 (Decay of the matrix powers forces the Schur condition)**.** *If $A^k \to 0$ as $k \to \infty$ then A is Schur (definition 451).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. Suppose $|\mu| \geq 1$ for an eigenpair (μ, v) of $A_{\mathbb{C}}$ with $v \neq 0$. Complexification is continuous, so $A_{\mathbb{C}}^k v \to 0$; but $A_{\mathbb{C}}^k v = \mu^k v$ (lemma 209) has norm $|\mu|^k \|v\| \geq \|v\| > 0$. □

Theorem 455 (Pointwise decay of the state forces the Schur condition)**.** *If $A^k x \to 0$ as $k \to \infty$ for every real state x, then A is Schur (definition 451).*

Proof. The standard basis vectors return the columns of A^k, so $A^k \to 0$; apply theorem 454. □

Theorem 456 (Schur is decay of the matrix powers)**.** *A real square matrix A is Schur (definition 451) if and only if $A^k \to 0$ as $k \to \infty$.*

Proof. Forwards, theorem 452 supplies a geometric envelope; backwards is theorem 454. □

Theorem 457 (Schur is a geometric envelope for the matrix powers)**.** *A real square matrix A is Schur (definition 451) if and only if there are $C > 0$ and $0 \leq \rho < 1$ with $\|A^k\| \leq C\rho^k$ for all k.*

Proof. Forwards is theorem 452; backwards, the envelope forces $A^k \to 0$ and theorem 456 applies. □

8.3.3 Hespanha's Theorem 8.3

Clauses (1) and (3) of the theorem, marginal stability and instability, ask that the Jordan blocks at eigenvalues of magnitude one be 1×1; as in continuous time they need a Jordan normal form and are not recorded here.

Theorem 458 (Asymptotic stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally asymptotically stable (definition 362) if and only if A is Schur (definition 451): every eigenvalue of A has magnitude strictly smaller than* 1*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. For constant A the state transition matrix is A^{t-t_0} (theorem 231), so theorem 446 turns the statement into one about $A^k x_0$. If the responses converge then A is Schur (theorem 455); conversely a Schur A has a geometric envelope $\|A^k x\| \leq C\rho^k \|x\|$ (theorem 453), which bounds the responses and makes them vanish. □

Theorem 459 (Exponential stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally exponentially stable (definition 367) if and only if A is Schur (definition 451).*

Together with theorem 458 this says the two notions coincide for a time-invariant system.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. Against theorem 448, which already asks for a geometric envelope $\|A^{t-t_0}x\| \leq c\rho^{t-t_0}\|x\|$. Forwards, at $t_0 = 0$ the envelope makes $A^k x \to 0$, so A is Schur (theorem 455); backwards is theorem 453. □

8.3.4 The discrete Lyapunov equation

The discrete-time counterpart of $PA + A^\mathsf{T}P = -Q$ is $A^\mathsf{T}PA - P = -Q$. Unlike continuous time, the solution needs no integral: it is the convergent series $\sum_k (A^\mathsf{T})^k Q A^k$.

Definition 460 (Discrete Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *discrete-time Lyapunov equation* for A with forcing Q when

$$A^\mathsf{T}PA - P = -Q.$$

Along $x(t+1) = Ax(t)$ the quadratic form $V(x) = x^\mathsf{T}Px$ then changes by $V(Ax) - V(x) = -x^\mathsf{T}Qx$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.4.

Lemma 461 (Decrease of the quadratic form along a discrete LTI system)**.** *If $A^\mathsf{T}PA - P = -Q$ (definition 460), then for every x,*

$$(Ax)^\mathsf{T}P(Ax) - x^\mathsf{T}Px = -x^\mathsf{T}Qx.$$

Proof. $(Ax)^\mathsf{T}P(Ax) = x^\mathsf{T}(A^\mathsf{T}PA)x$, and $A^\mathsf{T}PA = P - Q$. □

Lemma 462 (The transpose of a Schur matrix is Schur)**.** *If A is Schur (definition 451) then so is A^T.*

Proof. A and A^T have the same characteristic polynomial, hence the same spectrum over $\mathbb{C}$, and the spectrum is the set of eigenvalues (lemma 205). □

Theorem 463 (Discrete Lyapunov equation for a Schur matrix)**.** *If A is Schur (definition 451), then for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 460), and P is the unique matrix solution.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take $P = \sum_k (A^\mathsf{T})^k Q A^k$. It converges because both A and A^T are Schur (lemma 462), so their powers decay geometrically (theorem 452). Shifting the index, $A^\mathsf{T}PA = P - Q$. The first term is Q and the others are positive semidefinite, so P is positive definite (lemma 424). For uniqueness, a difference W of two solutions satisfies $W = A^\mathsf{T}WA$, hence $W = (A^\mathsf{T})^k W A^k \to 0$. □

Theorem 464 (A discrete Lyapunov certificate proves the spectrum lies in the unit disc)**.** *If P and Q are positive definite and $A^\mathsf{T}PA - P = -Q$, then A is Schur (definition 451).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Complexify the equation and pair it against an eigenvector v with eigenvalue μ: the left-hand side is $(|\mu|^2 - 1)\, v^*Pv$ and the right-hand side is $-v^*Qv$. Both quadratic forms are strictly positive (lemma 204), so $|\mu|^2 < 1$. □

Theorem 465 (The discrete Lyapunov inequality certifies the Schur condition)**.** *If P is positive definite and $-(A^\mathsf{T}PA - P)$ is positive definite, then A is Schur (definition 451).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take the forcing to be minus the residual and apply theorem 464. □

Theorem 466 (Schur is solvability of the discrete Lyapunov equation for every forcing)**.** *A real square matrix A is Schur (definition 451) if and only if for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 460), unique among all matrix solutions.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards is theorem 463. Backwards, instantiate the forcing at $Q = I$ and apply theorem 464. □

Theorem 467 (Schur is the existence of one discrete Lyapunov certificate)**.** *A real square matrix A is Schur (definition 451) if and only if some positive definite P makes $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards, solve the equation with $Q = I$ (theorem 463); the residual is then $-I$. Backwards is theorem 465. □

8.3.5 Hespanha's Theorem 8.4

Theorem 468 (Hespanha's Theorem 8.4)**.** *For the time-invariant system $x^+ = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 362).*
2. *The origin is exponentially stable (definition 367).*
3. *A is Schur (definition 451).*
4. *For every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 460), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Every clause is tied to clause 3: $1 \Leftrightarrow 3$ is theorem 458, $2 \Leftrightarrow 3$ is theorem 459, $3 \Leftrightarrow 4$ is theorem 466, and $3 \Leftrightarrow 5$ is theorem 467. □

Chapter 9

Lyapunov's indirect method

Near an equilibrium a nonlinear system behaves like its linearization: if the Jacobian is Hurwitz the equilibrium is locally exponentially stable, and if the Jacobian has an eigenvalue with positive real part it is unstable. Both branches run the same way. A quadratic certificate for the linear system (theorem 426, theorem 442) is a certificate for the nonlinear one as well, once the linearization error is absorbed near the equilibrium.

9.1 The linearization error

Lemma 469 (Absorbing the linearization remainder)**.** *Let f be C^1 with $Df(x_{\mathrm{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\mathrm{eq}}\| < r$,*

$$\left| D\,q_M(x)\,[\,f(x) - f(x_{\mathrm{eq}}) - A(x - x_{\mathrm{eq}})\,]\right| \le c\,\|x - x_{\mathrm{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\mathrm{eq}}),\, x - x_{\mathrm{eq}}\rangle$.

The bracket is the first-order remainder of f at x_{eq}. The point is that c is arbitrary*: near enough to the equilibrium the remainder perturbs $\dot q_M$ by less than any prescribed multiple of $\|x - x_{\mathrm{eq}}\|^2$, which is the order of $\dot q_M$ itself. This is why the linearization decides stability.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7 (the quadratic-Lyapunov proof of Lyapunov's indirect method).*

Proof. Write $y = x - x_{\mathrm{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \le \eta\|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \le 2\|M\|\,\|y\|\,\|e\|$, so the product is at most $2\|M\|\eta\,\|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \le c$. □

9.2 The stable branch

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 470 (Lyapunov's indirect method, stable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1, and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then x_{eq} is locally exponentially stable (definition 297) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P; after absorbing the $o(\|x - x_{\text{eq}}\|)$ linearization remainder on a small ball, $V(x) = (x - x_{\text{eq}})^{\mathsf{T}} P (x - x_{\text{eq}})$ is an exponential Lyapunov function with exponent 2 there, so theorem 336 applies. □

9.3 Time-varying systems

For a time-varying system the linearization is itself time-varying, and exponential stability transfers between the two in both directions. The converse Lyapunov function of the nonlinear system serves for its linearization; conversely, the solution of the differential Lyapunov equation of the linearization serves for the nonlinear system.

Theorem 471 (Exponential stability passes to the linearization)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\text{eq}}\| < r\}$, and $f(t, \cdot)$ and $\partial f / \partial x(t, \cdot)$ Lipschitz on D, uniformly in t. If x_{eq} is locally exponentially stable (definition 297), then it is a locally exponentially stable equilibrium of the linearization*

$$\dot{x} = A(t)(x - x_{\text{eq}}), \qquad A(t) = \frac{\partial f}{\partial x}(t, x_{\text{eq}}).$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15, "only if" direction. Khalil's bounded Jacobian is the Lipschitz bound on $f(t, \cdot)$.*

Proof. Let V be the converse Lyapunov function of the nonlinear system (theorem 339). Along the linearization,

$$\dot{V}_{\text{lin}} = \dot{V} + \frac{\partial V}{\partial x}(A(t)(x - x_{\text{eq}}) - f(t, x)) \leq -c_3 \|x - x_{\text{eq}}\|^2 + c_4 L_1 \|x - x_{\text{eq}}\|^3,$$

by **??**. On a small ball this is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$, so V is an exponential Lyapunov function for the linearization (theorem 336). □

Theorem 472 (Lyapunov's indirect method for time-varying systems)**.** *Under the hypotheses of theorem 471, if x_{eq} is a locally exponentially stable equilibrium of the linearization $\dot{x} = A(t)(x - x_{\text{eq}})$, $A(t) = \frac{\partial f}{\partial x}(t, x_{\text{eq}})$, then it is locally exponentially stable for $\dot{x} = f(t, x)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.13. Khalil's bounded Jacobian is the Lipschitz bound on $f(t, \cdot)$.*

Proof. In coordinates the linearization is $\dot{v} = A(t)v$ with A continuous and bounded, and exponentially stable (lemma 312). Theorem 399 with $Q = I$ gives a bounded $P(t)$ with $c_1\|x\|^2 \leq x^{\mathsf{T}} P(t) x \leq c_2 \|x\|^2$ and derivative $-x^{\mathsf{T}} x$ along the linearization. Along f, $V = (x - x_{\text{eq}})^{\mathsf{T}} P(t)(x - x_{\text{eq}})$ has derivative

$$\dot{V} \leq -c_3 \|x - x_{\text{eq}}\|^2 + 2\|P\| L_1 \|x - x_{\text{eq}}\|^3,$$

by **??**, which is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$ on a small ball; conclude by theorem 336. □

Theorem 473 (Exponential stability and the linearization)**.** *Under the hypotheses of theorem 471, x_{eq} is locally exponentially stable for $\dot{x} = f(t, x)$ if and only if it is for the linearization.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15.*

Proof. Theorem 471 and theorem 472. □

9.4 The unstable branch

The quadratic certificate is fed to Chetaev's theorem (theorem 357).

Theorem 474 (Lyapunov's indirect method, unstable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $f(x_{\text{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable (definition 289) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems *(Lyapunov's indirect method).*

Proof. Build the real quadratic certificate supplied by theorem 442, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. □

Theorem 475 (Instability of a linear system)**.** *If $A \in \mathbb{R}^{n\times n}$ has a complex eigenvalue μ with $\operatorname{Re}\mu > 0$, then the origin is unstable (definition 289) for any system on $\mathbb{R}^n$ whose state equation is $\dot{x} = Ax$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7.*

Proof. The field $x \mapsto Ax$ is C^1, vanishes at 0 and is its own Jacobian, so this is theorem 474 at $x_{\text{eq}} = 0$. □

Chapter 10

Controllability and observability

10.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state matrix A, input matrix B, and output matrix C, this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

10.1.1 Observability matrix

Definition 476 (Observability matrix)**.** The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n\times n}$ and $C \in \mathbb{F}^{p\times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C\,A \\ C\,A^2 \\ \vdots \\ C\,A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np)\times n}.$$

Rows are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, p$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Definition 477 (Observability)**.** A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C\,A^k\,x = 0 \qquad \text{for every } k = 0, 1, \ldots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 483 has real content.

Lemma 478 (Entries of the observability matrix)**.** *Block-row entry shape: at row (k, i) and column j, the observability matrix coincides with the (i, j) entry of $C\,A^k$:*

$$\mathcal{O}(A, C)_{(k,i),\,j} = (C\,A^k)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 476. □

Lemma 479 (Action of the observability matrix)**.** *For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \operatorname{Fin} n \times \operatorname{Fin} p$,*

$$(\mathcal{O}(A, C) \cdot x)_{(k,i)} = (C\, A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 483.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 476). □

10.1.2 Controllability matrix

Definition 480 (Controllability matrix)**.** The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} B & A\,B & A^2\,B & \cdots & A^{n-1}\,B \end{bmatrix} \in \mathbb{F}^{n\times(nm)}.$$

Columns are indexed by $\operatorname{Fin} n \times \operatorname{Fin} m$, so that A^k is available without casting $k : \operatorname{Fin} n$ through Fin.val.

Lemma 481 (Entries of the controllability matrix)**.** *Block-column entry shape: at row i and column (k, j), the controllability matrix coincides with the (i, j) entry of $A^k\,B$:*

$$\mathcal{C}(A, B)_{i,\,(k,j)} = (A^k\,B)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 480. □

Definition 482 (Controllability)**.** A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \ldots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k\,B\,u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 485 has real content.

10.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 483 (Observability via a trivial kernel)**.** *A finite-dimensional linear system (A, C) is observable in the sense of definition 477 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:*

$$\operatorname{IsObservable}(A, C) \iff (\forall x,\ \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge lemma 479, which identifies the (k, i)-coordinate of $\mathcal{O}(A, C) \cdot x$ with the i-coordinate of $(C\,A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. □

10.1.4 Rank characterizations

Theorem 484 (Observability rank test)**.** *A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.*

$$\operatorname{rank} \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 483 with the matrix-level bridge $(\forall x,\ M \cdot x = 0 \Rightarrow x = 0) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{O}(A, C)$. □

Theorem 485 (Controllability rank test)**.** *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\operatorname{rank} \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y,\ \exists x,\ M\,x = y) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{C}(A, B)$. □

10.1.5 Hautus observability

Definition 486 (Unobservable subspace)**.** The *unobservable subspace* of (A, C) is the A-invariant subspace

$$\mathcal{N}(A, C) \;=\; \{\, v \in \mathbb{C}^n \;:\; C\,A^k\,v = 0 \text{ for every } k = 0, 1, \dots, n-1 \,\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 487 reads as content.

Theorem 487 (Observability via the unobservable subspace)**.** *A finite-dimensional system (A, C) is observable in the sense of definition 477 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 486 against definition 477. □

Lemma 488 (A-invariance of the unobservable subspace)**.** *The unobservable subspace is closed under the action of A: for every $v \in \mathcal{N}(A, C)$, also $A\,v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C\,A^n\,v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$-linear combination of $C\,A^i\,v$ for $i < n$. Each of those is zero, so the combination is zero. □

Definition 489 (Hautus observability matrix)**.** The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) \;=\; \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p)\times n}.$$

Lemma 490 (Kernel of the Hautus matrix)**.** *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (A\,v = \mu\,v \ \wedge \ C\,v = 0).$$

Proof. □

Theorem 491 (Unobservability gives a Hautus rank drop)**.** *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$*.*

Proof. The unobservable subspace definition 486 is nontrivial by theorem 487. It is A-invariant by lemma 488, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via lemma 490. □

Theorem 492 (A Hautus rank drop gives unobservability)**.** *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k, $A^k v = \mu^k v$, so $C\,A^k\,v = \mu^k\,Cv = 0$ for every k. The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 493 (Hautus (PBH) test for observability)**.** *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\mathrm{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C},\ \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 491 (failure direction) and theorem 492 (converse). □

10.1.6 Hautus controllability

Definition 494 (Hautus controllability matrix)**.** The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H^{\mathrm{ctrl}}_{A,B}(\mu) \ = \ [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 495 (Hautus (PBH) test for controllability)**.** *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H^{\mathrm{ctrl}}_{A,B}(\mu)$ has full row rank:*

$$\mathrm{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C},\ \mathrm{rank}\, H^{\mathrm{ctrl}}_{A,B}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 493, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from 'MatrixAlgebra.Rank', and finally identify the transposed Hautus matrices. □

10.2 Reachable subspace and Kalman decomposition

Definition 496 (Reachable subspace)**.** The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Reference: Hespanha, *Linear Systems Theory*.

Lemma 497 (Membership in the reachable subspace)**.** *Let $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n\times(nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \quad \iff \quad \exists u : \operatorname{Fin}(n) \times \operatorname{Fin}(m) \to \mathbb{F}, \qquad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B)$ is the reachable subspace.

Proof. □

Theorem 498 (Controllability via the reachable subspace)**.** *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Lemma 499 (A-invariance of the reachable subspace)**.** *The reachable subspace is A-invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Definition 500 (Kalman decomposition)**.** A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A.

References: Hespanha, *Linear Systems Theory*; Kailath, *Linear Systems*.

Theorem 501 (Existence of the Kalman decomposition)**.** *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A.*

References: Hespanha, Linear Systems Theory*; Kailath,* Linear Systems*.*

Proof. □

Definition 502 (Adapted coordinate equivalence)**.** Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 503 (Block-zero pattern in adapted coordinates)**.** *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \qquad B' = \begin{bmatrix} * \\ \\ 0 \\ 0 \end{bmatrix}, \qquad C' = \begin{bmatrix} 0 & * & 0 & * \end{bmatrix}.$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A.

References: Hespanha, Linear Systems Theory*; Kalman, "Mathematical Description of Linear Dynamical Systems" (1963); Kailath,* Linear Systems*.*

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □

Chapter 11

Optimization

11.1 Preliminaries

11.2 Fixed-point flows